



MONITORING REPORT ITAÚBA REDD+

Document Prepared By CARBONEXT Tecnologia em Soluções Ambientais LTDA

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Prepared By	Carbonext Tecnologia em Soluções Ambientais LTDA
Validation/Verification Body	To be defined
GHG Accounting/ Crediting Period	14 December 2021 – 13 December 2051; 30-year total period
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1 SUMMARY OF PROJECT BENEFITS

1.1 Unique Project Benefits

Outcome or Impact	Achievements during the Monitoring Period	Section Reference	Achievements during the Project Lifetime
1) Conservation of 14,400.6 hectare of Amazon Rainforest	The Itaúba REDD+ Project was able to preserve 152 hectares of Amazon rainforest in this monitoring period. A Forest Integrity Plan was developed with all the strategies to combat illegal deforestation. Weekly patrol has been conducted in the project area (PA) since before start date and in November 2022 the app ArcGIS QuickCapture was implemented to support the patrol. Remote monitoring of PA has been done since 3 June 2022 by a monitoring and alerting system called MonitoraCarbon, Signs were installed in the vicinity of the project area informing that this is an area monitored by the Itaúba REDD+ Project. Firebreaks have been opened to prevent fires from spreading into the project area. Fire brigade training was scheduled to take place in November 2022, but was cancelled due to protests on roads in Mato Grosso State. This training has been rescheduled for April 2023.	2.2.7	The Itaúba REDD+ Project has been able to preserve 152 hectares of forest so far. A Forest Integrity Plan was developed with all the strategies to combat illegal deforestation. Weekly patrol has been conducted in the project area (PA) since before start date and in November 2022 the app ArcGIS QuickCapture was implemented to support the patrol. Remote monitoring of PA has been done since 3 June 2022 by a monitoring and alerting system called MonitoraCarbon, Signs were installed in the vicinity of the project area informing that it is an area monitored by the Itaúba REDD+ Project. Firebreaks was opened to prevent fires from spreading into the project area. Fire brigade training was scheduled to take place in November 2022, but was cancelled due to protests on roads in Mato Grosso State. This training has been rescheduled for April 2023.
2) Monitoring of deforestation and forest degradation via satellite	The project implemented a modern remote monitoring system to the PA, which works by artificial intelligence capable of processing a large volume of data and issuing alerts to	2.2.7	The project implemented a modern remote monitoring system to the PA, which works by artificial intelligence capable of processing a large volume of data and issuing alerts to the

	the monitoring agents. This system, called MonitoraCarbon, began to be developed in January 2022 and was effectively implemented in June 2022.		monitoring agents. This system, called MonitoraCarbon, began to be developed in January 2022 and was effectively implemented in June 2022.
3) Improvement in health and quality of life for the local community	Offering of the course “Cultivating Emotional Balance” for the teachers and managers of the Papa João Paulo II School.	4.3	Offering of the course “Cultivating Emotional Balance” for the teachers and managers of the Papa João Paulo II School.
4) Conducting interviews and workshops with the communities involved to present and design project activities	Onsite visits to understand the urgent demands and vulnerabilities of the communities; Build of good relationship with the communities’ schools; design and planning of the projects activities to improve quality of life and well-being.	4.3	Onsite visits to understand the urgent demands and vulnerabilities of the communities; Build of good relationship with the communities’ schools; design and planning of the projects activities to improve quality of life and well-being.

1.2 Standardized Benefit Metrics

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
GHG emission reductions & removals	Net estimated emission removals in the project area, measured against the without-project scenario	Not applicable	N/A	Not applicable
	Net estimated emission reductions in the project area, measured against the without-project scenario	53,913 tCO ₂ e	3.2.4	53,913 tCO ₂ e
Forest ¹ cover	For REDD ² projects: Number of hectares of reduced forest loss in the project area measured against the without-project scenario	152 hectares	3.2.1	152 hectares
	For ARR ³ projects: Number of hectares of forest cover increased in the project area measured against the without-project scenario	Not applicable	N/A	Not applicable
Improved land management	Number of hectares of existing production forest land in which IFM ⁴ practices have occurred as a result of the project's activities, measured against the without-project scenario	Not applicable	N/A	Not applicable
	Number of hectares of non-forest land in which improved land management has occurred as a result of the project's activities, measured against the without-project scenario	Not applicable	N/A	Not applicable

¹ Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (*VCS Program Definitions*)

² Reduced emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (*VCS Program Definitions*)

³ Afforestation, reforestation and revegetation (ARR) - Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing and/or restoring vegetative cover through the planting, sowing and/or human-assisted natural regeneration of woody vegetation (*VCS Program Definitions*)

⁴ Improved forest management (IFM) - Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood and fuelwood (*VCS Program Definitions*)

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
Training	Total number of community members who have improved skills and/or knowledge resulting from training provided as part of project activities	15	2.2.7 4.3.1	15
	Number of female community members who have improved skills and/or knowledge resulting from training provided as part of project activities	Not applicable	N/A	Not applicable
Employment	Total number of people employed in of project activities, ⁵ expressed as number of full time employees ⁶	Not applicable	N/A	Not applicable
	Number of women employed in project activities, expressed as number of full time employees	Not applicable	N/A	Not applicable
Livelihoods	Total number of people with improved livelihoods ⁷ or income generated as a result of project activities	Data not available	N/A	Data not available
	Number of women with improved livelihoods or income generated as a result of project activities	Data not available	N/A	Data not available
Health	Total number of people for whom health services were improved as a result of project activities, measured against the without-project scenario	14	4.3.1	14
	Number of women for whom health services were improved as a result of project activities, measured against the without-project scenario	14	4.3.1	14
Education	Total number of people for whom access to, or quality of, education was improved	Data not available	N/A	Data not available

⁵ Employed in project activities means people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, sub-contracted workers and community members that are paid to carry out project-related work.

⁶ Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region or economic territory (adapted from UN System of National Accounts (1993) paragraphs 17.14[15.102];[17.28])

⁷ Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. *The Sustainable Livelihood Approach to Poverty Reduction*. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
	as a result of project activities, measured against the without-project scenario			
	Number of women and girls for whom access to, or quality of, education was improved as a result of project activities, measured against the without-project scenario	Data not available	N/A	Data not available
Water	Total number of people who experienced increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	Not applicable	N/A	Not applicable
	Number of women who experienced increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	Not applicable	N/A	Not applicable
Well-being	Total number of community members whose well-being ⁸ was improved as a result of project activities	14	4.3.1	14
	Number of women whose well-being was improved as a result of project activities	14	4.3.1	14
Biodiversity conservation	Change in the number of hectares significantly better managed by the project for biodiversity conservation, ⁹ measured against the without-project scenario	14,400.60	2.1.1	14,400.60

⁸ Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g. Training, Employment, Health, Education, Water, etc.), but could also include other benefits such as empowerment of community groups, strengthened legal rights to resources, conservation of access to areas of cultural significance, etc.

⁹ Biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation.

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
	Number of globally Critically Endangered or Endangered species ¹⁰ benefiting from reduced threats as a result of project activities, ¹¹ measured against the without-project scenario	Data not available	N/A	Data not available

¹⁰ Per IUCN's Red List of Threatened Species

¹¹ In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit

2 GENERAL

2.1 Project Description

2.1.1 Implementation Description

The Itaúba REDD+ Project comprises an area of 14,400.60 hectares of Amazon rainforest located in the Itaúba and Marcelândia municipalities, northern of Mato Grosso State. The project reduces greenhouse emissions through forest conservation. The project area comprises 9 rural properties, 7 located in Marcelândia and 2 in Itaúba.

During this first monitoring period, the project has implemented activities to protect and preserve its area, as well as to promote the well-being of the community near the project area. The activities include:

- Implementation of the Forest Integrity Plan, with all the strategies to combat illegal deforestation and forest fires.
- Training in QuickCapture App, used to record the patrols in the project area and send evidence quickly.
- Implementation of MonitoraCarbon, a remote system that detects deforestation in the project area and issue alerts.
- Installation of signs in the vicinity of the project area informing that it is an area monitored by the Itaúba REDD+ Project.
- Opening of firebreaks to prevent fire spread inside the project area.
- Offering of the course "Cultivating Emotional Balance" for teachers and managers of the Papa João Paulo II School.

These activities were able to avoid any deforestation predicted to occur in the project area in this monitoring period, thus preserving 152 ha of forest in the Amazon Biome. The deforestation that occurred in the leakage belt was less than predicted for this period, which represents no activity displacement leakage.

The total GHG emission reductions generated in this monitoring period was 53,913 tCO_{2e}. It is a difference of 17.65% between ex-ante and ex-post emission reductions, since there was no emission from unavoided deforestation, forest fires and activity displacement leakage.

The project area has been monitored by surveillance patrols and remote sensing. The remote monitoring is done by MonitoraCarbon, an online system that weekly records satellite images, optical and radar sensors with the application of artificial intelligence to analyze large amounts of data and issue immediate alerts. The deforestation in the leakage belt has also been monitored by remote sensing, using optical images from Sentinel 2 orbital sensors. For periods of high incidence of clouds, the classification was made by radar images (SAR) available from the Sentinel 1 sensor and thus measured all changes in land use within the period defined for this MR.

The Leakage Management Area established in PD are the state schools in the municipalities of Itaúba and Marcelândia, which are considered community in the project. The first activity in this area was the offering of the course "Cultivating Emotional Balance" to teacher and managers of the Papa João Paulo II School, in Itaúba. This activity was considered a priority specially due to the post-covid emotional overload, being requested by the school. Other activities are planned to take place during the next MR, which will be more focused on prevention of activity displacement leakage.

Regarding non-permanence risks, reviewed for this MR, a decrease was identified due to the Natural Risk which is now 2 and the Internal Risk which is 15. The Natural Risk was considered lower because during the monitoring period there was no fire in the project area. The Internal Risk, on the other hand, was lower because of the decrease in the ratio of NPV_baseline/NPV_project_with_VCUs, which decreases the opportunity cost. This achievement in Natural Risk is the result of the set of actions of the Forest Integrity Plan, which consist in strategies to prevent and combat forest fires and illegal deforestation (further information in section 2.2.7). Among these strategies, patrols were carried out weekly by the local monitor with the QuickCapture App, which allows him to mark geographical coordinates and take pictures with his own cell phone. The data is sent quickly and efficiently, allowing effective action when it is needed. In addition, there is a WhatsApp group among neighbors, that helps communicate threats and join efforts to fight nearby fires. For the next monitoring period, the farm employees will be trained as firefighters. Another strategy to reduce risk is the remote monitoring of the project area by MonitoraCarbon, as explained previously.

2.1.2 Project Category and Activity Type

Project Category: Reducing Emissions from Deforestation and Degradation (REDD+).

Type of Activity: Avoided Unplanned Deforestation & Degradation (AUDD).

Climate, Community and Biodiversity (CCB – Version 3).

Reference of Methodology: Approved VCS Methodology for Avoided Unplanned Deforestation VM0015 V.1.1.

This is a grouped project.

2.1.3 Project Proponent(s)

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2.1.4 Other Entities Involved in the Project

There is no other entity involved in the project.

2.1.5 Project Start Date (G1.9)

Prior to the project, there was a single employee hired by the owners to patrol and carry out maintenance on the farms. The presence of an employee might inhibit illegal invasions, but it is not considered to be enough given the size of the area. In this way, the owners looked for alternatives to intensify the actions of monitoring and preservation of their forests through the execution of the carbon project.

On December 14th, 2021, the owners signed a partnership agreement with Carbonext, firming the intention of implementing a conservation project through a REDD initiative. Together with Carbonext, the owners assumed the commitment to preserve the forest area for 30 years, guaranteeing the reduction of GHG emissions through the intensification of monitoring actions.

Through the partnership with Carbonext, the properties are now monitored remotely by the MonitoraCarbon system, which analyzes satellite images frequently and issues fire and deforestation alerts, facilitating early detection of invasions.

2.1.6 Project Crediting Period (G1.9)

The ITAÚBA REDD+ project the duration of the project is 30 years, with crediting period from December 14, 2021, to December 13, 2051.

2.1.7 Project Location

The ITAÚBA REDD+ project is composed by 9 properties: São Judas Farm, Casa Blanca Farm, Natal Farm, Guanabara Farm, Vale Verde Farm, Ponderosa Farm, Santa Maria Farm, Julian Grimas Farm and Nossa Senhora Salete Farm. The area of this farms is distributed in two municipalities, Marcelândia and Itaúba, which belong to Mato Grosso state, in the central-western region of Brazil (Figure 1).

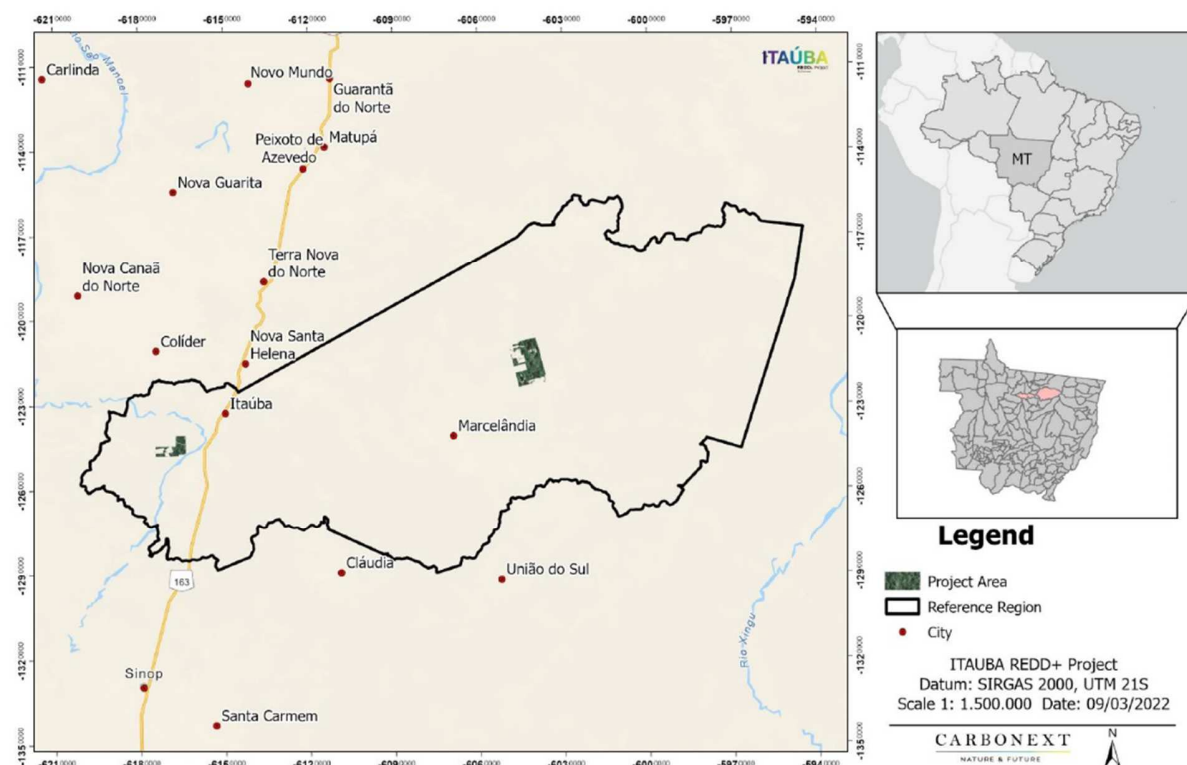


Figure 1: Project location with the regional division of the State of Mato Grosso.

2.1.8 Title and Reference of Methodology

The methodology used in the project is the Methodology for Avoided Unplanned Deforestation VM0015 version 1.1 (3 December 2012).

The tools VT0001 Tool for The Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities – Version 3.0, and AFOLU “Non-Permanence Risk Tool” VCS Version 4, Procedural Document, 19 September 2019, v4.0 were used.

2.1.9 Other Programs (G5.9)

The Itaúba REDD+ Project isn't part of any other programs, such as Emission Trading Programs and other Binding Limits, other Forms of Environmental Credit or Participation under other GHG Programs.

2.1.10 Sustainable Development

The Itaúba REDD+ Project complies with the environmental priorities defined by the Brazilian Federal Administration and its international commitments represented by the elements of the Brazilian NDC, adopted within the framework of the Paris Agreement (COP21). Ever since, Brazil has committed to reduce its emissions in 37% by 2025 and 43% by 2030, when compared with 2005 levels. It has also pledged to end illegal deforestation by 2030 [1]

Due to the increase in deforestation in the Legal Amazon, the Action Plan for the Prevention and Control of Deforestation in the Legal Amazon (PPCDAm; Plano de Ação para a Prevenção e Controle do Desmatamento na Amazônia Legal) came into effect, starting mitigation and ongoing actions to reduce deforestation [2].

The PPCDAm goals are as follow [3]:

- 1) Promote land tenure regularization;
- 2) Promote territorial planning, strengthening protected areas;
- 3) Promote accountability for environmental crimes and infractions;
- 4) Implement shared forest management;
- 5) Prevent and combat the occurrence of forest fires;
- 6) Improve and strengthen forest cover monitoring;
- 7) Promote sustainable forest management;
- 8) Promote the sustainability of agricultural production systems;
- 9) Implement normative and economic instruments to control illegal deforestation.

To attain these international commitments and national goals, an essential component is to ally government initiatives with independent actions, such as the present project. The Itaúba REDD+ Project directly contributes with goals 5), 6) and 7).

The Itaúba Project is also aligned with the Brazilian commitments to the UN Sustainable Development Goals (SDGs), being the more direct contributions to the agenda described as it follows:

Table 1. Project Activities and the SDGs.

SDG	Project Benefits
4.7 - By 2030, enable that students and school staff acquire the knowledge and skills necessary to promote sustainable development, including, among others, through education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship and appreciation of cultural diversity and the contribution of culture to sustainable development.	Expected Community Benefits: The project is expected to have a positive impact on local communities, strengthening awareness and capacity building in activities related to the reduction of deforestation and activities aimed at the conservation of natural resources, such as environmental education activities. Overall, the project aims to empower local communities and promote the principles of forest valuation and natural resource use. Section Reference: 4

13.3 - Improve education, raise awareness and human and institutional capacity on mitigation, adaptation, impact reduction and early warning of climate change	Expected Climate Benefits: the project aims to climate change mitigation by avoiding deforestation of 2,919 hectares in the Amazon Biome, mainly through local surveillance, remote satellite monitoring, and a system of deforestation alerts in the Project Area for early identification of illegal trespassing. Section Reference: 3
15.1.2 - Proportion of important sites for terrestrial and freshwater biodiversity covered by protected areas, by ecosystem type 15.2.1 - Progress in sustainable forest management	Expected Biodiversity Benefits: the project guarantees constant monitoring of the forest integrity inside the Project Area and ensures the conservation of local habitats and biodiversity. Section Reference: 5

The achievements to those goals were monitored during this MR and will continue to be monitored in the next verifications. The results will be publicly available at Verra and Carbonext's website, and will be presented to the communities and other stakeholders during the consultations, respecting the language and cultural diversity.

2.2 Project Implementation Status

2.2.1 Implementation Schedule (G1.9)

The key dates and milestones during the first monitoring period are presented below (Table 2).

Table 2: Key dates and milestone of the ITAÚBA REDD+ Project up to the first Monitoring period.

Date	Milestone(s) in the project's development and implementation
14 th December 2021	Project Start Date – signature of the contract between the proponents
14-18 th February 2022	First meeting with the stakeholders and start of socioeconomic diagnosis in schools and farms
03 th June 2022	Implementation of MonitoraCarbon System
5-11 th June 2022	Second meeting with proponents in the farms and stakeholders (E. E. Paulo Freire, E. E. Pedro Bianchini, E. E. João Paulo II, Marcelândia and Itaúba city hall) to present the project and explain about the validation process
June 2022	Implementation of the relationship channels
17-24 th July 2022	Validation audit on-site
13-19 th October 2022	Third meeting with proponents in the farms and stakeholders – EMBRAPA, EMPAER, SEMA, UFMT, Marcelândia and Itaúba city hall and municipal secretariats, E. E. Paulo Freire, E. E. Pedro Bianchini, E. E. João Paulo II

15 th October 2022	Presentation of Forest Integrity Plan to the employee's farm, which guarantees the integrity of the project areas through the implementation of preventive, combat and mitigation measures for events that can cause GHG emissions
15-16 th October 2022	Implementation of infrastructure, waste management and social diagnosis in the farms
17 th October - 28 th November 2022	Realization of the course "Cultivating Emotional Balance" for teachers and managers of the Papa João Paulo II School, which was carried out in 6 modules (one module per week)
17 th October 2022	Interview with Brazil nut sellers
20-25 th November 2022	Fourth meeting with proponents in the farms and stakeholders - E. E. Paulo Freire, E. E. Pedro Bianchini, E. E. João Paulo II, Brazil Nut Collectors Association, SENAR, SEBRAE
23 th November 2022	Presentation of the Project to the Association of Brazil Nut Collectors in Itaúba-MT, along with preliminary information gathering
24 th November 2022	Training the forest monitor and proponents to use the ArcGIS QuickCapture App
29 th November 2022	Implementation of terrestrial monitoring through the QuickCapture App – first ground patrol using the system
24 th November 2022	Diagnosis for the elaboration of the occupational health and safety risk matrix

2.2.2 Methodology Deviations

Not applicable – No methodology deviations.

2.2.3 Minor Changes to Project Description (*Rules 3.5.6*)

The project design is aligned with the ongoing activities of the Itaúba REDD+ Project Description, presenting until this moment no changes regarding the project description on community, climate, and biodiversity.

2.2.4 Project Description Deviations (*Rules 3.5.7 – 3.5.10*)

Not applicable – No project description deviations.

2.2.5 Grouped Projects

The Itaúba REDD+ Project is a grouped project and does not have new project areas and it is still composed by the 9 properties first described in the PD: São Judas Farm, Casa Blanca Farm, Natal Farm, Guanabara Farm, Vale Verde Farm, Ponderosa Farm, Santa Maria Farm, Julian Grimas Farm and Nossa Senhora Salete Farm.

1) New Project Areas and Communities (G1.13)

On this first monitoring period, no new project areas, stakeholders, and communities have been included in the project, other than the list already identified in the PD.

2) Removed Project Areas and Communities (G1.13)

On this first monitoring period, no project areas, stakeholders nor communities have been removed from list of communities presented, thus this list has not changed.

3) Eligibility Criteria for Grouped Projects (G1.14)

Not applicable.

4) Scalability Limits for Grouped Projects (G1.15)

Not applicable.

5) Risk Mitigation for Grouped Projects (G1.15)

Not applicable.

6) Project Zone Map (G1.13)

Not applicable.

7) Changes to Management (G4.1)

Not applicable.

2.2.6 Risks to the Project (G1.10)

The risks to the Project taking into consideration climate, community and biodiversity remain the same, as indicated in item 2.1.18. in the Project Description, and are listed in the table below:

Table 3: The Project Risk Table

Identify Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed to mitigate the risk	Actions implemented during this monitoring period
Project Non-Permanence Risk	Discontinuity of the project activities on climate, community and biodiversity	To ensure the continuity of the project some actions were taken: - Contract between the project proponents signing the commitment with the project	- The contract between the project proponents was signed to guarantee the commitment with the project activities during its entire lifetime; - The team of specialists worked in the field 4 times during this monitoring period.

		activities during its entire lifetime; - Engagement with the community to maintain their participation in the project management; - Project proponents have the necessary funds to maintain the project activities until the start of the GHG revenue generation; - Team of specialists working on field and in the project PD and verifications.	All onsite visits involved engagement activities with the community and other stakeholders.
Illegal deforestation inside the project area/ Invasion of project area by outsiders	Deforestation inside the project area	Ground patrols and remote sensing monitoring of the project area in order to identify illegal deforestation as soon as possible and take action.	- A Forest Integrity Plan was developed with all the strategies to combat illegal deforestation. For this monitoring period, no deforestation and activity displacement leakage were identified. - Weekly patrols have been carried out in the vicinity of the PA and recorded in the app ArcGIS QuickCapture. - Signs were installed in the vicinity of the PA informing that it is an area monitored by the Itaúba REDD+ Project. - The PA has been monitored via satellite through MonitoraCarbon since June 3, 2022. - Firebreaks were opened to prevent fires from spreading into the PA. - Fire brigade training was scheduled to take place in November 2022 but was cancelled due to protests on roads in Mato Grosso State. This training has been rescheduled for April 2023.

Inefficiency of project activities	Poor results on climate community, or biodiversity aspects, differing from what was expected previously as a result from project activities	Guarantee that the local specific needs are being considered when designing and proposing activities; monitoring of indicators to quantify and qualify obtained results; collect feedback from stakeholders involved.	-Onsite visits to understand the urgent demands and vulnerabilities of the communities; Build of good relationship with the communities' schools; design and planning of the projects activities to improve quality of life and well-being. Offering of the course "Cultivating Emotional Balance" for the teachers and managers of the Papa João Paulo School, which 14 people completed it.
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2.2.7 Benefit Permanence (G1.11)

To maintain and enhance climate, community and biodiversity benefits beyond the project lifetime, a set of actions will be taken over 30-years, such as trainings, biodiversity inventories, educational workshops, among other activities as described in the PD. For this monitoring period, the following activities have been carried out:

Climate

The Itaúba REDD+ Project has developed a set of strategies to maintain the forest integrity of the PA, through intense forest monitoring that avoids deforestation, fires, and inhibit illegal invasions. These strategies compose the "Forest Integrity Plan", which was structured to standardize the activities, establish the flows of actions that must be taken in the field, how the identified occurrences must be reported and how the records must be stored.

The project monitors the area on two fronts. The first is through surveillance patrols that have been made by a local employee (forest monitor) using a motorcycle, and this patrolling has been done since the beginning of the project. Since November 2022, these patrols have been taking place weekly/monthly (depending on the rainy season) and with the help of an application called ArcGIS QuickCapture, which allows the employee in the field to mark georeferenced points of interest (fire outbreaks, deforestation, natural events and invasion) and take pictures with his own cell phone. This data can be collected offline, and on the first internet connection this data is sent to ArcGIS. In this way, Carbonext receives field data immediately, and in case of occurrences it can act more quickly and efficiently. The Operational Procedure "Use of QuickCapture for Monitoring" was developed to implement the use of this tool.

In November 2022, the forest monitor and the landowners were trained to use that app on their cell phones to record the periodic rounds on the property. The training was held by a GIS specialist and MRV manager at Carbonext. The focus of the training was implementing the usage of the app and reinforcing the communication channels between the property workers and the project proponents through this technology.

Orientations about the frequency and path of the monitoring according to the season or summer, accessibility and accident prevention were given.



Figure 2. Training on the ArcGIS QuickCapture app that is helping the monitoring patrols and doing their registration.

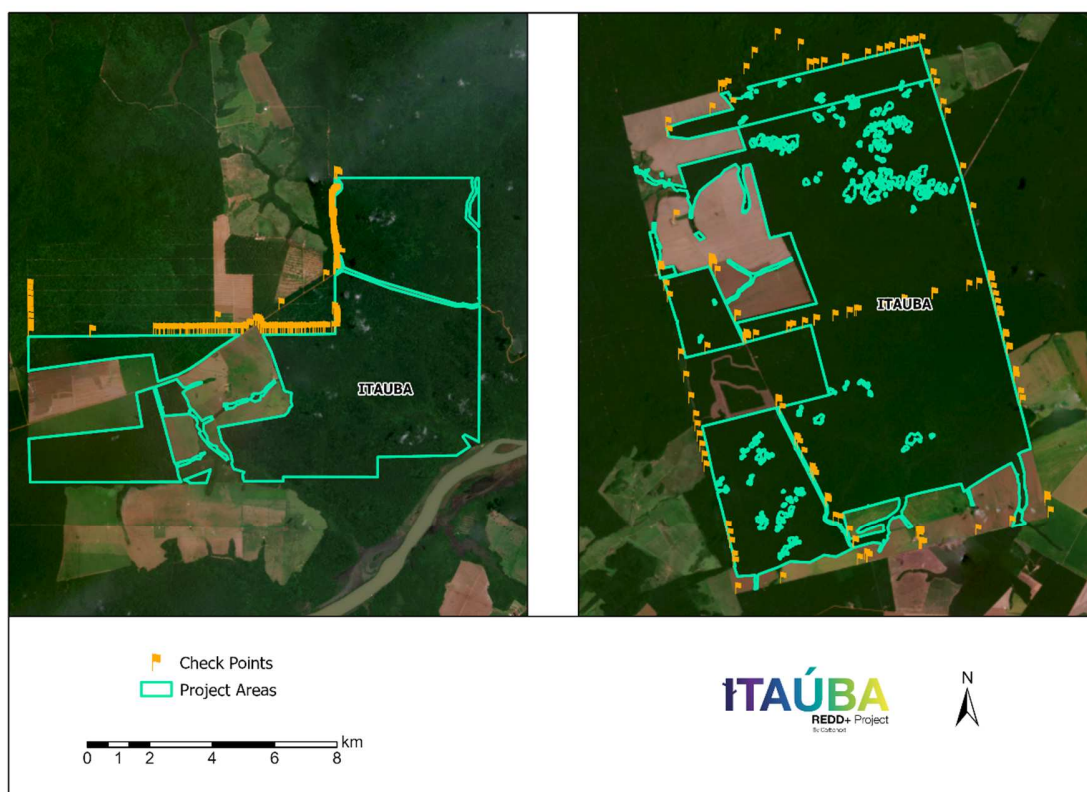


Figure 3. Check points made by the forest monitor during land patrols using the QuickCapture app in the Project Area.

Table 4. Dates of land patrols using the QuickCapture app in the Project Area.

Project Area in Itaúba	Project Area in Marcelândia
29/11/2022	02/12/2022
27/12/2022	06/12/2022
	07/12/2022
	23/12/2022
	28/12/2022
	29/12/2022

The project area has also been monitored by remote sensing with high-resolution satellite images. In June 2022, Carbonext implemented a deforestation alert system called MonitoraCarbon, which detects probable changes in land use and issues alerts of deforestation and fire quickly. This system consists of artificial intelligence to process optical and radar images from sensors such as Landsat and Sentinel (1 and 2) that detect any change in forest cover on a weekly basis and trigger alerts for monitoring teams to check and take necessary actions to contain the deforesting agent. Additionally, MonitoraCarbon automatically integrates data from other deforestation systems in the Amazon such as Deter, SAD, Queimadas, MapBiomas Alerta and Global Forest Watch in order to provide more robustness and assertiveness in detections. In the first half of 2022, this deforestation alert system was being developed and calibrated, so it was effectively implemented in June/2022.

The project proponents installed signage near the project area, informing that it is a private area monitored by the Itaúba REDD+ Project. The signs also inform the project ID and the total area in hectare.



Figure 4. Sign informing about the Itaúba REDD+ Project.

The project will provide for constant forest fire brigade training, which will be offered to the rural workers of the project farms and neighboring farms. The first training was scheduled to take place in November 2022, however it was cancelled due to security issues in the region related to the protests that took place on the roads in Brazil after the presidential elections. This training has been rescheduled to take place in April 2023 and will be reported in the next MR. Moreover, there is a Whatsapp group among neighbors to communicate about possible fires and illegal land invasions, so that they can act quickly in case of an incident. Neighbors can also share resources, such as a water truck for fighting fires.

Community

Both in the internal (farm) and external (schools) community of the project, lectures were held to introduce the Itaúba REDD+ Project and its impacts, clarify doubts about the project and other related issues, such as how the carbon credit market works and the benefits of forest preservation. These activities aimed to engage the whole community in the project's goals and to introduce them to environmental issues.

Regarding the external community, in addition to the lectures, questionnaires were applied to gather more specific information about the teachers and students in the schools.

Moreover, the course "Cultivating Emotional Balance" was offered to teachers and managers of the Papa João Paulo II School, in Itaúba. This activity was considered a priority since the school itself requested it due to the post-covid emotional overload. Further information is in section 4. Community.

Biodiversity

The monitoring of forest integrity, through land patrol and the MonitoraCarbon system described previously in this section, is linked to the maintenance of the vegetation cover and the biodiversity present in the project area. Thus, fragmentation, loss of biodiversity and habitat are prevented, and consequently the fauna and flora species are protected as well.

2.3 Stakeholder Engagement

2.3.1 Stakeholder Access to Project Documents (G3.1)

During 2022, the Itaúba REDD+ Project team made four visits in the project region. The visits had the objective of presenting the project to public school students and teachers, and also to other stakeholders, such as secretaries of municipal public institutions of social assistance, secretaries of education, secretaries of health, secretaries of development, among others; and federal public and private institutions, besides the local association of Brazil nut collectors.

During the visits in February and June project summaries were made available physically in booklets, as well as virtually via e-mail.

The full project documentation was made available during the visits in October and November. This material was made available to the municipal secretaries, schools, UFMT, Embrapa and EMPAER (Figure 5). All project information material is written in Portuguese, the main language spoken among the parties involved in the project. Doubts were also clarified, explaining the objective of the Itaúba REDD+ project and the CCB standards.



Figure 5: Dissemination and distribution of summaries of the Itaúba REDD+ project to communities and other stakeholders: (A) Federal University of Mato Grosso (UFMT); (B) EMPAER-MT; (C) City Hall of Marcelândia; (D) Brazilian Agricultural Research Company – EMBRAPA; (E) Papa João Paulo II State School.

During all site visits, 494 people received a presentation of the project goals and activities. The table below details the number stakeholders, people and departments that received information from the project.

Table 5. Number stakeholders, people and departments that received information from the project.

Number of Stakeholders	Number of People	Number of Departments*
11	494	47

*Departments include secretaries and number of school classes

2.3.2 Dissemination of Summary Project Documents (G3.1)

Communication mechanisms for dissemination of project information are detailed in Section 2.3.2 of the Project Document (PD).

During the project formulation process summary documents referring to the stages and objectives were generated as support materials to ensure that the project information was clear and accessible. Folders, banners, and oral presentations were made to facilitate the understanding process. Printed project summaries were distributed among the communities and other stakeholders.

The dissemination of this monitoring results will be done during a visit that is schedule to take place in April 2023.

2.3.3 Informational Meetings with Stakeholders (G3.1)

Detailed information on all meetings held by the project (previous meetings and stakeholder consultation) is described in Section 2.3.3 of the Project Document (PD). This section provides relevant information regarding the preparation of meetings, outcomes and information provided to stakeholders.

The table below shows the total number of people who participated in the awareness process.

Table 6. Year 2022 - Total of stakeholders that received presentation of the Itaúba REDD+ Project.

Stakeholder	Departments*	People consulted	Female	Male
Municipal Government of Itaúba	7	7	5	2
Municipal Government of Marcelândia	8	8	5	3
Papa João Paulo II State School	12	241	126	115
Paulo Freire State School	7	133	69	64
Pedro Bianchini State School	5	89	45	44

IBAMA - Brazilian Institute for the Environment and Renewable Natural Resources	2	4	2	2
EMBRAPA - Brazilian Public Agricultural Research Company	1	1	0	1
EMPAER - Mato-Grossense Research, Assistance and Rural Extension Company	1	4	2	2
UFMT - Federal University of Mato Grosso	2	2	0	2
Association of Brazil Nut Collectors from Itaúba	1	2	2	0
Chestnut sellers from Itaúba	1	3	2	1
Total	47	494	258	236

* Departments include secretaries and number of school classes

2.3.4 Community Costs, Risks, and Benefits (G3.2)

As mentioned in section 2.3.4 of the PD, during the consultations community members were informed that there wouldn't be any costs related to project activity, since it is funded by the proponents. It was also reinforced that the activities are planned to increase their well-being, and yet they still have the freedom to choose if they want to participate. Also, the project risks and strategies to avoid them were explained. The consultation was held by a social specialist, who communicated in a manner that all presents could understand.

The activities proposed by the project come from diagnoses, feedbacks, and go through a validation of the communities involved.

2.3.5 Information to Stakeholder on Verification Process (G3.3)

During stakeholder consultation meetings, communities and local authorities were informed about the general functioning of the carbon markets, the standards available to obtain certification, the nature of the carbon credits, and the verification process required to obtain these credits (see Information to Stakeholders on Validation and Verification Process – Section 2.3.5 PD).

2.3.6 Site Visit Information and Opportunities to Communicate with Auditor (G3.3)

The method of communication about the validation and verification process with the communities is mainly through oral presentation, where they are informed of the dates in advance to respect their daily activities and to allow their participation in the process (see Stakeholder information about the validation and verification process - Section 2.3.5 PD).

2.3.7 Stakeholder Consultation (G3.4)

Since the beginning of the project, the proponents (Carbonext and Landowners) have put in place a stakeholder engagement and project management structure to ensure the active involvement of all stakeholders in the planning and execution of project activities. During four field visits that took place in February, June, October, and November 2022, the project team met with municipal governments, municipal secretaries, and state schools in the municipalities of Itaúba and Marcelândia, as well as the farms staff. The team also met with environmental inspection institutions (SEMA) and research institutions (UFMT,

EMPAER and EMBRAPA) located in the municipality of Sinop, which also operate in the regions of Itaúba and Marcelândia. Meetings were also held with the Association of Brazil Nut Sellers in the municipality of Itaúba. These meetings aimed to understand in depth the regional context and how the project could address focal issues related to climate, community and biodiversity.

The meetings and onsite visits, together with the questionnaires applied with the local communities were very important to identify characteristics and vulnerabilities by the Carbonext team, making a socioeconomic diagnosis on each community and farm. This valuable information supports the development of effective project activities with partners, with the goal of improving the quality of life and well-being of the local population.

The activities were designed by Carbonext team, considering the results and observations obtained in the meetings and the socioeconomic diagnosis. The communities were directly consulted about all the activities and their participation was fundamental to implement an effective action plan that could benefit them. The actions were implemented considering the suggestions and necessities appointed to. In this sense, the team develops, adapts, and makes changes according to the feedbacks received.



Figure 6: Some examples of meetings and onsite visits that took place in 2022. A) Meeting with the municipal secretaries at the Marcelândia city hall; B) Onsite visit to the headquarters of Nossa Senhora Salete Farm; C) Meeting with a teacher and a coordinator of the Paulo Freire school. D) Focus group discussion with students of Papa João Paulo II school.

As an example of the influence of the communities in the project activities design, from October to November 2022 the project promoted the course "Cultivating Emotional Balance" for teachers and managers of the Papa João Paulo II State School, in Itaúba. This activity was developed as a suggestion

from the school workers, who reported problems with interpersonal relationships at work and mental overload.



Figure 7: First class of the course "Cultivating Emotional Balance" started in October 2022.

The following activities will be carried out by using the same effective communication with schools and other stakeholders, so that the activities will be well aligned among partners and engage the community members according to their respective differences and vulnerabilities.

2.3.8 Continued Consultation and Adaptive Management (G3.4)

The project is working on an effective ongoing communication and consultation channel between the project proponents, communities, and other stakeholders, with the development and implementation of the Communication Operational Procedure¹². On this first monitoring period, the communication happened through the project team in the field visits, meetings and during implementation of activities, as well as through the communication channels indicated between the communities/other stakeholders with the Carbonext social team. In the office, the Carbonext team can also be reached by email and telephone at any time.

Before every verification, the communities and other stakeholders will be contacted and consulted to receive updates on the project activities and results, but also to hear suggestions, complaints, and feedbacks in case of actions that weren't effective. The Carbonext teams is constantly monitoring if there is any negative impact or urgent matter. All their considerations on the project are heard and, if applicable, will be essential to adapt activities and mitigate impacts.

¹² PO-GQ-0003_Relacionamento com Stakeholders and PO-GQ-0002_Atendimento de Reclamação.

The process of adaptative management of the project activities is crucial to create a successful project, providing benefits to the local people, to the climate and to the biodiversity as well. It begins with communication, trust, proximity and alignment with the communities and other stakeholders interested. This generates information about their main vulnerabilities, the characteristics of the region, suggestions, and opinions. The activities and actions are then adapted and modified if not in accordance with this information, and only then, implemented. The feedbacks and observation about the actions from the local people and from the implementation team from Carbonext are collected and reported in the field reports that are available for auditing, so there can be more adaptations in the project.

2.3.9 Stakeholder Consultation Channels (G3.5)

All the consultations and participatory processes in the first monitoring period have been undertaken directly with the communities and other stakeholders. A first contact was made by email and/or telephone to introduce the project in advance, except for the farms, where the first contact was made in person.

The institutions were first contacted by email, and a summary of the project was sent. The Carbonext team also reached these institutions by telephone to notify the email sent about the Itaúba REDD+ Project. The project summary contains the project's objectives, introduction, location, and the contact address. Subsequently, an in-person or remote presentation was scheduled for those institutions that showed interest.

During the onsite visits, flyers with a brief summary of the project (Figure 8) were shared with all communities and stakeholders, including students from the state schools of Itaúba and Marcelândia. In all the institutions visited, printed versions of the project summary were made available with the main information about the project, such as its objectives, location, timeline, and channels for relationships and denunciations. These materials were shared in printed form at the following institutions: Paulo Freire State School; Pedro Bianchini State School; Papa João Paulo II State School; City Hall and Municipal Secretaries of Itaúba and Marcelândia; Environmental Secretariat of Sinop - SEMA; UFMT; EMPAER; EMBRAPA; Association of Brazil Nut Sellers.



Figure 8: Flyer with a brief summary of the project that were shared during the onsite visits.



Figure 9: Students from the Paulo Freire State School reading the flyer shared with them.



Figure 10: Printed project summary shared with all stakeholders.

A poster in Portuguese was carried out by the Carbonext team to assist the meetings and present the project. It contains some information about the project, such as location of area, contact information, and a brief explanation about the acronyms REDD and CCB. One copy was left at each farm headquarters involved in the project. It is important to emphasize that the materials developed contained Carbonext contact information, meaning that each community and stakeholders could send suggestions, complaints, and denunciations related to the Itaúba REDD+ Project.

CARBONEXT
NATURE & FUTURE

ITAÚBA

Projeto REDD+

ORIGEM DO NOME
Árvore-pedra em Tupi.

TIPO DE PROJETO
REDD+ (Redução de Emissões provenientes de Desmatamento e Degradação florestal) CCB (Comunidade, Clima, Biodiversidade)

BENEFÍCIOS

- Treinamentos de brigadas de incêndio.
- Capacitação técnica para os professores e alunos.
- Monitoramento terrestre e remoto.
- Inventários de biodiversidade.

Itaúba e Marcelândia MT, Brasil

175 espécies protegidas

Evitar o desmatamento de 2.916 ha de floresta

Atividades sociais

14.388 hectares preservados

30 anos de duração

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SAIBA MAIS
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limite da propriedade área do projeto

Figure 11: Poster of the Itaúba REDD+ Project presented by Carbonext team on the public consultations and meetings with the communities and workers of the properties involved in the project.

On the farms, the forest monitor and the landowners were trained on the procedures of the forest integrity plan, such as the actions to be taken in case of illegal invasion or fire outbreaks. A poster with a summary of these procedures was left at each farm for consultation. In addition to the poster, a copy of the detailed operating procedure and a handout with information about the Itaúba project was made available for easy access.



Figure 12: A) A poster about the project summary and another poster about the forest integrity plan were fixed in each farmhouse; B) A copy of the procedure was fixed on the wall in the employee living room.

2.3.10 Stakeholder Participation in Decision-Making and Implementation (G3.6)

The consultations with the community and stakeholders were organized ensuring adherence and a full understanding of the project and activities. The meetings with the communities were scheduled according to their availability. The meetings and activities took place directly in each community or on site at each institution, and were conducted by Carbonext's social, technical, and local specialists, in order to communicate the content in a culturally appropriated way.

The members of the communities, men, women, and children were invited to comment and to question the project, to make suggestions and explain their vulnerabilities. The consultation does not discriminate any stakeholders, reaching adult men and women of all ages, religion, and ethnicity. The Table 6 shows how many people were consulted and their respective gender.

As described in section 2.3.7, an example of the influence of the communities in the project activities implemented, the activity implemented in the school during October and November 2022 was suggested by teachers and managers of the Papa João Paulo II State School, who reported problems with interpersonal relationships at work and mental overload.

On the farms, the rural workers were encouraged to engage in the project's activities and monitoring. The team introduced them to the project and collected information to understand the local threats to the forest and biodiversity. This step was very important for the development of the Forest Integrity Plan (see section 2.2.7), in such a way that it was possible to elaborate strategies more consistent with the local reality.

2.3.11 Anti-Discrimination Assurance (G3.7)

The proponents of the project, possess the principles of legality, impersonality, and morality, which enshrine the principles of anti-discrimination based on race, color, sex, or religion. In addition, the project proponents are in compliance with workers' rights and safety that are guaranteed by well-defined laws, such as the Consolidation of Labor Laws (CLT) Law No. 5,452/43, recently amended by law No. 13,467/2017 (labor reform), which provides for anti-discrimination in employment and occupation. Implementation reports are used to ensure transparency and avoid any form of discrimination.

Carbonext has a channel for anonymous complaints and relationship channels, which were presented during the field visits. The main relationship channels with project representatives are:

Email: comunidades@carbonext.com.br

WhatsApp: +55 (91) 99314-0724.

Other contacts:

Email: red.d.itauba@carbonext.com.br

Phone numbers:

(91) 3118-2040 – Belém/PA Office

(11) 3168-8521 – São Paulo/SP Office

Site: www.carbonext.com.br

For denunciations: compliance@carbonext.com.br

The channels should be used for any complaints from any stakeholder to ensure that the project will not be involved or be complicit in any form of discrimination or sexual harassment.

2.3.12 Grievances (G3.8)

No grievances have been received during this monitoring period.

2.3.13 Worker Training (G3.9)

The implementation of each activity of the project begins with an initial training where the proposal of the activity and its characteristics are shown. The objective of each activity is to build capacities and relationships with the project communities.

The trainings carried out in the monitoring period 2021-2022 regarding the project activities were:

- Monitoring and control: Field visit and training to introduce the ArcGIS QuickCapture app to the forest monitor and the landowners to help monitor patrols and make their registration. They can be replicators of this knowledge, therefore the local capacity is not lost.

Because the period for carrying out the activities coincided with the violent protests that closed roads and set cars on fire during the presidential elections, the fire brigade training that were to be held during this monitoring period were postponed preserving the physical integrity of the project team and its service providers.

2.3.14 Community Employment Opportunities (G3.10)

During the verified period, no permanent job positions were created. For the project activity carried out, the "Cultivating Emotional Balance" course, we prospected people who had the necessary training to conduct the course. It was conducted by a woman (see section 2.4.3). The opportunity to conduct this training was offered without any discrimination of sex, age, religion, marital status, or ethnicity. For all project activities that require contracting, priority will be given, where possible, to local people who have the appropriate expertise, according to the premises established in section 2.3.15 of the PD.

2.3.15 Relevant Laws and Regulations Related to Worker's Rights (G3.11)

In Brazil, workers' rights are guaranteed by well-defined laws, such as the Consolidation of Labor Laws (CLT) Law No. 5.452/43, recently amended by law No. 13.467/2017 (labor reform). It is also worth to mention the existence of Law 5,889/73 which regulates specifically the rules applicable for rural employees.

The CLT has Brazilian Regulatory Norms (also known as NR) that are cited in Chapter V, Title II, and were approved by Ordinance no. 3214, in June 1978. The Brazilian Regulatory Norms (NR) regulate and provide guidance on mandatory procedures related to occupational safety and health.

The main NR regulations for the project are the following:

- NR 4- Specialized Occupational Health and Safety Services (Last update: MTPS Ordinance No. 510 of April 29, 2016);
- NR 5- Internal Commission for Accident Prevention. (Last updated: SIT Ordinance No. 247 of July 12, 2011);
- NR 6: Personal Protective Equipment – PPE;
- NR 28: Deals with the inspection and penalties that can be applied to companies that do not comply with the other rules established by the Federal Government
- NR31: Safety and health at Work in Agriculture, Livestock forestry, forestry and Aquaculture [DO|C2]
- ILO Declaration on Fundamental Principles and Rights at Work;
- Conventions 87 and 98: Dealing with union freedom and protection of the right to union and collective bargaining.
- Convention 11: Right to Organize in Agriculture;

The project proponent complies with laws and regulations (as declared in the Partnership Agreement).

In addition, during the Due Diligence carried out by the legal team, it is analyzed with the Public Ministry of Labor if there are any lawsuits and labor debts of the owners. The information was reviewed again by the legal department for this monitoring period.

2.3.16 Occupational Safety Assessment (G3.12)

It was considered in the Project Description that all project works are required to receive training before they take up their roles in the project. The training is designed to inform workers of any potential risks they might face, and how to reduce them. All necessary trainings in health and safety will be provided by specialized professionals of the areas, in agreement with the Brazilian Regulatory Norms (NR). The main Brazilian Regulatory Norms related to the project, that regulate and provide guidance on mandatory procedures related to occupational safety and health in Brazil, are the following:

- NR4 - Specialized Occupational Health and Safety Services;
- NR5 - Internal Commission for Accident Prevention;
- NR6 - Personal Protective Equipment – PPE;
- NR20 - Occupational Safety and Health with Flammable and Combustible
- NR23 - Fire Protection, and
- NR31 - Safety and health at Work in Agriculture, Livestock forestry, forestry and Aquaculture

The project implementation predicts risks during firefighting, ground monitoring and maintenance activities. To minimize the risks to workers, a fire brigade training was scheduled to take place in November 2022, however it was cancelled due to security issues in the region related to the protests that took place on the roads in Brazil after the presidential elections. This training has been rescheduled to take place in April 2023.

In addition to the measures to minimize the risks, standard operating procedures have been implemented, containing the guidelines that shall be followed in the events of fires, invasions, or in case of emergency.

The Carbonext's Occupational Safety specialist has prepared a risk matrix of the team's work in the field to prevent them. Moreover, the entire project development team received the trainings "First Aid in Remote Areas" and "Rules of Conduct and Safety Procedures in Remote Areas". For all field work the appropriate PPE is carried, such as helmet, boots, long sleeve shirt, and repellent.

2.4 Management Capacity

2.4.1 Required Technical Skills (G4.2)

Carbonext's technical team has experienced professionals in several areas of knowledge necessary for the planning and monitoring of the project, such as forestry engineers, social workers, biologists, oceanographers, geologist, geoprocessing analysts and other related areas, as described in the section below.

Carbonext is carrying out the monitoring of carbon stocks from deforestation rates and greenhouse gas emissions produced or avoided during the project, which is being developed by a specialized team that has

previous experience with other REDD projects. In addition, it has a team of social specialists, who have the necessary skills to engage the community and stakeholders, so the team is able to ensure community participation, designing activities and monitoring socioeconomic indicators.

In June 2022, Carbonext implemented a deforestation alert system called MonitoraCarbon™, which consists of processing optical and radar images from sensors such as Landsat and Sentinel (1 and 2) that detect any change in forest cover on a weekly basis. So, this system detects probable changes in land use, validates and sends alerts to monitoring agents to confirm and notify the competent authorities to take appropriate actions.

In November 2022, the monitoring agent (property employee) received a training, which aimed to enable him to handle the monitoring data collection application (QuickCapture) and make him a replicator of this knowledge. This App allows the monitoring agent to mark points during patrol where deforestation, invasion, fire or natural events occur, which helps in a quick decision-making process.

For the fauna and flora inventory, a third-party team will be hired to collect data in the field and generate the diagnostic report. Preference will be given to local teams that has knowhow of local conditions.

2.4.2 Management Team Experience (G4.2)

Carbonext was founded in 2010 with the objective of preserving the Amazon Forest through the elaboration of REDD+ carbon credit projects. Since then, we have designed three REDD+ projects according to the VCS standard, with new projects in progress. In addition, Carbonext has a multidisciplinary technical team, composed of forest engineers, biologists, ecologists, geographers, social workers, and others that integrate a skilled team of researchers and specialists able to conduct the project activities.

For more information about Carbonext, visit our website: <https://carbonext.com.br/>.

The summaries of the expertise of the technical team dedicated to the validation and monitoring of the ITAÚBA REDD+ Project are shown below:

Name	Janaina Dallan
Position	CEO at Carbonext
Education	Forestry Engineer by the Luiz de Queiroz College of Agriculture (ESALQ) and a Master's in business of the environment.
Resume summary	Janaína has been working with the carbon market since 2001 on projects around the world and, as CEO of Carbonext, she is directly involved in coordinating the development and implementation of carbon credit projects. Throughout her career, she has managed UNFCCC CDM (Clean Development Mechanisms) projects through several international companies such as Golder Associates, Global Energy Partners, Orbeo / Société Generale, One Carbon and Ecofys.

	She is currently a member of the United Nations Framework Convention on Climate Change (UNFCCC), in which she has been a member of the Registration and Issuance Team (RIT) since 2013, she is president of the Brazil Alliance for Nature-Based Solutions and is a member of the experts for the Taskforce for Scaling Voluntary Carbon Markets (TSVCM).
Contact Information	janaina@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/janainadallan/

Name	Luiz Fernando de Moura
Position	Technical Director at Carbonext
Education	Forestry Engineer by the Luiz de Queiroz College of Agriculture (ESALQ), M.Sc. and Ph.D. in Wood technology by the Université Laval (Quebec, Canada).
Resume summary	He is responsible for coordinating the technical group at Carbonext, working with projects for the Carbon Markets, mainly Forestry projects. Dr. de Moura had participation in the preparation of "Energia Verde Carbonization Project - Mitigation of Methane Emissions in the Charcoal Production of Grupo Queiroz Galvão, Maranhão, Brazil", registered on March 21, 2011. Dr. de Moura has also participation as project designer in Florestal Santa Maria REDD+ Project, Fortaleza Ituxi REDD+ Project, UNITOR REDD+ Project and Evergreen REDD+ Project.
Contact Information	luiz.moura@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/luiz-fernando-de-moura-6077089/

Name	Jéssica Lima Guin
Position	Project Coordinator
Education	Degree in Forestry Engineering from Luiz de Queiroz College of Agriculture (ESALQ) and University of Kentucky. Specialization in Conservation, Restoration and Sustainable Use of Tropical Forests from Yale School of Forestry & Environmental Studies.
Resume summary	She has experience in developing REDD+ projects, having participated in the elaboration of project descriptions and monitoring reports. She also has previous experiences in strategic forestry planning, FSC certification and elaboration of a Conservation Unit Management Plan.

Contact Information	jessica.guin@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/jessica-guin/

Name	Thiago Bopp Resnitzky
Position	REDD+ Analyst
Education	Currently on studying in Economic Sciences in the University of São Paulo (bachelor's degree).
Resume summary	Experience with socioenvironmental entrepreneurship, conservation of protected areas, nature-based solutions, and climate change. Thiago previously worked in the implementation of the Acceleration Program for Municipal Conservation Units (2021), implemented by the German Agency for International Cooperation (GIZ) in partnership with ICLEI South America and IUCN, which promoted effectiveness in the management of six Brazilian protected areas. He is a member of the Youth Climate Leaders (YCL) and Climate Reality (CR) networks and has been part of organizations such as Enactus and Group Turing of Artificial Intelligence of the University of São Paulo.
Contact Information	thiago.bopp@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/thiago-bopp

Name	Raisa Fagundes de Figueiredo
Position	REDD+ Analyst
Education	Geologist by the Federal University of Roraima – UFRR, M.Sc. and Ph.D. in Geology and Natural Resources by the University of Campinas – UNICAMP. Post-doctoral research at Luiz de Queiroz College of Agriculture – ESALQ/USP.
Resume summary	Raisa has experience with geological mapping in Amazon, techniques of passive inorganic carbon capture, and evaluation of the agronomic potential of rock dust. She has worked in research focused on carbon dioxide storage through mineral carbonation in mine tailings and soils.
Contact Information	raisa.figueiredo@carbonext.com.br
LinkedIn	https://br.linkedin.com/in/raisa-fagundes-de-figueiredo-geologa

Name	Jane Dones
Position	REDD+ Social Analyst
Education	Degree in Environmental Engineering from Faculdade Oswaldo Cruz
Resume summary	Experience with a social humanization project, inspired by Anthroposophy, at Monte Azul (São Paulo, Brazil). Experience with sociocultural projects, in traditional Brazilian cultures, developing workshops on musical experience and cultural training, at the AMJUS Institute (Rio Grande do Norte, Brazil). Experience in reforestation project with agroforestry systems and organic food plantation. Jane also created and coordinated a social action project for children in social vulnerability, at the AMJUS Institute.
Contact Information	jane.dones@carbonext.com.br
LinkedIn	www.linkedin.com/in/jane-dones-

Name	Marcos Breno Lopes Marques
Position	REDD+ Social Analyst
Education	Degree in Forestry Engineering and Master's in forest and environment science from Federal University of Amazonas (UFAM). Currently studying Project Management from Luiz de Queiroz College of Agriculture (ESALQ) (post-graduation/ MBA).
Resume summary	Experience with socio-environmental activities with traditional communities, biomass and carbon stock quantification, forest inventory, environmental dynamics and ESG practices in Central Amazon Marcos holds certificates of independent courses in Environmental Management Systems, by Senac São Paulo, and Ecosystemic Services Integration in Planning Processes by IPÊ – Instituto de Pesquisas Ecológicas. He is also a member of the Sustainable Development Solutions Network Amazonia (SDSN Amazonia).
Contact Information	marcos.marques@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/marcos-marquess/

Name	Laissa Ferreira Carvalho
Position	REDD+ Analyst
Education	Degree in Forestry Engineering and Master's in Forestry Science from Federal University of Viçosa (UFV).
Resume summary	Experience with wood technology, forest quality, conservation units, environmental education, and urban forestry.
Contact Information	laissa.carvalho@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/laissacarvalho/

Name	Daniela Paes da Rocha
Position	Remote Sensing & GIS Analyst
Education	Geographic Information Systems and Remote Sensing from the Public University of Navarra and master in biosystems engineering from the Fluminense Federal University, in natural systems with an emphasis on microclimate. Forestry Engineer graduated from the Federal Rural University of Rio de Janeiro.
Resume summary	Experience with remote sensing and routines GIS, processes and solutions for conservation projects, land use and land cover management, forest management and forest integrity monitoring.
Contact Information	daniela.rocha@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/danielapaesdarocha/

Name	Mateus da Silva Trez
Position	Remote Sensing & GIS Leader
Education	Environmental and Sanitary Engineer by the Engineering School of Piracicaba.
Resume summary	Experience with utilization of Geographic Information Systems and softwares such as ArcGIS, Spring, Quantum Gis, Saga Gis, GvSig and Kosmo for monitoring quality and quantity of water in hydrographic basins, developing hydrological models, climate monitoring, erosion estimations, identification of priority areas for conservation (AAVC), territorial planning for areas of permanent preservation and legal reserves, evolution of land use changes

	for prediction of future changes, consultancies for acquisition of environmental and forestry certifications such as FSC, CERFOR and 14000, remote sense imagery treatment and analysis for land use mapping, productivity assessments.
Contact Information	mateus.trez@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/mateustrez/

2.4.3 Project Management Partnerships/Team Development (G4.2)

For this monitoring period, two organizations were hired to support the implementation of the project activities. These partnerships were established for the course "Cultivating Emotional Balance" and for the fire brigade training, since the courses need capacitated and specialized professionals. Even though the fire brigade training was not carried out during this monitoring period due to force majeure, this partnership has already been established for the next monitoring period. The partners are:

Gabriella Araújo: She has a degree in Social Communication / Journalism from PUC - SP, attended the specialization course in art therapy, at Alquim Art - Learning Research Center in São Paulo. She has a post-graduate degree in Emotional Management in Organizations, from the Instituto Israelita de Ensino Albert Einstein and training as an instructor in the Cultivating Emotional Balance - CEB program, from the Santa Barbara Institute. Currently, she is dedicated to integrating CEB learning and practices into her life and studies related to contemplative skills, human flourishing, and emotional balance.

Biopreve Gestão Integrada: Company specialized in courses for firefighters, including forest fire brigade training, as well as other occupational health and safety courses. The company is located in the capital of the state of Mato Grosso, Cuiabá, and counts on professionals as civilian firefighters in its staff.

The implementation for the next project activities is ongoing and other partnerships are being planned for the future. However, since the partnerships are in process of officialization, these partners will not be presented. As mentioned in section 2.4.2 of the PD, for the fauna and flora inventory, a third-party company or independent specialists will be hired to carry out data collection in the field, giving preference to local teams that know the local reality.

2.4.4 Financial Health of Implementing Organization(s) (G4.3)

Carbonext has a financial department that is responsible for taking care of the company's financial health, so that it can run its operation on a regular basis in addition to being able to execute projects without running financial risks. To this end, the company has an annual budgeting process that is periodically reviewed in order to give managers visibility of what the main expenses will be for the period, including expenses related to the execution of projects. In this way, a monetary amount is set aside for each project that will be executed, in order to avoid any cash flow risk.

Supporting evidence of the financial health of both proponents will be made available to the audit team and reported in the Non-Permanence Risk Report.

2.4.5 Avoidance of Corruption and Other Unethical Behavior (G4.3)

In Brazil, the two main legal instruments that aim to prevent acts of corruption and unethical attitudes are (i) Law 12,846/2013 (Anti-Corruption Law); and (i) Law 8,429/1992 (Administrative Misconduct Law).

Law no. 12,846/2013, known as the Anti-Corruption Law, establishes rules regarding the administrative liability and civil liability of legal entities in the case of practice of acts against the public administration, national and/or foreign. Under Law 8,429/1992, administrative improbity is an immorality qualified by the dishonesty and acts of bad faith by the public agent.

Both practices, in addition to the legal risks, could represent enormous reputational risk to the project and to all that are involved. Due to this risks, Carbonext Legal Due Diligence aims to verify and, if possible, mitigate any involvement of the landowner (partner) with such practices.

The clearance certificates examined by Carbonext's legal team (applicable for all the project proponents/partners and real estate assets) during the Legal Due Diligence process cover a large spectrum of jurisdictional and administrative matters, such as, for example, consultation before State Courts of Justice and Superior Courts, State and Federal Public Prosecutors, Environmental Agencies, Protest Registry Offices, Real Estate Registry Offices, etc., significantly reducing the chances of both debts and/or acts of corruption (regarding the landowner partner or the real estate assets) go unnoticed. Eventual discovery of relevant debts and/or acts of corruption involving the landowner (partner), is one of the many reasons for the termination of the intended partnership, as provided for in the contract.

Carbonext also has an internal Anti-Corruption Policy and an internal Code of Ethics and Organizational Conduct, in order to ensure non-involvement with bad practices or corruption.

To register your denunciation, send an e-mail to compliance@carbonext.com.br.

Anti-corruption policy and code of ethics and organizational conduct in force: <https://carbonext.com.br/governanca-corporativa/>

2.4.6 Commercially Sensitive Information (Rules 3.5.13 – 3.5.14)

Some information required by the VCS and CCB standards is considered confidential, commercially sensitive or intellectual property of the company and cannot be made public by the project proponent. This information will be supplied to the audit team, but not available for the public. The documents are:

- Agreements and contracts between the parties involved;
- Financial evidence of the proponent's funds for the project;
- Property Right Documents;
- Project spreadsheets.

2.5 Legal Status and Property Rights

2.5.1 Recognition of Property Rights (G5.1)

All the real estate assets in which the Itaúba REDD+ Project was developed, the private property rights are recognized, respected, and supported by Brazilian Law. Information regarding the ownership and number of registration (real estate record file) of each real estate asset is listed in the PD, item 2.5.2. Documentation will be made available to the auditors.

2.5.2 Free, Prior and Informed Consent (G5.2)

The project does not encroach or affect other private or public real estate, traditional communities, or environmental protection areas, as described in section 2.3.5 of the Project Description.

It is worth noting that there are no traditional communities within the project area, however the project was presented to employees and other stakeholders for their knowledge, who signed a term of knowledge of its existence. Active communication and respect for the particularities of the communities joining the project have been adopted.

2.5.3 Property Right Protection (G5.3)

The Project activities did not cause involuntary removal or relocation of property rights holders from their lands or territories, neither forced right holders to relocate activities important to their culture or livelihood.

Furthermore, the proponents are keeping constant communication with the stakeholders through the communication channel established between them, in order to protect the property rights and the culture of livelihood in the project area and its surroundings.

2.5.4 Identification of Illegal Activity (G5.4)

In the baseline scenario, illegal deforestation and forest fires were predicted as deforestation drivers in the reference region and inside the project area. For this verification period, by contrast, it is verified that no deforestation occurred inside the project area. During the same period, the deforestation of 24,918 ha in the reference region for agricultural activity indicates the dimension of the deforestation pressure as well as the effectiveness of the project activities concerning deforestation avoidance. The ex-ante assessment for the reference region previewed 21.268 ha of forest conversion to other land uses, so the actual deforestation outside the project area was even higher than the estimates.

There are no project benefits that could be associated with illegal activities. To prevent deforestation, the project strengthens the local monitoring system by maintaining an employee responsible for monitoring possible invasions and the forest integrity in each of the two fragments of the project area.

As mentioned in section 2.3.13, a monitoring and control training was held by Rafael Ruas Martins, Geographical Information Systems specialist and MRV manager at Carbonext. The focus of the training was implementing the usage of the ArcGIS Quick Capture App and reinforcing the communication channels between the property workers and the project proponents through this technology. Orientations about the

frequency and path of the monitoring according to the season or summer, accessibility and accident prevention were given.

Furthermore, the project implemented Carbonext's MonitoraCarbon system, which consists of remote monitoring by satellite images from different sensors (Sentinel and Landsat) and radar images (Sentinel), with a maximum frequency of 7 days. This allows the project proponents to identify any illegal trespassing and act in time if necessary.

The project activity with regard to forest integrity is based on Carbonext's operational procedure for Integrity Control in REDD+ Project (PO-GQ-0001_INTEGRI FLORESTAL E CONTR PATRIMONIAL), which establishes control and surveillance mechanisms, and determines which monitoring activities will be carried out during the project lifetime.

2.5.5 Ongoing Disputes (G5.5)

In the project area, no conflicts or disputes over property rights, use of real estate, or access to natural resources have been identified, nor disputes with traditional third parties or squatters that may abrogate the right of ownership of the Itaúba REDD+ project area, as detailed above. As a preventive measure, the project has a specific channel for complaints, as well as one for technical questions and for stakeholder relationships.

In relation to public entities, whether state, municipal or federal, of direct or indirect administration, the certificates analyzed indicate the legitimacy of the ownership, ruling out the possibility of irregular "dominion" over public lands (*grilagem*), a fact that would make the development of the project impossible, because regardless of the domain duration, it could not be regularized (public lands cannot undergo usucapion), except with the effective participation of the public entity that owns the area, through some legislative measure (law, decree, ordinance, etc.) that expressly legalizes this domain.

Under Brazilian legislation, we can say that the certificates from environmental, administrative, jurisdictional, and certifying bodies indicate that both possession and property belong to the proponent, being tame, peaceful, stabilized, fully legal and extensively documented and proven. In other words, the owner has full right to use, enjoy and dispose of the properties that make up the project, having legitimacy to allow the project to be developed.

In this way, since there are no unresolved disputes relevant to the project, the project activities will not bring any type of prejudice.

2.5.6 National and Local Laws (G5.6)

Since the beginning of the Itaúba REDD+ Project, no national and local laws and regulations in the host country have gone into effect, changed, or been eliminated that are relevant to the project activities. The laws and regulations can be accessed in the sector 2.5.7 from the Project Description.

The project follows all national and local laws and regulations in Brazil, developing and implementing the project aligned with the laws, hiring services and material only from reliable sources, guaranteeing the quality and legality of the project.

3 CLIMATE

3.1 Monitoring GHG Emission Reductions and Removals

3.1.1 Data and Parameters Available at Validation

Data / Parameter	CF_{tree}
Data unit	tC/tdm
Description	Default value of carbon fraction of biomass for tree species
Source of data	From scientific literature: 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 AFOLU
Value applied	0.47
Justification of choice of data or description of measurement methods and procedures applied	The default value was used for conservativeness purposes. The same value was applied in the "Third Brazilian Inventory", which is the source of the biomass data.
Purpose of Data	• Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage
Comments	N/A

Data / Parameter	CF_{litter}
Data unit	tC/tdm
Description	Carbon fraction for litter
Source of data	From scientific literature: 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 AFOLU
Value applied	0.47
Justification of choice of data or description of measurement methods and procedures applied	The default IPCC value was applied in the "Third Brazilian Inventory", which is the source of the biomass data.
Purpose of Data	• Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage

Comments	N/A
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Data / Parameter	CE_{ab,icl}
Data unit	Dimensionless
Description	Average combustion efficiency of above-ground biomass
Source of data	IPCC 2003. Good practice guidance for land use, land-use change and forestry. Kanagawa: IGES, 2003. Available at: < https://www.ipcc-nggip.iges.or.jp/public/gpglulucf/gpglulucf_files/Chp3/Anx_3A_1_Data_Tables.pdf >
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	The IPCC default value was used for conservativeness purposes. According to the methodology VM0015 v1.1, pg 83, "If no appropriate combustion efficiency can be used, the IPCC default of 0.5 should be used".
Purpose of Data	• Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage
Comments	N/A

Data / Parameter	CE_{cl,icl}
Data unit	Dimensionless
Description	Average combustion efficiency of litter
Source of data	Kauffman, B. J., Cummings, D. E., Ward, D. L., & Babbitt, R., 1995. Fire in the Brazilian Amazon: 1. Biomass, nutrient pools, and losses in slashed primary forests. Oecologia, 104, 397–408. Available at: https://link.springer.com/article/10.1007/BF00341336
Value applied	0.99
Justification of choice of data or description of measurement methods and procedures applied	The study carried out by Kauffman et al. (1995) highlights the combustion efficiency for litter in the Amazon.
Purpose of Data	• Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage
Comments	N/A

Data / Parameter	44/12
Data unit	Dimensionless
Description	Carbon mass to CO ₂ e mass conversion factor
Source of data	From scientific literature: 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 AFOLU
Value applied	44/12
Justification of choice of data or description of measurement methods and procedures applied	Conversion from C to CO ₂ based on molecular weights
Purpose of Data	• Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage
Comments	IPCC standard value

Data / Parameter	R
Data unit	t root d.m.t ⁻¹ shoot d.m.
Description	Root to shoot ratio appropriate to species or forest type biome.
Source of data	The "2006 IPCC Guidelines for National Greenhouse Gas Inventories", V. 4, Ch. 4, AFOLU, pg. 4.49, Table 4-4.
Value applied	0.37
Justification of choice of data or description of measurement methods and procedures applied	Local values are not known, and the IPCC is a conservative value.
Purpose of Data	• Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage
Comments	This parameter is applied in the biomass calculations of the Third Brazilian Inventory of Anthropogenic Greenhouse Gas Emissions and Removals.

Data / Parameter	BCEF
Data unit	Dimensionless
Description	Biomass Conversion and Expansion Factor
Source of data	According to "Brown, S., A. J. R. Gillespie, and A. E. Lugo, 1989. Biomass estimation methods for tropical forests with applications to forest inventory data. Forest Science, 35:881-902". (Table 4; pg. 890; minimum value deducted from lowest limit.: $1.743 - 0.083 = 1.66$)
Value applied	1.66
Justification of choice of data or description of measurement methods and procedures applied	BCEF was applied for conversion of merchantable volume to total aboveground tree biomass
Purpose of Data	• Determination of baseline scenario (AFOLU projects only) • Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage
Comments	N/A

Data / Parameter	Cab_{icl}
Data unit	tCO ₂ e/ha
Description	Average carbon stock per hectare in the above-ground biomass carbon pool of initial forest class icl
Source of data	Terceiro Inventário Brasileiro de Emissões e Remoções Antrópicas de Gases do Efeito Estufa (Third Brazilian Inventory of Anthropogenic Greenhouse Gas Emissions and Removals), 2015, pg. 107, table 14. The conversion from total biomass to above-ground biomass is explained in Section 3.2.1. The explanation of the use of these data is in section 3.2.2.
Value applied	355.5 for F1 and 106.4 for F2
Justification of choice of data or description of measurement methods and procedures applied	Other secondary data was consulted, however the data from the Third Brazilian Inventory was chosen for being the most conservative.

Purpose of Data	Calculation of baseline emissions
Comments	The values used for F1 is the average of As and Fs, once there is no data for Evergreen Seasonal Forest, which is a composition between Ombrophylous Forest and the Seasonal Forest. Conservative lower limit to be used in calculations, based on uncertainties in source values.

Data / Parameter	Cbb_{icl}
Data unit	tCO ₂ e/ha
Description	Average carbon stock per hectare in the below-ground biomass carbon pool of initial forest class icl
Source of data	Third Brazilian Inventory of Anthropogenic Greenhouse Gas Emissions and Removals, pg. 107, table 14.
Value applied	31.0 for F1 and 23.4 for F2
Justification of choice of data or description of measurement methods and procedures applied	Other secondary data was consulted, however the data from the Third Brazilian Inventory was chosen for being the most conservative.
Purpose of Data	Calculation of baseline emissions
Comments	The values used for F1 is the average of As and Fs, once there is no data for Evergreen Seasonal Forest, which is a composition between Ombrophylous Forest and the Seasonal Forest Conservative lower limit to be used in calculations, based on uncertainties in source values.

Data / Parameter	Cl_{icl}
Data unit	tCO ₂ e/ha
Description	Average carbon stock per hectare in the litter biomass carbon pool of initial forest class icl
Source of data	Third Brazilian Inventory of Anthropogenic Greenhouse Gas Emissions and Removals, pg. 107, table 14.

Value applied	18.3 for F1 and 17.5 for F2
Justification of choice of data or description of measurement methods and procedures applied	Other secondary data was consulted, however the data from the Third Brazilian Inventory was chosen for being the most conservative.
Purpose of Data	Calculation of baseline emissions
Comments	The values used for F1 is the average of As and Fs, once there is no data for Evergreen Seasonal Forest, which is a composition between Ombrophylous Forest and the Seasonal Forest. Conservative lower limit was used in calculations, based on uncertainties in source values.

Data / Parameter	Cabfcl_{cropland}
Data unit	tCO ₂ e/ha
Description	Average carbon stock per hectare in the above-ground biomass carbon pool of final post-deforestation class fcl for cropland conversion
Source of data	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, V. 4, Chapter 5: Cropland, pg. 5.41, Table 5.9 (for Annual cropland Tropical, moist)
Value applied	23.70
Justification of choice of data or description of measurement methods and procedures applied	Conservative default value from IPCC, to estimate post-deforestation land use carbon stock.
Purpose of Data	Calculation of baseline emissions
Comments	Conservative upper limit was used in calculations, based on uncertainties in source values.

Data / Parameter	Cabfcl_{grassland}
Data unit	tCO ₂ e/ha

Description	Average carbon stock per hectare in the above-ground biomass carbon pool of final post-deforestation class fcl for grassland conversion
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, V. 4, Chapter 6: Grassland, pg. 6.27, Table 6.4 (for Pasture)
Value applied	15.63
Justification of choice of data or description of measurement methods and procedures applied	Conservative default value from IPCC, to estimate post-deforestation land use carbon stock.
Purpose of Data	Calculation of baseline emissions
Comments	Conservative upper limite was used in calculations, based on uncertainties in source values.

Data / Parameter	Cbbfcl_{cropland}
Data unit	tCO ₂ e/ha
Description	Average carbon stock per hectare in the below-ground biomass pool of final post-deforestation class fcl for cropland conversion
Source of data	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, V. 4, Chapter 5: Cropland, pg. 5.41, Table 5.9 (for Annual cropland Tropical, moist) calculated using the root-shoot-ratio 0.175 described in [61].
Value applied	4.15
Justification of choice of data or description of measurement methods and procedures applied	Conservative default value from IPCC, to estimate post-deforestation land use carbon stock.
Purpose of Data	Calculation of baseline emissions
Comments	Conservative upper limit to be used in calculations, based on uncertainties in source values.

Data / Parameter	Cbbfcl_{grassland}
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Data unit	tCO ₂ e/ha
Description	Average carbon stock per hectare in the below-ground biomass pool of final post-deforestation class fcl for grassland conversion
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, V. 4, Chapter 6: Grassland, pg. 6.27, Table 6.4 (for Pasture)
Value applied	24.96
Justification of choice of data or description of measurement methods and procedures applied	Conservative default value from IPCC, to estimate post-deforestation land use carbon stock.
Purpose of Data	Calculation of baseline emissions
Comments	Conservative upper limite to be used in calculations, based on uncertainties in source values.

Data / Parameter	C_lfcl_{cropland}
Data unit	tCO ₂ e/ha
Description	Average carbon stock per hectare in the litter biomass pool of final post-deforestation class fcl for cropland conversion
Source of data	Page 150 of VM0015 Methodology v1.1: "In some forest ecosystem litter carbon stocks in the litter carbon pool can be a significant component of the total carbon stock while in anthropogenic ecosystem, particularly in agricultural or pastoral systems, litter is almost absent."
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	According to VM0015 Methodology v1.1 and other sources consulted, litter is nearly absent in the post-deforestation scenario.
Purpose of Data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	Cifcl_{grassland}
Data unit	tCO ₂ e/ha
Description	Average carbon stock per hectare in the litter biomass pool of final post-deforestation class fcl for grassland conversion
Source of data	Page 150 of VM0015 Methodology v1.1: "In some forest ecosystem litter carbon stocks in the litter carbon pool can be a significant component of the total carbon stock while in anthropogenic ecosystem, particularly in agricultural or pastoral systems, litter is almost absent."
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	According to VM0015 Methodology v1.1 and other sources consulted, litter is nearly absent in the post-deforestation scenario.
Purpose of Data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	EI
Data unit	N/A
Description	Ex ante estimated effectiveness index
Source of data	Local assessment
Value applied	0.9
Justification of choice of data or description of measurement methods and procedures applied	The project design team conservatively considers that surveillance activities are able to attain 90% of effectiveness in avoiding unplanned deforestation inside the Project Area.
Purpose of Data	Calculation of project emissions
Comments	This value is an ex-ante estimate. Accurate and actual values will be monitored and reported in verification periods.

Data / Parameter	DLF
Data unit	N/A
Description	Displacement Leakage Factor
Source of data	Page 100 of VM0015 v1.1 (footnote #45).
Value applied	0.05
Justification of choice of data or description of measurement methods and procedures applied	If deforestation agents do not participate in leakage prevention activities and project activities, the Displacement Factor shall be 100%. Where leakage prevention activities are implemented, the factor shall be equal to the proportion of the baseline agents estimated to be given the opportunity to participate in leakage prevention activities and project activities. The project design team estimates that 100% of potential deforestation agents in the Reference Region will be given the opportunity to participate in leakage prevention activities. Projections of deforestation within the Leakage Belt have duly been made as part of the present project, according to projections of deforestation presented in Section 3.2.3. The project proponent has conservatively allowed 5% leeway on this estimate, which is unlikely to be surpassed. If this allowance is surpassed, an assessment of how much is due to the project will be conducted during monitoring. Thus, the "Displacement Leakage Factor" (DLF) was conservatively defined as 0.05. Leakage fighting activities are also listed in this PD.
Purpose of Data	Calculation of leakage
Comments	This value is an ex-ante estimate. Accurate and actual values will be monitored and reported in verification periods

Data / Parameter	Deforestation
Data unit	ha
Description	Maps of forest cover areas converted into non-forest areas
Source of data	Measured through data from MapBiomass project (if available) or supervised classification.
Value applied	Yearly variable: deforestation values are presented for the Reference Region, Leakage Belt and Project Area (projections) in this PD
Justification of choice of data or description of	The project area is located within a region that is subject to a monitoring program that is approved or sanctioned by the national government (<i>MapBiomass</i>). The data generated by this program is used in this project. <i>MapBiomass</i> data are applicable for use in this project, according to the criteria

measurement methods and procedures applied	listed below (Methodology VM0015 v1.1): i) <i>MapBiomass</i> monitoring occurs in the entire project area and leakage belt. ii) <i>MapBiomass</i> monitoring occurs in the entire reference region and covers the beginning, middle and end of the fixed baseline period. iii) <i>MapBiomass</i> monitors conversion of forest land to non-forest land. iv) Monitoring occurred during the entire fixed baseline period. v) data accuracies are acceptable according to the methodology criteria. An order of preference of data applies to the available sources of data: 1) MapBiomass in first order of preference; 2) PRODES in second place; and 3) place self-classification of images.
Purpose of Data	Calculation of baseline emissions
suComments	N/A

Data / Parameter	LDF
Data unit	tCO ₂ / m ³
Description	Logging damage factor
Source of data	Calculated considering the study from Jennifer S. Johns, Paulo Barreto, Christopher Uhl, Logging damage during planned and unplanned logging operations in the eastern Amazon, Forest Ecology and Management, Volume 89, Issues 1–3, 1996, Pages 59-77, ISSN 0378-1127 ¹³. Explanation of the calculation in the Error! Reference source not found.
Value applied	0.28 tCO₂ / m³ for Itaúba municipality 0.77 tCO₂ / m³ for Marcelândia municipality
Justification of choice of data or description of measurement methods and procedures applied	The study carried out by Johns et al, 1996 analyzes the impact of different forest management scenarios in the Amazon. For the calculation of the logging damage factor of the project, low impact forest management (<30m ³ /ha) and the use of skidder with tire, which is used within the project properties, were considered.
Purpose of Data	Calculation of project emissions
Comments	N/A

¹³ Johns et al, 1996 <<https://www.sciencedirect.com/science/article/pii/S0378112796038698>>

Data / Parameter	LIF
Data unit	tCO ₂ / m ³
Description	Logging infrastructure factor
Source of data	Decree Nº 1313 of March 11, 2022 ¹⁴ that regulates that “§ 2 The sum of the opening areas of planned infrastructure cannot exceed 2% of the total area of the Forest Management Unit”.
Value applied	0.47 or 0.59 tCO ₂ / m ³ for Itaúba municipality 0.52 tCO ₂ / m ³ for Marcelândia municipality
Justification of choice of data or description of measurement methods and procedures applied	The state decree that regulates the % of area that can be opened for infrastructure in forest management areas was considered.
Purpose of Data	Calculation of project emissions
Comments	N/A

Data / Parameter	PCOMi
Data unit	Dimensionless
Description	Commercial volume as a percent of total aboveground volume in stratum i.
Source of data	For calculation of this parameter, the volume of merchantable wood in exploitation (in m ³ /ha) has been based on “Florestal Santa Maria Project ¹⁵ ” (35 m ³ /ha), whose Project Area is located in Colniza Municipality, Mato Grosso State. As recommended in VMD0005 v1.1, this data can be chosen from “ <i>Forest inventory from a proxy area in the same region, representing the same forest type and age class, distinguishing commercially viable stocks on the basis of species and tree size</i> ”. Both projects have the forest type “As”, Submontane Open Ombrophilous Forest. Age classes are assumed to be similar, as both

¹⁴ Decree Nº 1313 of March 11, 2022. <SEFAZ MT> Accessed August 30,2022

¹⁵ Florestal Santa Maria Project <<https://registry.verra.org/app/projectDetail/VCS/875>> Accessed March 3,2023

	sites were primary forests before forest management. In this context, it is assumed that the composition of species and tree sizes share similarities. In the "Florestal Santa Maria Project" aboveground biomass ranges from 224 to 268 t/ha, while the average aboveground biomass in this project ranges from 210.8 t/ha (F1) and 70.8 (F2). In other words, given that the biomass in the Itauba project area is smaller, it can be concluded that a volume of commercial wood greater than that inventoried in the Santa Maria project would be unfeasible.
Value applied	F1: 0.10 F2: 0.31
Justification of choice of data or description of measurement methods and procedures applied	As prescribed in CP-W, a forest inventory from a proxy area in the same region, representing the same forest type and age class was chosen for calculation of this parameter. This parameter is updated at baseline renewal when aboveground biomass is re-inventoried (at least every 6 years).
Purpose of Data	Calculation of baseline emissions
Comments	Updated at the time of baseline revision (at least every 10 years). Application of the commercial percentage of total volume introduces the simplifying assumption (and conservative, as it is only used in the ex-ante baseline calculations) that all commercial stocks are extracted (i.e., perfect efficiency).

Data / Parameter	STF_w
Data unit	Dimensionless
Description	Fraction of wood products and waste that will be emitted to the atmosphere within 3 years; all carbon shall be assumed to be lost immediately;
Source of data	Pereira, D., Santos, D., Vedoveto, M., Guimarães, J., & Veríssimo, A., 2010. Fatos Florestais da Amazônia 2010. 1st ed., Vol. 1, Belém: Imazon, 124p. Table 9, pg 33, data for Mato Grosso State.
Value applied	0.55
Justification of choice of data or description of measurement methods and procedures applied	This parameter was calculated based on a sawmill processing yield of 44.8% for the Mato Grosso State, a study conducted by Pereira et al., (2010)
Purpose of Data	Calculation of baseline emissions

Comments	N/A
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Data / Parameter	Wood density
Data unit	t/m ³
Description	Average wood density for each forest class from Mato Grosso State
Source of data	Nogueira, E. M., Fearnside, P. M., Nelson, B. W., & França, M. B. 2007. Wood density in forests of Brazil's "arc of deforestation": Implications for biomass and flux of carbon from land-use change in Amazonia. Forest Ecology and Management, 248(3), 119–135. https://doi.org/10.1016/j.foreco.2007.04.047
Value applied	F1: 0.645 F2: 0.634
Justification of choice of data or description of measurement methods and procedures applied	The study carried out by Nogueira et al. (2007) bring out values of wood density for forest classes from different parts of Amazon, including the Mato Grosso State.
Purpose of Data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	Rate of harvest
Data unit	m ³ /ha
Description	Rate of forest harvest carried out in the Itaúba and Marcelândia municipalities.
Source of data	Annual Production Units (UPA in Portuguese) for Sustainable Forest Management
Value applied	15.6 m ³ /ha for F1 (Marcelândia municipality) 17.2 m ³ /ha for F2 (Itaúba municipality)
Justification of choice of data or description of measurement methods and procedures applied	The rates of harvest were calculated based on the real volume per hectare managed in the past and the area authorized by the environmental agency for each municipality. These data were extracted from old "Annual Production Units", documents required for Sustainable Forest Management.

Purpose of Data	• Calculation of project emissions
Comments	N/A

Data / Parameter	Regeneration growth rate
Data unit	tCO ₂ /ha/year
Description	Annual increments in aboveground and belowground biomass in Itaúba and Marcelândia municipalities
Source of data	West, T.A.P, Vidal, E., Putz, E.F. 2014. "Forest biomass recovery after conventional and reduced-impact logging in Amazonian Brazil." Forest Ecology and Management 314: 59-63.
Value applied	3.0 tCO ₂ /ha/year for F1 (Marcelândia) 1.0 tCO ₂ /ha/year for F2 (Itaúba)
Justification of choice of data or description of measurement methods and procedures applied	West et al. (2014) carried out a study on post-logging biomass dynamics in the Brazilian Amazon. The studied forest is Evergreen, similar to the Marcelândia forest class.
Purpose of Data	• Calculation of project emissions
Comments	N/A

Data / Parameter	GWP
Data unit	Dimensionless
Description	GWP of CH ₄ and N ₂ O
Source of data	Sixth Report IPCC (AR6) IPCC, 2022: Climate Change 2022: Impacts, Adaptation, and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama (eds.)]. Cambridge University Press. Cambridge University Press, Cambridge, UK and New York, NY, USA, 3056 pp., doi:10.1017/9781009325844.

Value applied	29.8 for CH ₄ 273 for N ₂ O
Justification of choice of data or description of measurement methods and procedures applied	The IPCC default value was used for conservativeness purposes
Purpose of Data	Calculation of baseline emissions
Comments	N/A

3.1.2 Data and Parameters Monitored

Data / Parameter	ABSLKt
Data unit	ha
Description	Annual area of baseline deforestation within the leakage belt at year t
Source of data	Remote sensing data and GIS
Description of measurement methods and procedures to be applied	Deforestation in the leakage belt area will be considered activity displacement leakage. Activity data for the leakage belt area will be determined using the same methods applied to monitoring deforestation activity data in the project area.
Frequency of monitoring/recording	Annually
Value applied	2021 = 8 ha 2022 = 164 ha
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	Best practices in remote sensing and GIS: Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, as closer as possible to the end of the monitoring period.

	2) Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images. 3) For analysis of areas with cloud cover, visual interpretation of radar image is performed. 4) Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping should be 90%.
Purpose of data	• Calculation of leakage
Calculation method	Analysis of satellite images and maps
Comments	Deforestation occurred in 2021 is insignificant according to the "Tool for testing significance of GHG emissions in A/R CDM project activities", representing 2% of total GHG emission for this MR. Thus it has not been accounted for.

Data / Parameter	ABSLPA_t
Data unit	<i>ha</i>
Description	Annual area of baseline deforestation in the project area at year t
Source of data	Remote sensing data and GIS
Description of measurement methods and procedures to be applied	Forest cover change due to deforestation is monitored through periodic assessment of classified satellite imagery covering the project area.
Frequency of monitoring/recording	Annually
Value applied	2021 = 0 ha 2022 = 0 ha
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	Best practices in remote sensing and GIS: 1) Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, as closer as possible to the end of the monitoring period.

	2) Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images. 3) For analysis of areas with cloud cover, visual interpretation of radar image is performed. 4) Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping should be 90%.
Purpose of data	Calculation of project emissions Calculation of baseline emissions
Calculation method	Analysis of satellite images and maps
Comments	N/A

Data / Parameter	$\Delta CADLK_t$
Data unit	tCO ₂ e
Description	Total decrease in carbon stocks due to displaced deforestation at year t
Source of data	Remote sensing data and GIS
Description of measurement methods and procedures to be applied	Deforestation in the leakage belt area will be considered activity displacement leakage. Activity data for the leakage belt area will be determined using the same methods applied to monitoring deforestation activity data in the project area.
Frequency of monitoring/recording	Annually
Value applied	2021 = 0 2022 = 0
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	Best practices in remote sensing and GIS: 1) Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, as closer as possible to the end of the monitoring period. 2) Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images.

	3) For analysis of areas with cloud cover, visual interpretation of radar image is performed. 4) Evaluation of classification accuracy is performed by analysing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping should be 90%.
Purpose of data	Calculation of leakage
Calculation method	Emissions from deforestation are estimated by multiplying the detected area of forest loss by the average forest carbon stock per unit area.
Comments	N/A

Data / Parameter	ΔCPAdPA_t
Data unit	tCO ₂ e
Description	Total decrease in carbon stock due to all planned activities at year t in the project area.
Source of data	Documents, remote sensing and GIS.
Description of measurement methods and procedures to be applied	The planned activities in the project area that result in carbon stock decrease will be subject to monitoring, when significant.
Frequency of monitoring/recording	Annually.
Value applied	2021 = 0 2022 = 0
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	Best practices in remote sensing and GIS: 1) Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, as closer as possible to the end of the monitoring period. 2) Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images. 3) For analysis of areas with cloud cover, visual interpretation of radar image is performed.

	4) Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping should be 90%.
Purpose of data	Calculation of project emissions
Calculation method	Emissions from all planned activities are estimated by the sum of the emissions from planned deforestation and emissions from planned logging activities.
Comments	N/A

Data / Parameter	ΔCUDdPA_t
Data unit	tCO ₂ e
Description	Total actual carbon stock change due to unavoided unplanned deforestation at year t in the project area
Source of data	Remote sensing and GIS
Description of measurement methods and procedures to be applied	Forest cover change due to unplanned deforestation is monitored through periodic assessment of classified satellite imagery covering the project area.
Frequency of monitoring/recording	Annually
Value applied	2021 = 0 2022 = 0
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	Best practices in remote sensing and GIS: 1) Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, as closer as possible to the end of the monitoring period. 2) Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images. 3) For analysis of areas with cloud cover, visual interpretation of radar image is performed. 4) Evaluation of classification accuracy is performed by analysing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping should be 90%.

Purpose of data	Calculation of project emissions
Calculation method	Emissions from unavoided unplanned deforestation are estimated by multiplying the detected area of forest loss by the average forest carbon stock per unit area.
Comments	N/A

Data / Parameter	EBBPSPAt
Data unit	tCO2e
Description	Sum of (or total) actual non-CO2 emissions from forest fire at year t in the project area
Source of data	Remote sensing data and GIS, - Supervisor reports.
Description of measurement methods and procedures to be applied	If forest fires occur, these non-CO2 emissions will be subject to monitoring and accounting, when significant.
Frequency of monitoring/recording	Annually
Value applied	2021 = 0 2022 = 0
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	Best practices in remote sensing and GIS: 1) Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, as closer as possible to the end of the monitoring period. 2) Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images. 3) For analysis of areas with cloud cover, visual interpretation of radar image is performed. 4) Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping should be 90%.
Purpose of data	Calculation of project emissions

Calculation method	Analysis of satellite images and maps to determine the incidence of deforestation and multiplying it by the respective emission factors.
Comments	If forest fires occur, these non-CO2 emissions will be subject to monitoring and accounting, when significant.

Data / Parameter	EADLK_t
Data unit	tCO ₂ e
Description	Total ex post increase in GHG emissions due to displaced forest fires at year t
Source of data	Remote sensing data and GIS
Description of measurement methods and procedures to be applied	When significant, GHG emissions due displaced forest fires will be monitored.
Frequency of monitoring/recording	Annually
Value applied	2021 = 0 2022 = 0
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	Best practices in remote sensing and GIS: 1) Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, as closer as possible to the end of the monitoring period. 2) Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images. 3) For analysis of areas with cloud cover, visual interpretation of radar image is performed. 4) Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping should be 90%.
Purpose of data	Calculation of leakage
Calculation method	Analysis of satellite images and maps to determine the incidence of deforestation and multiplying it by the respective emission factors.

Comments	Where strong evidence can be collected that forest fires in the leakage belt is attributable to deforestation agents that are not linked to the project area, the detected deforestation will not be attributed to the project activity, thus not considered leakage.
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Data / Parameter	RFt
Data unit	%
Description	Risk factor used to calculate VCS buffer credits
Source of data	- VCS Non-Permanence Risk Report (v4.0), - Remote sensing data and GIS, - Supervisor report. - Literature data.
Description of measurement methods and procedures to be applied	All sources of data from the VCS Non-Permanence Risk Report will be used to measure the various risk factors.
Frequency of monitoring/recording	Annually
Value applied	19%
Monitoring equipment	VCS-approved AFOLU Non-Permanence Risk Tool
QA/QC procedures to be applied	Literature data from reputed sources will be used and critically checked. When possible, the average of two or more sources will be used.
Purpose of data	Calculation of VCS buffer credits
Calculation method	All the risk factors described in the VCS Risk Report were assessed.
Comments	N/A

Data / Parameter	Deforestation in the project area and leakage belt
Data unit	ha
Description	Forest cover areas converted into non-forest areas inside the Project Area and Leakage Belt

Source of data	Calculated through remote sensing images
Description of measurement methods and procedures to be applied	The monitoring of the forest cover in the Project Area and Leakage Belt will be done through satellite image analysis. When data from the MapBiomass system are not available, the forest cover monitoring will be carried out by automatic classification and visual interpretation of images from other optical sensors or SAR data.
Frequency of monitoring/recording	Annually
Value applied	Project area = 0 Leakage belt = 172 ha
Monitoring equipment	Remote sensing images digital processing program, geographic information systems
QA/QC procedures to be applied	Best practices in remote sensing and GIS: 1) Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, as closer as possible to the end of the monitoring period. 2) Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images. 3) For analysis of areas with cloud cover, visual interpretation of radar image is performed. 4) Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping should be 90%.
Purpose of data	Calculation of project emissions Calculation of leakage
Calculation method	Analysis of satellite images and maps
Comments	N/A

3.1.3 Monitoring Plan

This Monitoring Plan was developed according to Methodology VM0015 "Methodology for Avoided Unplanned Deforestation", v1.1. The methodology encompasses three main monitoring tasks:

- i) Monitoring of actual carbon stock changes and GHG emissions within the project area;
- ii) Monitoring of leakage; and
- iii) Ex post calculation of net anthropogenic GHG emission reduction.

This Monitoring Plan describes how these tasks were implemented. For each task, the monitoring plan included the following aspects:

a) *Technical description of the monitoring tasks.*

The mapping of land use and land cover is carried out using satellite images and remote sensing techniques, with the objective of identifying and quantifying the deforestation that occurred during the monitoring period in the reference region, leakage belt and project area. Satellite analysis of land uses was based on primary information due to the unavailability of secondary data such as Prodes or Mapbiomas data.

Based on secondary data from the database of the BD/Queimadas (INPE) system, we evaluated the outbreaks of fires that occurred during the monitoring period from the reference region.

In order to monitor the project areas, we use a deforestation alert detection system called MonitoraCarbon. Using optical and radar images, this system continuously monitors the areas, with the aim of identifying and mitigating deforestation problems quickly and effectively.

b) *Data to be collected.*

For the measurement of deforestation in the MR period, images from the Sentinel-2 satellite were used, as there were no images from the end of the monitoring with low cloud cover, it was necessary to look at earlier dates. To cover possible deforestation after the dates of the optical images, until the closure of the MR, we revisited, from visual inspection, the entire area with radar images from the Sentinel-1 satellite. In this context, the deforestation that occurred in the period from 14/12/2021 (December 14, 2021) to 31/12/2022 (December 31, 2022) was mapped through semi-automatic classification of supervised multispectral optical images and with images of radar from visual inspection.

For the semi-automatic classification, the first step was to generate a mosaic of images from Sentinel 2 – MultiSpectral Instrument (MSI) dated from images from October 8, 2022, and November 19, 2022, which would cover the entire reference region with the smallest clouds cover index possible for the period. The composition chosen was False Color, combining bands R8 G4 B3 to highlight vegetation through band 8, near infrared. With the aid of high-resolution imagery (WMS imaging services such as ESRI basemaps and Planet), LU/LC samples were collected for forest, deforestation, water and rocky outcrops across the entire reference region. In addition to the multispectral optical image, an NDVI (Normalized Difference Vegetation Index) image was created through the algebra of the bands $(B8-B4)/(B8+B4)$, where band 4 is RED and band 8 is INFRARED to feed the classifier with one more variable and enhance the separation of the forestry class from any other non-forestry class.

The conclusion of the deforestation measurements for the monitoring period until 31/12/2022 (December 31, 2022) was carried out through the visual interpretation of the SAR radar (Synthetic Aperture Radar),

operating band C (between 8 and 4 GHz or 3.8 - 7.5 cm), double polarization VV and VH, spatial resolution 20 meters, from the Sentinel 1 satellite (European Space Agency) with images from 01/01/2023 and 08/01/2023. This strategy was necessary to map deforestation in a period when clouds predominate and block the use of optical images to classify land use. Microwave remote sensors like Sentinel 1 (C-band) are widely adopted for this type of mapping because they can penetrate clouds and help identify deforestation.

The mapping of fire outbreaks was carried out from the database of the BD/Queimadas (INPE) system, where we exported data on fire outbreaks for the entire reference region corresponding to the monitoring period.

c) *Overview of procedures.*

The classification of the Sentinel-2 optical image was done using the Random Forest algorithm, in the Google Earth Engine (GEE) environment. Samples for training the Random Forest classifier were divided into two blocks, with 30% for training and evaluation of the classifier; and 70% for use in the classification process. The classifier was run with 100 decision trees.

The initial result of the classification was treated in a GIS environment (ArcGIS), in order to reduce residuals and errors of the classifier. Pixel filters such as Majority (ArcGIS) was applied twice to reduce the so-called “salt and pepper” effect of the classification, identifying isolated pixels and recategorizing it according to the majority value of its eight neighbors. In addition, a visual inspection of the classification product overlaid on the image was performed with the objective of correcting main errors of the classifier via visual interpretation and vector scanning.

The acquisition and processing of Sentinel-1 radar images was performed in the GEE environment, generating products with VV and VH polarization. These images were used in the GIS environment (ArcGIS) for vectorization by visual interpretation of the increment of deforested areas omitted from the visual images due to the high cloud cover until the end of the monitoring period.

The mapping of semi-automatic deforestation and through visual interpretation was carried out through the following steps:

Target identification: non-forest areas;

Training: deforestation areas observed in optical satellite images, of equal or better resolution, are overlaid to identify patterns of hue, texture/roughness, shape and location of deforested areas;

Photointerpretation: identification and vectorization on the radar image of deforestation areas omitted by clouds on the optical image, using as a mask all deforestation already mapped;

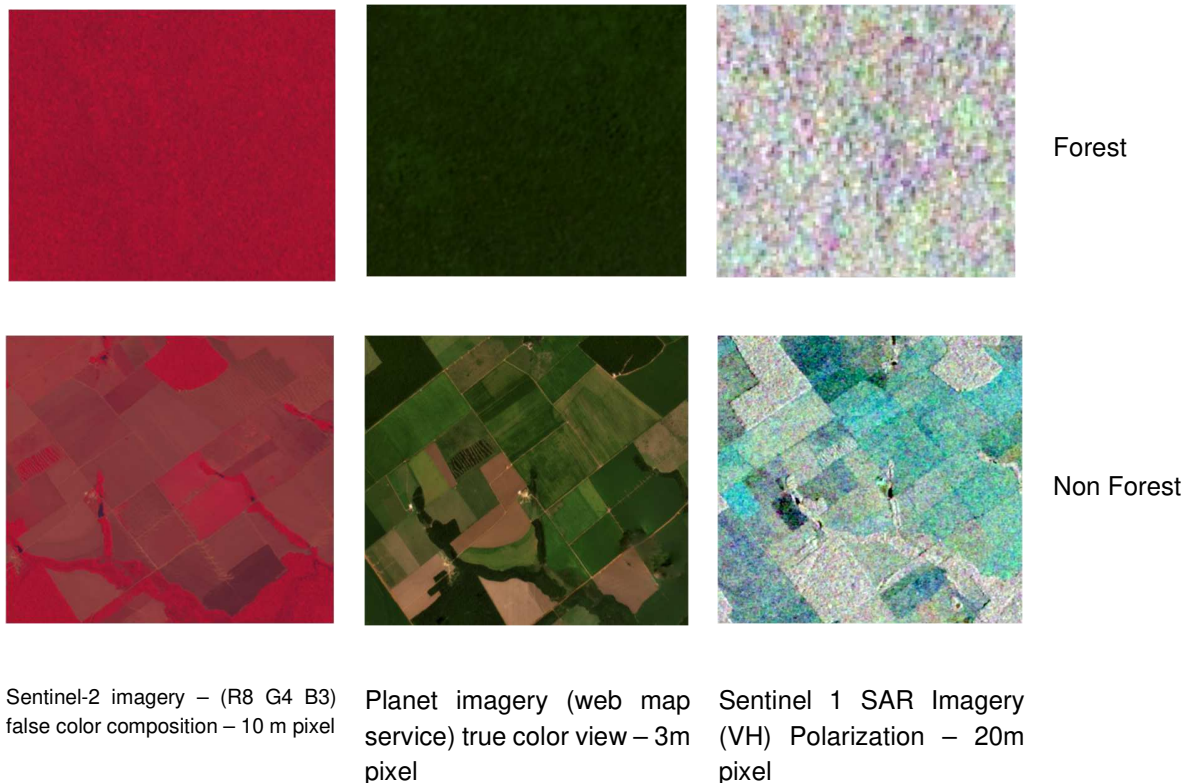


Figure 13: Land use sampling frame for photointerpretation and deforestation measurement

The data containing the fire areas were specialized in ArcGIS and superimposed on the limits of the project areas, with the objective of identifying the occurrence of fires and possible losses in the forest cover of the project areas.

All fire points found within the project areas were analyzed using high and medium resolution images available and nearest as possible of the likely fire event date. This verification is done by analyzing images in a closed before and another one closed after the hot focus detection data aid to detect any change over forest cover in terms of color, hue, texture and roughness.

d) Quality control and quality assurance procedures.

The validation of land-use data used for modelling of land use is performed using a confusion matrix, in order to calculate the overall index of success by period and by class. Four specific classes will be used: forest, deforestation, water, rocky outcrops.

The analysis of the accuracy of the Sentinel 2 image classification was performed by means of the confusion matrix, overall accuracy and Kappa index. The sampling was stratified, proportional by area, based on a total of 316 points randomly distributed across the reference region, leakage belt and project area by using the *Create Accuracy Assessment Points* tool on ArcGIS.

The accuracy assessment points map assigns value to the land use map generated in the classification process in one column. This column is omitted, and the interpreter populates the *Ground Truth* field based

on the interpretation of the reference data used in such as raw optical image itself and web service maps from ESRI and Planet with higher spatial resolution than the input image. So, the accuracy assessment points table is the input for *Compute Confusion Matrix* tool on ArcGIS.

Table 7. Confusion Matrix and Kappa Accuracy Assessment - Sentinel imagery.

REFERENCE							
CLASSIFIED		Water	Forest	Deforestation	Rocky Outcrops	Total	User Accuracy
	Water	10	0	0	0	10	100%
	Forest	0	180	4	1	185	97.30%
	Deforestation	0	5	106	0	111	95.50%
	Rocky Outcrops	0	1	0	9	10	90%
	Total	10	186	110	10	316	Kappa
	Prod. Accuracy	100%	96.77%	96.40%	90%		93.44%

The overall classification accuracy for all land use and land cover classes was 93.44% in the reference region, higher than the 90% suggested by the VM0015 v1.1 methodology. The specific accuracy for the Forest and Deforestation classes were 97.3 and 95.5% respectively.

The accuracy analysis of the classification of the radar image was performed in the same way as the validation of the product classified from the optical image.

Table 8: Confusion Matrix and Kappa Accuracy Assessment - SAR (radar) imagery

REFERENCE							
CLASSIFIED		Water	Forest	Deforestation	Rocky Outcrops	Total	User Accuracy
	Water	10	0	0	0	10	100%
	Forest	0	58	2	0	60	96.67%
	Deforestation	0	1	38	0	39	97.44%
	Rocky Outcrops	0	0	0	10	10	100%
	Total	10	59	40	10	119	Kappa
	Prod. Accuracy	100%	98.31%	95%	100%		95.97%

e) *Results.*

The results obtained by the procedures carried out in item 3.1.3 were superimposed and cropped at the limits of the PA and LB to identify forest losses in the respective areas. In this way, we obtained the following deforestation results in the monitoring period for these areas.

Table 9: Total deforested area found in PA and LB

Period	Deforested area (ha)	
	PA	LB
14 December 2021 to 31 December 2022	0	172

No deforestation was identified in the project areas for the monitoring period. For the LB, 172 ha of deforestation were identified in the LB area during the monitoring period.

Five fire focus points were identified within the PA areas, but when analyzing images after the period of occurrence, we did not identify any changes in the vegetation cover of the area. Since the spatial resolution of the sensors that make the detection (1km or more) and these focusses are close to the edges of the project areas (less than 1 km), these occurrences may be outside the PA area or be a false positive.



Figure 14: Example of a hot spot identified on the edge of the PA.

The focus of heat detected by the sensor was allocated within the PA, at a distance from the edge of just 102m, and in the adjacent area, agricultural management is a relevant cause of hot spots in the soil. It is

understood that due to the low spatial resolution of the heat sensor, as already mentioned, the points at the edges of the PA tend to be false-positive.

Monitora Carbon System

As a complementary strategy for monitoring deforestation on a continuous basis, Carbonext has implemented a deforestation detection and warning system called MonitoraCarbon. The adoption of this system increases the chances of detecting possible changes in land use (deforestation), because it is programmed to perform continuous analysis as soon as an optical or radar image from the sentinel 2 e sentinel 1 satellites respectively, of the project area is available. Because it is an alert system, its process is designed to be sensitive enough to detect any change in the forest, undergo a rapid internal validation process, and issue alerts for monitoring agents to confirm and take appropriate action to contain the problem as efficiently as possible. This system also integrates sources of secondary deforestation alerts, such as DETER, SAD Imazon, Mapbiomas Alertas, RADD that help complement the remote monitoring of areas. There were no deforestation detection alerts for the RM period.

f) Data archiving.

All maps and records generated during project implementation, as well as invoices, contracts and reports related to activities implemented were conserved and will be made available during verification audits. All documents and records will be kept in a secure retrievable manner for at least two years after the end of the project crediting period. The landowners are responsible to archive the physical documents.

g) Organization and responsibilities of the parties involved in all the above.

The project proponents are responsible to implement this monitoring item. All satellite imagery assessments were performed by Carbonext GIS expert team. Carbonext was responsible for reporting and data archiving (digital version of the original documents), according to VM0015 v1.1, and for providing assistance during verification audits. The landowners were responsible for implementing the surveillance rounds and were co-responsible for the acquisition of data in loco.

h) Non-conformances

No nonconformities were identified during this monitoring period.

EX POST NET ANTHROPOGENIC GHG EMISSION REDUCTION:

a) Technical description of the monitoring tasks.

The calculation of ex post net anthropogenic GHG emission reductions is similar to the ex-ante calculation with the only difference that ex post estimated carbon stock changes and GHG emissions must be used in the case of the project scenario and leakage.

b) Data to be collected.

Table 10: Data collected to monitor net ex-post GHG reductions for the Itaúba REDD+ project.

Parameter	Unit	Source	Frequency
Reduction of net GHG emissions attributable to the project activities at AUD in the year	tCO ₂ e	Calculated	Annually
Number of Verified Carbon Units (VCUs) to be made available for sale in the year	tCO ₂ e	Calculated	Annually

c) *Overview of data collection procedures.*

This step involves compilation of data from previous procedures to calculate ex post net anthropogenic GHG emission reduction.

d) *Quality control and quality assurance procedures.*

All tasks and tools indicated in part 2 of the Methodology VM0015 v1.1 were used to ensure that the data is adequate for the verification process and the number of Verified Carbon Units is reliable.

e) *Data archiving.*

All data and records generated during project implementation will be kept in a secure retrievable manner for at least two years after the end of the project crediting period. All documents relating to project monitoring are made available to the verification body at each verification event.

f) *Organization and responsibilities of the parties involved in all the above.*

Carbonext was responsible for reporting and data archiving, and for providing assistance during verification audits. The Project Proponent was co-responsible for data archiving.

g) *Non-conformances.*

No nonconformities were identified during this monitoring period.

3.1.4 Dissemination of Monitoring Plan and Results (CL4.2)

Monitoring plan and results will be publicly available at Verra and Carbonext's website. A summary of the results will be presented for the communities and other stakeholders during the consultations in an appropriate manner, respecting the language and cultural diversity.

3.2 Quantification of GHG Emission Reductions and Removals

3.2.1 Baseline Emissions

Appropriate data sources were used to analyze land-use and land-cover change during the historical reference period within the reference region and project area, as described in PD. The following table shows further information about the data collected:

Table 11: Data used for historical LU/LC change analysis (Table 5 of the VM0015 v1.1).

Vector (Satellite or Aeroplane)	Sensor	Resolution		Coverage km	Acquisition date (YYYY)	Scene or Point identifier	
		Spatial	Spectral			Path / Latitude	Row / Longitude
Satellite	Landsat 8 OLI	30 X 30 m	0.43 - 2.26 µm	170 x 183	2013	1	66
Satellite	Landsat 8 OLI	30 X 30 m	0.43 - 2.26 µm	170 x 183	2013	1	67
Satellite	Landsat 8 OLI	30 X 30 m	0.43 - 2.26 µm	170 x 183	2013	233	66
Satellite	Landsat 8 OLI	30 X 30 m	0.43 - 2.26 µm	170 x 183	2013	233	67
Satellite	Landsat 8 OLI	30 X 30 m	0.43 - 2.26 µm	170 x 183	2016	1	66
Satellite	Landsat 8 OLI	30 X 30 m	0.43 - 2.26 µm	170 x 183	2016	1	67
Satellite	Landsat 8 OLI	30 X 30 m	0.43 - 2.26 µm	170 x 183	2016	233	66
Satellite	Landsat 8 OLI	30 X 30 m	0.43 - 2.26 µm	170 x 183	2016	233	67
Satellite	Landsat 8 OLI	30 X 30 m	0.43 - 2.26 µm	170 x 183	2018	1	66
Satellite	Landsat 8 OLI	30 X 30 m	0.43 - 2.26 µm	170 x 183	2018	1	67
Satellite	Landsat 8 OLI	30 X 30 m	0.43 - 2.26 µm	170 x 183	2018	233	66
Satellite	Landsat 8 OLI	30 X 30 m	0.43 - 2.26 µm	170 x 183	2018	233	67

The accumulated baseline deforestation projected to occur within the project area over the 30-year project lifetime was estimated at 2,919 hectares. The estimated average annual rate of deforestation is 97.3 hectares for the project area.

For this monitoring period (2021 to 2022), the expected deforestation to occur in the absence of the Itaúba REDD+ Project was 152 hectares. Nevertheless, the project was able to avoid all deforestation predicted for this period and no deforestation took place in the project area.

The project area presents two forest classes. The Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs) compose class F1, while Savanna Woodland (Sd) make up class F2.

Table 12: Annual areas of baseline deforestation in the project area (Table 9.b of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year	Total - PA			
		F1 ha	F2 ha	annual ABSLPA _t ha	cummulative ABSLPA ha
2021	0	0.0	0.0	0	0
2022	1	127	25	152	152
2023	2	108	38	146	298
2024	3	89	41	130	428
2025	4	110	32	142	570
2026	5	103	16	119	689
2027	6	243	56	299	988
2028	7	105	65	170	1,158
2029	8	52	41	93	1,251
2030	9	35	37	72	1,323
2031	10	58	39	97	1,420
2032	11	98	32	130	1,550
2033	12	79	16	95	1,645
2034	13	46	45	91	1,736
2035	14	52	47	99	1,835
2036	15	36	85	121	1,956
2037	16	29	31	60	2,016
2038	17	53	89	142	2,158
2039	18	58	24	82	2,240
2040	19	52	19	71	2,311
2041	20	83	15	98	2,409
2042	21	72	39	111	2,520
2043	22	78	26	104	2,624
2044	23	54	41	95	2,719
2045	24	63	38	101	2,820
2046	25	58	41	99	2,919
2047	26	0	0	0	2,919
2048	27	0	0	0	2,919
2049	28	0	0	0	2,919
2050	29	0	0	0	2,919
2051	30	0.0	0.00	0	2,919
TOTAL				2,919.0	

Table 13: Total ex-ante and ex-post deforestation in the project area (Table 9.b of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year	Total Ex-ante - PA				Total Ex-post - PA			
		F1	F2	annual ABSLPA _t	cummulative ABSLPA	F1	F2	annual ABSLPA _t	cummulative ABSLPA
		ha	ha	ha	ha	ha	ha	ha	ha
2021	0	0	0	0	0	0	0	0	0
2022	1	127	25	152	152	0	0	0	0
TOTAL				152				0	

The accumulated baseline deforestation projected to occur within the Leakage Belt over the 30-year project lifetime was estimated at 2,640 hectares. The estimated average annual rate of deforestation is 88 hectares.

For this monitoring period, the expected deforestation to occur in the leakage belt was 278 hectares.

Table 14: Annual areas of baseline deforestation in the leakage belt (Table 9.c of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year	Total - LK			
		F1	F2	annual ABSLPA _t	cummulative ABSLPA
		ha	ha	ha	ha
2021	0	0.0	0.0	0	0
2022	1	224	54	278	278
2023	2	101	24	125	403
2024	3	136	36	172	575
2025	4	220	10	230	805
2026	5	167	23	190	995
2027	6	189	31	220	1,215
2028	7	56	14	70	1,285
2029	8	53	10	63	1,348
2030	9	69	18	87	1,435
2031	10	96	14	110	1,545
2032	11	78	9	87	1,632
2033	12	51	6	57	1,689

Year	Project year	Total - LK			
		F1	F2	annual ABSLPA _t	cummulative ABSLPA
		ha	ha	ha	ha
2034	13	68	10	78	1,767
2035	14	93	18	111	1,878
2036	15	48	4	52	1,930
2037	16	93	29	122	2,052
2038	17	73	12	85	2,137
2039	18	60	10	70	2,207
2040	19	74	4	78	2,285
2041	20	62	21	83	2,368
2042	21	58	16	74	2,442
2043	22	58	4	62	2,504
2044	23	75	10	85	2,589
2045	24	33	18	51	2,640
2046	25	0	0	0	2,640
2047	26	0	0	0	2,640
2048	27	0	0	0	2,640
2049	28	0	0	0	2,640
2050	29	0	0	0	2,640
2051	30	0.0	0.00	0	2,640
TOTAL				2,640.0	

Table 15. Total ex-ante and ex-post deforestation in the leakage belt (Table 9.c of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semideciduous Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year	Total Ex-ante - LK				Total Ex-post - LK			
		F1	F2	annual ABSLLK _t	cummulative ABSLLK	F1	F2	annual ABSLLK _t	cummulative ABSLLK
		ha	ha	ha	ha	ha	ha	ha	ha
2021	0	0	0	0	0	0	8	8	8
2022	1	224	54	278	278	2	162	164	172
TOTAL				278				172	

The accumulated baseline deforestation projected to occur for forest classes F1 and F2 within the Reference Region was estimated as 357,652 hectares over the 30-year project lifetime, corresponding to an estimated average annual rate of deforestation is 11,921 hectares.

The ex-ante estimated deforestation to occur in this monitoring period was 21,268 ha. However, the real deforestation in the reference region was 24,918 ha, being 17% higher than the previously projected. This fact corroborates the importance of the Itaúba REDD+ Project for the conservation of the forest in the region.

Table 16: Annual areas of baseline deforestation in the reference region (Table 9.a of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semideciduous Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year	Total - RR			
		F1 (ha)	F2 (ha)	annual ABSLRR _t ha	cummulative ABSLRR ha
2021	0	815	126	941	941
2022	1	17,205	3,122	20,327	21,268
2023	2	20,464	1,235	21,699	42,967
2024	3	20,416	2,561	22,977	65,944
2025	4	21,277	2,879	24,156	90,101
2026	5	22,488	2,745	25,233	115,334
2027	6	24,048	2,156	26,204	141,537
2028	7	21,783	2,456	24,239	165,776
2029	8	19,270	3,120	22,390	188,166
2030	9	18,224	2,421	20,645	208,811
2031	10	16,480	2,516	18,996	227,807
2032	11	14,947	2,487	17,434	245,242
2033	12	13,986	1,965	15,951	261,193
2034	13	12,685	1,854	14,539	275,732
2035	14	11,581	1,610	13,191	288,923
2036	15	10,361	1,540	11,901	300,824
2037	16	9,198	1,465	10,663	311,487
2038	17	7,892	1,578	9,470	320,957
2039	18	6,998	1,320	8,318	329,276
2040	19	6,246	956	7,202	336,478
2041	20	5,262	854	6,116	342,594
2042	21	4,272	785	5,057	347,651
2043	22	3,365	654	4,019	351,670
2044	23	2,430	568	2,998	354,668
2045	24	1,529	462	1,991	356,659
2046	25	672	321	993	357,652
2047	26	0	0	0	357,652
2048	27	0	0	0	357,652
2049	28	0	0	0	357,652
2050	29	0	0	0	357,652

2051	30	0	0	0	357,652
TOTAL				357,652	

Table 17: Total ex-ante and ex-post deforestation in the reference region (Table 9.a of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year	Total Ex-Ante - RR				Total Ex-Post - RR				percentage difference (δ)
		F1	F2	annual ABSLRR _t ha	cummulative ABSLRR ha	F1	F2	annual ABSLRR _t ha	cummulative ABSLRR ha	
2021	0	815	126	941	941	852	319	1,171	1,171	24%
2022	1	17,205	3,122	20,327	21,268	17,274	6,473	23,747	24,918	17%
TOTAL				21,268				24,918		17%

Table 18. Annual areas deforested per forest class icl within the project area in the baseline case (baseline activity data per forest class) (Table 11.b of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	ID _{icl} >	1	2	Total baseline deforestation in the project area	
	Name >	F1	F2	annual ABSLPA _t	cummulative ABSLPA
	Project year <i>t</i>	ha	ha	ha	ha
2021	0	0	0	0	0
2022	1	127	25	152	152
2023	2	108	38	146	298
2024	3	89	41	130	428
2025	4	110	32	142	570
2026	5	103	16	119	689
2027	6	243	56	299	988
2028	7	105	65	170	1158
2029	8	52	41	93	1251
2030	9	35	37	72	1323
2031	10	58	39	97	1420
2032	11	98	32	130	1550
2033	12	79	16	95	1645
2034	13	46	45	91	1736
2035	14	52	47	99	1835
2036	15	36	85	121	1956
2037	16	29	31	60	2016
2038	17	53	89	142	2158
2039	18	58	24	82	2240

2040	19	52	19	71	2311
2041	20	83	15	98	2409
2042	21	72	39	111	2520
2043	22	78	26	104	2624
2044	23	54	41	95	2719
2045	24	63	38	101	2820
2046	25	58	41	99	2919
2047	26	0	0	0	2919
2048	27	0	0	0	2919
2049	28	0	0	0	2919
2050	29	0	0	0	2919
2051	30	0	0	0	2919
TOTAL				2,919	

Table 19. Total ex-ante and ex-post annual areas deforested per forest class icl within the project area in the baseline case (baseline activity data per forest class) (Table 11.b of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Area deforested per forest class icl within the project area								
	Total Ex-ante - PA					Total Ex-post - PA			
	ID _{icl} >	1	2	Total baseline deforestation in the PA		1	2	Total baseline deforestation in the PA	
	Name >	F1	F2	annual ABSLPA _t	cummulative ABSLPA	F1	F2	annual ABSLPA _t	cummulative ABSLPA
	Project year <i>t</i>	ha	ha	ha	ha	ha	ha	ha	ha
2021	0	0	0	0	0	0	0	0	0
2022	1	127	25	152	152	0	0	0	0
TOTAL		127	25	152				0	

Table 20. Annual areas deforested per forest class icl within the leakage belt area in the baseline case (baseline activity data per forest class) (Table 11.c of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Area deforested per forest class icl within the leakage belt area				
	ID _{icl} >	1	2	Total baseline deforestation in the leakage belt area	
	Name >	F1	F2	annual ABSLLK _t	cummulative ABSLLK
	Project year <i>t</i>	ha	ha	ha	ha
2021	0	0	0	0	0
2022	1	224	54	278	278

Year	Area deforested per forest class icl within the leakage belt area				
	ID _{icl} >	1	2	Total baseline deforestation in the leakage belt area	
	Name >	F1	F2	annual ABSLLKt	cummulative ABSLLK
	Project year <i>t</i>	ha	ha	ha	ha
2023	2	101	24	125	403
2024	3	136	36	172	575
2025	4	220	10	230	805
2026	5	167	23	190	995
2027	6	189	31	220	1,215
2028	7	56	14	70	1,285
2029	8	53	10	63	1,348
2030	9	69	18	87	1,435
2031	10	96	14	110	1,545
2032	11	78	9	87	1,632
2033	12	51	6	57	1,689
2034	13	68	10	78	1,767
2035	14	93	18	111	1,878
2036	15	48	4	52	1,930
2037	16	93	29	122	2,052
2038	17	73	12	85	2,137
2039	18	60	10	70	2,207
2040	19	74	4	78	2,285
2041	20	62	21	83	2,368
2042	21	58	16	74	2,442
2043	22	58	4	62	2,504
2044	23	75	10	85	2,589
2045	24	33	18	51	2,640
2046	25	0	0	0	2,640
2047	26	0	0	0	2,640
2048	27	0	0	0	2,640
2049	28	0	0	0	2,640
2050	29	0	0	0	2,640
2051	30	0	0	0	2,640
TOTAL				2,640	

Table 21. Ex-ante and ex-post annual areas deforested per forest class icl within the leakage belt area in the baseline case (baseline activity data per forest class) (Table 11.c of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

	Total Ex-ante - LB				Total Ex-post - LB		
	ID _{icl} >	1	2	Total baseline deforestation in the leakage belt area	1	2	Total baseline deforestation in the leakage belt area

Year	Name >	F1	F2	annual ABSLLK _t	cummulative ABSLLK	F1	F2	annual ABSLLK _t	cummulative ABSLLK
	Project year <i>t</i>	ha	ha	ha	ha	ha	ha	ha	ha
2021	0	0	0	0	0	0	8	8	8
2022	1	224	54	278	278	2	162	164	172
TOTAL		224	54	278				172	

Calculation of baseline activity data per forest class

To project the LU/LC classes that will replace forests in the baseline case, Method 1 “Historical LU/LC-change” was chosen, because historical LU/LC-changes are assumed to be representative of future trends. Hence, post-deforestation land-uses are allocated to the projected areas of annual deforestation in same proportions as those observed on lands deforested during the historical reference period in the Reference Region.

Pasture and soy farming are responsible for most of the occupation of deforested land in the project region. The post-deforestation carbon stocks were determined by the historical area-weighted average carbon stock of pasture and soy farming (cropland).

The following tables shows the results.

Table 22: Annual areas deforested in each zone within the project area in the baseline case (baseline activity data zone) (Table 13.b of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semideciduous Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	ID _z >	1	Total baseline deforestation in the project area	
	Name >	Pasture and soy plantation		
	Project year <i>t</i>	ha	annual ABSLPA _t ha	cummulative ABSLPA ha
2021	0	0	0	0
2022	1	152	152	152
2023	2	146	146	298
2024	3	130	130	428
2025	4	142	142	570
2026	5	119	119	689
2027	6	299	299	988
2028	7	170	170	1,158
2029	8	93	93	1,251
2030	9	72	72	1,323
2031	10	97	97	1,420

Year	ID _z >	1	Total baseline deforestation in the project area	
	Name >	Pasture and soy plantation		
	Project year <i>t</i>	ha	annual ABSLPA _t ha	cummulative ABSLPA ha
2032	11	130	130	1,550
2033	12	95	95	1,645
2034	13	91	91	1,736
2035	14	99	99	1,835
2036	15	121	121	1,956
2037	16	60	60	2,016
2038	17	142	142	2,158
2039	18	82	82	2,240
2040	19	71	71	2,311
2041	20	98	98	2,409
2042	21	111	111	2,520
2043	22	104	104	2,624
2044	23	95	95	2,719
2045	24	101	101	2,820
2046	25	99	99	2,919
2047	26	0	0	2,919
2048	27	0	0	2,919
2049	28	0	0	2,919
2050	29	0	0	2,919
2051	30	0	0	2,919
TOTAL		2,919	2,919.0	

Table 23: Ex-ante and ex-post annual areas deforested in each zone within the project area in the baseline case (baseline activity data zone) (Table 13.b of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Total Ex-ante - PA				Total Ex-post - PA		
	ID _z >	1	Total baseline deforestation in the project area		1	Total baseline deforestation in the project area	
	Name >	Pasture and soy plantation			Pasture and soy plantation		
	Project year <i>t</i>	ha	annual ABSLPA _t ha	cummulative ABSLPA ha	ha	annual ABSLPA _t ha	cummulative ABSLPA ha
2021	0	0	0	0	0	0	0
2022	1	152	152	152	0	0	0
TOTAL		152	152				

Table 24: Annual areas deforested in each zone within the leakage belt area in the baseline case (baseline activity data per zone) (Table 13.c of VM0015 v1.1). F1 = Average between Submontane Open

Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	ID _z >	1	Total baseline deforestation in the leakage belt	
	Name >	Pasture and soy plantation		
	Project year <i>t</i>	ha	annual ABSLLK _t ha	cummulative ABSLLK ha
2021	0	0.00	0	0
2022	1	278	278	278
2023	2	125	125	403
2024	3	172	172	575
2025	4	230	230	805
2026	5	190	190	995
2027	6	220	220	1,215
2028	7	70	70	1,285
2029	8	63	63	1,348
2030	9	87	87	1,435
2031	10	110	110	1,545
2032	11	87	87	1,632
2033	12	57	57	1,689
2034	13	78	78	1,767
2035	14	111	111	1,878
2036	15	52	52	1,930
2037	16	122	122	2,052
2038	17	85	85	2,137
2039	18	70	70	2,207
2040	19	78	78	2,285
2041	20	83	83	2,368
2042	21	74	74	2,442
2043	22	62	62	2,504
2044	23	85	85	2,589
2045	24	51	51	2,640
2046	25	0	0	2,640
2047	26	0	0	2,640
2048	27	0	0	2,640
2049	28	0	0	2,640
2050	29	0	0	2,640
2051	30	0.00	0	2,640
TOTAL		2,640	2,640	

Table 25. Ex-ante and ex-post annual areas deforested in each zone within the leakage belt area in the baseline case (baseline activity data per zone) (Table 13.c of VM0015 v1.1). F1 = Average between

Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Total Ex-ante - LB				Total Ex-post - LB		
	ID _z >	1	Total baseline deforestation in the leakage belt		1	Total baseline deforestation in the leakage belt	
	Name >	Pasture and soy plantation			Pasture and soy plantation		
	Project year <i>t</i>	ha	annual ABSLLK _t ha	cummulative ABSLLK ha	ha	annual ABSLLK _t ha	cummulative ABSLLK ha
2021	0	0	0	0	8	8	8
2022	1	278	278	278	164	164	172
TOTAL		278			172		

Estimation of baseline carbon stock changes

The biomass values used for aboveground, belowground and litter were extracted from the 3rd National Climate Change Report [5], the "Third Brazilian Inventory of Anthropogenic Greenhouse Gas Emissions and Removals", an official document of the Brazilian State to meet the commitment assumed in the United Nations Framework Convention on Climate Change (UNFCCC), adopted at the Earth Summit, in Rio de Janeiro, in 1992. This document includes values of GHG emissions for the sectors defined by the Intergovernmental Panel on Climate Change (IPCC), including the Agriculture, Forestry and Other Land Use (AFOLU) sector, which use Tier 2 as estimation of associated carbon stocks (Emission Factor).

For the region in which the project area is located, this work calculated the biomass values for all the vegetation present, including those present in the project area, namely, Submontane Open Ombrophilous Forest and Submontane Semidecidual Seasonal Forest¹⁶ (project area from Marcelândia) and Savana Wooded (project area from Itaúba).

The project considers two forest classes for means of calculations. Class F1 represents the vegetation present in the PA from Marcelândia that has biomass data available in literature. It is the average between the Submontane Open Ombrophilous Forest (As) and the Submontane Semidecidual Seasonal Forest (Fs), which results in 404.7 tCO₂/ha of total biomass. Savanna Woodland (Sd) make up class F2, present in the PA from Itaúba, with 147.3 tCO₂/ha of total biomass. To obtain these values in tCO₂/ha, the biomass data in tC/ha present in the Third Brazilian Inventory were multiplied by the molecular weight of CO₂ (44/12). The carbon fraction used in this inventory is the IPCC value default CF=0.47 (IPCC, 2006). The lower value of biomass was adopted for F1 and F2 classes conservatively, once the uncertainty of the biomass data is 12.85% (see Figure 29 of the Third Brazilian Inventory [5]).

¹⁶ The project area from Marcelândia is part of the Evergreen Seasonal Forest, whose biomass value has not yet been estimated. In this way, taking into account the principle of prudence and conservatism, the average between the Submontane Decidual Seasonal Forest and the Submontane Open Forest was used, as explained in section 2.1.5.6 of PD.

The table below shows the carbon stock per hectare of Submontane Open Ombrophilous Forest (As), Submontane Semidecidual Seasonal Forest (Fs) and Savanna woodland (Sd) present in the project area and leakage belt.

Table 26: Carbon stocks per hectare of initial forest classes *icl* existing in the project area and leakage belt: Estimated values (Table 15.a of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year t	Initial forest class <i>icl</i>													
		Name: F1 ID <i>icl</i> 1 Average carbon stock per hectare $\pm 90\%$ CI													
		Cab <i>icl</i>		Cbb <i>icl</i>		CI <i>icl</i>		Cwp <i>icl</i>						Ctot <i>icl</i>	
		C stock	$\pm 90\%$ CI	C stock	$\pm 90\%$ CI	C stock	$\pm 90\%$ CI	short lived		medium lived		long lived		C stock	$\pm 90\%$ CI
		t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹
2021	0	407.9	355.5	35.5	31.0	21.0	18.3	0	0	0	0	10.5	10.5	453.9	394.2
2022	1		460.3		40.1		23.7		0		0		10.5		513.6

Year	Project year t	Initial forest class <i>icl</i>													
		Name: F2 ID <i>icl</i> 2 Average carbon stock per hectare $\pm 90\%$ CI													
		Cab <i>icl</i>		Cbb <i>icl</i>		CI <i>icl</i>		Cwp <i>icl</i>						Ctot <i>icl</i>	
		C stock	$\pm 90\%$ CI	C stock	$\pm 90\%$ CI	C stock	$\pm 90\%$ CI	short lived		medium lived		long lived		C stock	$\pm 90\%$ CI
		t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹
2021	0	122.1	106.4	26.8	23.4	20.1	17.5	0	0	0	0	10.3	10.3	158.7	137.0
2022	1		137.7		30.3		22.7		0		0		10.3		180.4

90% Confidence Intervals have been used to define whether the most suitable choice would be the average or the lower limit of the range, to mitigate uncertainties in estimates for above biomass, below biomass, litter and wood product carbon pools. The lower limit of the interval was used for the calculations, for conservativeness purposes and mitigation of uncertainties.

Table 27: Carbon stocks per hectare of initial forest classes *icl* existing in the project area and leakage belt: Values to be used after discounts due to uncertainties (Table 15.b of VM0015 v.1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year t	Initial forest class <i>icl</i>													
		Name: F1 ID <i>icl</i> 1 Average carbon stock per hectare $\pm 90\%$ CI													
		Cab <i>icl</i>		Cbb <i>icl</i>		CI <i>icl</i>		Cwp <i>icl</i>						Ctot <i>icl</i>	
		C stock	C stock change	C stock	C stock change	C stock	C stock change	C stock	C stock change	C stock	C stock change	C stock	C stock change	C stock	C stock change
		t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹
2021	0														
2022	1														
2023	2	355.5	-52.4	31.0	-4.6	18.3	-2.7	0.0	0.0	0.0	0.0	10	0.0	394.2	-59.7
...	...														
2051	30														

Year	Project year t	Initial forest class <i>icl</i>													
		Name: F2													
		ID <i>icl</i> 2													
		Average carbon stock per hectare ± 90% CI													
		Cab <i>icl</i>		Cbb <i>icl</i>		Cl <i>icl</i>		Cwp <i>icl</i>						Ctot <i>icl</i>	
C stock	C stock change	C stock	C stock change	C stock	C stock change	short lived		medium lived		long lived					
						C stock	C stock change	C stock	C stock change	C stock	C stock change	C stock	C stock change	C stock	C stock change
t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹	t CO ₂ e ha ⁻¹
2019	1	106.4	-15.7	23.4	-3.4	17.5	-2.6	0	0	0	0	10	0	137.0	-21.7
2020	2														
2021	3														
...	...														
2049	31														

The carbon stock change factors shown below were calculated based on VM0015 v1.1 premises, using Method 2, as activity data are available for categories. First the carbon stock change factors for each category were calculated as indicated in table 20.c from VM0015 v1.1 and then the baseline carbon stock changes were calculated for project area and leakage belt area by multiplying activity data with their corresponding emission factors.

Table 28: Carbon stock change factors for land-use change categories (*ct* or *ctz*) (Method 2) (Table 20.c of VM0015 1.1) in the Forest Class F1. F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs)

Year after deforestation		$\Delta C_{ab,ctz,t}$	$\Delta C_{bb,ctz,t}$	$\Delta C_{l,ctz,t}$	$\Delta C_{wp,ctz,t}$ long-lived
0	2021	-353.71	-1.19	-18.28	0
1	2022	1.79	-1.19	0.00	0
2	2023	1.79	-1.19	0.00	0
3	2024	1.79	-1.19	0.00	0
4	2025	1.79	-1.19	0.00	0
5	2026	1.79	-1.19	0.00	0
6	2027	1.79	-1.19	0.00	0
7	2028	1.79	-1.19	0.00	0
8	2029	1.79	-1.19	0.00	0
9	2030	1.79	-1.19	0.00	0
10	2031	0	0	0.00	0
11	2032	0	0	0.00	0
12	2033	0	0	0.00	0
13	2034	0	0	0.00	0
14	2035	0	0	0.00	0
15	2036	0	0	0.00	0
16	2037	0	0	0.00	0
17	2038	0	0	0.00	0
18	2039	0	0	0.00	0
19	2040	0	0	0.00	0
20	2041	0	0	0.00	0
21	2042	0	0	0.00	0
22	2043	0	0	0.00	0
23	2044	0	0	0.00	0
24	2045	0	0	0.00	0
25	2046	0	0	0.00	0
26	2047	0	0	0.00	0
27	2048	0	0	0.00	0
28	2049	0	0	0.00	0
29	2050	0	0	0.00	0
30	2051	0	0	0.00	0

Table 29: Carbon stock change factors for land-use change categories (ct or ctz) (Method 2) (Table 20.c of VM0015 v1.1) in the Forest Class F2. F2 = Savanna woodland (Sd).

Year after deforestation		$\Delta C_{ab,ctz,t}$	$\Delta C_{bb,ctz,t}$	$\Delta C_{l,ctz,t}$	$\Delta C_{wp,ctz,t}$ long-lived
0	2021	-104.59	-1.19	-17.54	0
1	2022	1.79	-1.19	0.00	0
2	2023	1.79	-1.19	0.00	0
3	2024	1.79	-1.19	0.00	0
4	2025	1.79	-1.19	0.00	0
5	2026	1.79	-1.19	0.00	0
6	2027	1.79	-1.19	0.00	0
7	2028	1.79	-1.19	0.00	0
8	2029	1.79	-1.19	0.00	0
9	2030	1.79	-1.19	0.00	0
10	2031	0	0	0.00	0
11	2032	0	0	0.00	0
12	2033	0	0	0.00	0
13	2034	0	0	0.00	0
14	2035	0	0	0.00	0
15	2036	0	0	0.00	0
16	2037	0	0	0.00	0
17	2038	0	0	0.00	0
18	2039	0	0	0.00	0
19	2040	0	0	0.00	0
20	2041	0	0	0.00	0
21	2042	0	0	0.00	0
22	2043	0	0	0.00	0
23	2044	0	0	0.00	0
24	2045	0	0	0.00	0
25	2046	0	0	0.00	0
26	2047	0	0	0.00	0
27	2048	0	0	0.00	0
28	2049	0	0	0.00	0
29	2050	0	0	0.00	0
30	2051	0	0	0.00	0

The following tables show the calculation of baseline carbon stock changes in above-ground biomass, below-ground biomass, litter, and wood products in the project area, using the carbon stock change factors presented in the tables immediately above.

Table 30: Baseline carbon stock change in the above-ground biomass in the project area (Table 22.b.1 of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year t	Activity data per category x Carbon stock change factor for above-ground biomass in the project area				Total baseline carbon stock change in the project area	
		F1		F2		annual	cumulative
		ABSLPA _{ct,t}	ΔCab _{ct,t}	ABSLPA _{ct,t}	ΔCab _{ct,t}	ΔCabBSLPA _t	ΔCabBSLPA
		ha	tCO ₂ -e ha ⁻¹	ha	tCO ₂ -e ha ⁻¹	tCO ₂ -e	tCO ₂ -e
2021	0	0	0	0	0	0	0
2022	1	127	-353.71	25	-104.59	-47,536	-47,536

Table 31: Baseline carbon stock change in the below-ground biomass in the project area (Table 22.b.2 of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year t	Activity data per category x Carbon stock change factor for below-ground biomass in the project area				Total baseline carbon stock change in the project area	
		F1		F2		annual	cumulative
		BBSLPA _{ct,t}	ΔCbb _{ct,t}	BBSLPA _{ct,t}	ΔCbb _{ct,t}	ΔCbbBSLPA _t	ΔCbbBSLPA
		ha	tCO ₂ -e ha ⁻¹	ha	tCO ₂ -e ha ⁻¹	tCO ₂ -e	tCO ₂ -e
2021	0	0	0.00	0	0.00	0	0
2022	1	127	-1.19	25	-1.19	-181	-181

Table 32: Baseline carbon stock change in the litter biomass in the project area (Table 22.b.4 of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year t	Activity data per category x Carbon stock change factor for litter biomass in the project area				Total baseline carbon stock change in the project area	
		F1		F2		annual	cumulative
		CLSLPA _{ct,t}	ΔCl _{ct,t}	CLSLPA _{ct,t}	ΔCl _{ct,t}	ΔCLBSLPA _t	ΔCLBSLPA
		ha	tCO ₂ -e ha ⁻¹	ha	tCO ₂ -e ha ⁻¹	tCO ₂ -e	tCO ₂ -e
2021	0	0	0.00	0	0.00	0	0
2022	1	127	-18.28	25	-17.54	-2,760	-2,760

Table 33: Baseline carbon stock change in the wood products in the project area (Table 22.b.6 of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year t	Activity data per category x Carbon stock change factor for wood products biomass in the project area				Total baseline carbon stock change in the project area	
		F1		F2		annual	cumulative
		WPSLPA _{ct,t}	ΔCwp _{ct,t}	WPSLPA _{ct,t}	ΔCwp _{ct,t}	ΔCwpBSLPA _t	ΔCwpBSLPA
		ha	tCO ₂ -e ha ⁻¹	ha	tCO ₂ -e ha ⁻¹	tCO ₂ -e	tCO ₂ -e
2021	0	0	0	0	0	0	0
2022	1	127	0	25	0	0	0

The following tables show the calculation of baseline carbon stock changes in above-ground biomass, below-ground biomass, litter and wood products in the Leakage Belt, using the carbon stock change factors.

Table 34: Baseline carbon stock change in the above-ground biomass in the leakage belt area (Table 22.c.1 of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year t	Activity data per category x Carbon stock change factor for above-ground biomass in the leakage belt				Total baseline carbon stock change in the leakage belt	
		F1		F2		annual	cumulative
		ABSLLK _{ct,t}	ΔCab _{ct,t}	ABSLLK _{ct,t}	ΔCab _{ct,t}	ΔCabBSLLK _t	ΔCabBSLLK
		ha	tCO ₂ -e ha ⁻¹	ha	tCO ₂ -e ha ⁻¹	tCO ₂ -e	tCO ₂ -e
2021	0	0	0	0	0	0	0
2022	1	224	-353.71	54	-104.59	-84,878	-84,878

Table 35: Baseline carbon stock change in the below-ground biomass in the leakage belt area (Table 22.c.2 of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year t	Activity data per category x Carbon stock change factor for below-ground biomass in the leakage belt				Total baseline carbon stock change in the leakage belt	
		F1		F2		annual	cumulative
		BBSLLK _{ct,t}	ΔCbb _{ct,t}	BBSLLK _{ct,t}	ΔCbb _{ct,t}	ΔCbbBSLLK _t	ΔCbbBSLLK
		ha	tCO ₂ -e ha ⁻¹	ha	tCO ₂ -e ha ⁻¹	tCO ₂ -e	tCO ₂ -e
2021	0	0	0	0	0	0	0
2022	1	224	-1.19	54	-1.19	-331	-331

Table 36: Baseline carbon stock change in the litter biomass in the leakage belt area (Table 22.c.4 of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semideciduous Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year t	Activity data per category x Carbon stock change factor for litter biomass in the leakage belt				Total baseline carbon stock change in the leakage belt	
		F1		F2		annual	cumulative
		CLSLLK _{ct,t}	ΔCl _{t,t}	CLSLLK _{ct,t}	ΔCl _{ct,t}	ΔCIBSLLK _t	ΔCIBSLLK
		ha	tCO ₂ -e ha ⁻¹	ha	tCO ₂ -e ha ⁻¹	tCO ₂ -e	tCO ₂ -e
2021	0	0	0.00	0	0.00	0	0
2022	1	224	-18.28	54	-17.54	-5,042	-5,042

Table 37: Baseline carbon stock change in the wood products in the leakage belt area (Table 22.c.6 of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semideciduous Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year t	Activity data per category x Carbon stock change factor for wood products biomass in the leakage belt area				Total baseline carbon stock change in the leakage belt area	
		F1		F2		annual	cumulative
		WPSLLK _{ct,t}	ΔCwp _{ct,t}	WPSLLK _{ct,t}	ΔCwp _{ct,t}	ΔCwpBSLLK _t	ΔCwpBSLLK
		ha	tCO ₂ -e ha ⁻¹	ha	tCO ₂ -e ha ⁻¹	tCO ₂ -e	tCO ₂ -e
2021	0	0	0	0	0	0	0
2022	1	224	0	54	0	0	0

Baseline non-CO₂ emissions from forest fires

The use of fire for deforestation in the Amazon is already a well-established and well-known practice. As described in Barlow et al. (2020)[6], fire is used to clear the primary forest and then prepare the area for agriculture (e.g., pasture, soybean). In addition, data published by INPE¹⁷ reveal that the greatest focus of fires occurs in areas that are recently deforested. The data show an increase in the number of fire alerts in the RR over the historical reference period.

To estimate non-CO₂ emissions from forest fires, the average percentage of the deforested area in which fire was used, the average proportion of mass burnt in each carbon pool (P_{burnt,p}), and the average combustion efficiency of each pool (CE_p) were estimated. These average percentage values were estimated for each forest class (icl) and are assumed to remain the same in the future.

In accordance with the VM0015 v1.1, GHG emissions from biomass burning were estimated based on revised IPCC 1996 GL LULUCF guidance, as follows.

¹⁷ <https://queimadas.dgi.inpe.br/queimadas/bdqueimadas>

$$EBB_{tot_{icl,t}} = EBB_{N20_{icl,t}} + EBB_{CH4_{icl,t}}$$

Where:

$EBB_{tot_{icl,t}}$	Total GHG emission from biomass burning in forest class icl at year t; tCO ₂ -e ha ⁻¹
$EBB_{N20_{icl,t}}$	Total GHG emission from biomass burning in forest class icl at year t; tCO ₂ -e ha ⁻¹
$EBB_{CH4_{icl,t}}$	CH ₄ emission from biomass burning in forest class icl at year t; tCO ₂ -e ha ⁻¹

$$EBB_{N20_{icl,t}} = EBB_{CO2_{icl,t}} * 12/44 * NCR * ER_{N20} * 44/28 * GWP_{N20}$$

$$EBB_{CH4_{icl,t}} = EBB_{CO2_{icl,t}} * 12/44 * ER_{CH4} * 16/12 * GWP_{CH4}$$

Where:

$EBB_{CO2_{icl,t}}$	Per hectare CO ₂ emission from biomass burning in slash and burn in forest class icl at year t; tCO ₂ -e ha ⁻¹
$EBB_{N20_{icl,t}}$	Per hectare N ₂ O emission from biomass burning in slash and burn in forest class icl at year t; tCO ₂ -e ha ⁻¹
$EBB_{CH4_{icl,t}}$	Per hectare CH ₄ emission from biomass burning in slash and burn in forest class icl at year t; tCO ₂ -e ha ⁻¹
NCR	Nitrogen to carbon ratio (IPCC default value = 0.01); dimensionless
ER_{N20}	Emission ratio for N ₂ O (IPCC default value = 0.007)
ER_{CH4}	Emission ratio for CH ₄ (IPCC default value = 0.012)
GWP_{N20}	Global Warming Potential for N ₂ O (IPCC AR6 default value = 273)
GWP_{CH4}	Global Warming Potential for CH ₄ (IPCC AR6 default value = 29.8)

$$EBBCO2_{icl,t} = Fburnt_{icl} * \sum_{p=1}^p (C_{p,icl,t} * Pburnt_{p,icl} * CE_{p,icl})$$

Where:

- EBBCO2_{icl,t}** Per hectare CO₂ emission from biomass burning in slash and burn in forest class icl at year t; tCO₂-e ha⁻¹
- Fburnt_{icl}** Proportion of forest area burned during the historical reference period in the forest class icl; %
- C_{p,icl,t}** Average carbon stock per hectare in the carbon pool p burnt in the forest class icl at year t; tCO₂-e ha⁻¹
- Pburnt_{p,icl}** Average proportion of mass burnt in the carbon pool p in the forest class icl; %
- CE_{p,icl}** Average combustion efficiency of the carbon pool p in the forest class icl; dimensionless
- p** Carbon pool that could burn (above and below - ground biomass, litter)
- icl** 1, 2, 3, ... Icl (pre-deforestation) forest classes
- t** 1, 2, 3 ... T, a year of the proposed project crediting period; dimensionless

The non-CO₂ emissions from forest fires were calculated and are shown in the following tables. The values used for each parameter are detailed in the section 3.1.1 Data and Parameters Available at Validation.

Table 38: Parameters used to calculate non-CO₂ emissions from forest fires (Table 23 of VM0015 v1.1) for F1 (above) and F2 (below). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Initial Forest Class		Parameters												
		<i>Fburnt_{icl}</i>	<i>Cab</i>	<i>Cl</i>	<i>Pburnt_{ab,icl}</i>	<i>Pburnt_{cl,icl}</i>	<i>CEab,icl</i>	<i>CEcl,icl</i>	<i>ECO2-ab</i>	<i>ECO2-l</i>	<i>EBBCO2-tot</i>	<i>EBBN2O_{icl}</i>	<i>EBBCH4_{icl}</i>	<i>EBBtot_{icl}</i>
<i>IDcl</i>	Name	%	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹	%	%	%	%	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹
1	F1	1.0	355.5	18.3	0.9	1.0	0.5	1.0	166.0	18.1	184.1	1.5	23.9	25.5

Initial Forest Class		Parameters												
		$F_{burnt_{(cl)}}$	C_{ab}	C_l	$P_{burnt_{ab,(cl)}}$	$P_{burnt_{cl,(cl)}}$	$C_{Eab,icl}$	$C_{Ecl,icl}$	$ECO2-ab$	$ECO2-l$	$EBBCO2-tot$	$EBBN2O_{(cl)}$	$EBBCH4_{(cl)}$	$EBBtot_{(cl)}$
IDcl	Name	%	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹	%	%	%	%	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹
2	F2	1.0	106.4	17.5	0.8	1.0	0.5	1.0	41.7	17.4	59.0	0.5	7.7	8.2

Table 39: Baseline non-CO2 emissions from forest fires in the project area (Table 24 of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Fs); F2 = Savanna woodland (Sd).

Year	Project year t	Emissions of non-CO2 gasses from baseline forest fires		Emissions of non-CO2 gasses from baseline forest fires		Total baseline non-CO2 emissions from forest fires in the project area	
		F1		F2		annual	cumulative
		ABSLPA _{icl,t}	EBBBSL _{tot,icl}	ABSLPA _{icl,t}	EBBBSL _{tot,icl}	EBBBSLPA _t	EBBBSLPA
		ha	tCO ₂ -e ha ⁻¹	ha	tCO ₂ -e ha ⁻¹	tCO ₂ -e	tCO ₂ -e
2021	0	0	25.45	0	8.16	0	0
2022	1	127	25.45	25	8.16	3,436	3,436

3.2.2 Project Emissions

The Itaúba REDD+ Project foresees, over a 30-year period, emissions related to planned controlled timber logging (sustainable forest management) and unavoided deforestation. Certain discrete areas of forest within the project area will be subject to project activities that will change the carbon stocks of these areas compared to the baseline. Afterthought that the logging that will take place in the project area is classified as low-impact, natural regeneration was considered to calculate the project emissions over a 30-year period.

The following tables shows the results of carbon decrease and increase due to logging activities in the project area over the project lifetime (Source: PD).

Table 40: Ex ante estimated actual carbon stock decrease due to planned logging activities in the project area (Table 25b of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Semidecidual Seasonal Forest (Sd); F2 = Savanna woodland (Sd)

Year	Project year t	Areas of planned logging activities x Carbon stock change (decrease) in the project area		Areas of planned logging activities x Carbon stock change (decrease) in the project area		Total carbon stock decrease due to planned logging activities	
		IDcl = 1		IDcl = 2		annual	cumulative
		APLPA _{icl,t}	ΔC _{tot,icl,t}	APLPA _{icl,t}	ΔC _{tot,icl,t}	ΔCPLdPA _t	ΔCPLdPA
		ha	tCO ₂ -e ha ⁻¹	ha	tCO ₂ -e ha ⁻¹	tCO ₂ -e	tCO ₂ -e

2021	0	0.0	0.00	0.00	0.00	0	0
2022	1	0.0	0.00	0.00	0.00	0	0
2023	2	0.0	0.00	1240.00	11.97	14,841	14,841
2024	3	0.0	0.00	0.00	0.00	0	14,841
2025	4	0.0	0.00	0.00	0.00	0	14,841
2026	5	0.0	0.00	0.00	0.00	0	14,841
2027	6	0.0	0.00	0.00	0.00	0	14,841
2028	7	0.0	0.00	0.00	0.00	0	14,841
2029	8	0.0	0.00	0.00	0.00	0	14,841
2030	9	0.0	0.00	0.00	0.00	0	14,841
2031	10	0.0	0.00	0.00	0.00	0	14,841
2032	11	0.0	0.00	0.00	0.00	0	14,841
2033	12	0.0	0.00	0.00	0.00	0	14,841
2034	13	0.0	0.00	0.00	0.00	0	14,841
2035	14	0.0	0.00	0.00	0.00	0	14,841
2036	15	0.0	0.00	0.00	0.00	0	14,841
2037	16	0.0	0.00	0.00	0.00	0	14,841
2038	17	0.0	0.00	0.00	0.00	0	14,841
2039	18	0.0	0.00	0.00	0.00	0	14,841
2040	19	0.0	0.00	0.00	0.00	0	14,841
2041	20	0.0	0.00	0.00	0.00	0	14,841
2042	21	0.0	0.00	0.00	0.00	0	14,841
2043	22	624.9	20.13	870.90	12.92	23,835	38,677
2044	23	1,567.1	20.13	0.00	0.00	31,546	70,222
2045	24	0.0	0.00	0.00	0.00	0	70,222
2046	25	865.7	20.13	914.80	12.92	29,251	99,473
2047	26	0.0	0.00	0.00	0.00	0	99,473
2048	27	1,407.6	20.13	0.00	0.00	28,336	127,809
2049	28	808.0	20.13	0.00	0.00	16,266	144,075
2050	29	1,211.0	20.13	0.00	0.00	24,378	168,452
2051	30	393.5	20.13	0.00	0.00	7,921	176,373

Table 41: Ex ante estimated carbon stock increase following planned logging activities in the project area (Table 26b of VM0015 v1.1). F1 = Average between Submontane Open Ombrophilous Forest (As) and Submontane Deciduous Seasonal Forest (Sd); F2 = Savanna woodland (Sd).

Year	Project year t	Areas of planned logging activities x Carbon stock change (increase up to maximum long-term average)				Total carbon stock increase due to planned logging activities	
		IDcl = 1		IDcl = 2		annual	cumulative
		APLPA _{icl,t}	ΔC _{toticl,t}	APLPA _{icl,t}	ΔC _{toticl,t}	ΔCPLiPA _t	ΔCPLiPA
		ha	tCO ₂ -e ha ⁻¹	ha	tCO ₂ -e ha ⁻¹	tCO ₂ -e	tCO ₂ -e
2021	0	0.0	0.00	0.00	0.00	0	0
2022	1	0.0	0.00	0.00	0.00	0	0
2023	2	0.0	0.00	1,240.00	1.02	1,265	1,265
2024	3	0.0	0.00	0.00	1.02	1,265	2,530
2025	4	0.0	0.00	0.00	1.02	1,265	3,795

2026	5	0.0	0.00	0.00	1.02	1,265	5,060
2027	6	0.0	0.00	0.00	1.02	1,265	6,325
2028	7	0.0	0.00	0.00	1.02	1,265	7,589
2029	8	0.0	0.00	0.00	1.02	1,265	8,854
2030	9	0.0	0.00	0.00	1.02	1,265	10,119
2031	10	0.0	0.00	0.00	1.02	1,265	11,384
2032	11	0.0	0.00	0.00	1.02	1,265	12,649
2033	12	0.0	0.00	0.00	1.02	1,265	13,914
2034	13	0.0	0.00	0.00	1.02	1,265	15,179
2035	14	0.0	0.00	0.00	1.02	1,265	16,444
2036	15	0.0	0.00	0.00	1.02	1,265	17,709
2037	16	0.0	0.00	0.00	1.02	1,265	18,974
2038	17	0.0	0.00	0.00	1.02	1,265	20,238
2039	18	0.0	0.00	0.00	1.02	1,265	21,503
2040	19	0.0	0.00	0.00	1.02	1,265	22,768
2041	20	0.0	0.00	0.00	1.02	1,265	24,033
2042	21	0.0	0.00	0.00	1.02	1,265	25,298
2043	22	624.9	3.04	870.90	1.02	4,052	29,350
2044	23	1,567.1	3.04	0.00	1.02	8,812	38,162
2045	24	0.0	3.04	0.00	1.02	8,812	46,975
2046	25	865.7	3.04	914.80	1.02	12,376	59,350
2047	26	0.0	3.04	0.00	1.02	12,376	71,726
2048	27	1,407.6	3.04	0.00	1.02	16,652	88,378
2049	28	808.0	3.04	0.00	1.02	19,107	107,484
2050	29	1,211.0	3.04	0.00	1.02	22,785	130,270
2051	30	393.5	3.04	0.00	1.02	23,981	154,251

As shown in previously tables, no forest logging occurred during this monitoring period. It is expected to occur during the next one. Thus, there were no project emissions related to this topic.

In the PD it was considered that some unplanned deforestation could happen in the project area despite the AUD project activity. To allow ex ante projections to be made, the project proponent conservatively considered that surveillance activities would be able to attain 90% of effectiveness in avoiding unplanned deforestation inside the project area. Then, the Effectiveness Index (EI) was estimated to be 90%. The estimated value of EI was used to multiply the baseline projections by the factor (1 - EI) and the result was considered the ex-ante estimated emissions from unplanned deforestation in the project case.

$$\Delta\text{CUDdPA}_t = \Delta\text{CBSL}_t * (1 - \text{EI})$$

Where:

ΔCUDdPA_t Total ex ante actual carbon stock change due to unavaoided unplanned deforestation at year t in the project area; tCO₂-e GHG emission from biomass burning in forest class icl at year t; tCO₂-e ha⁻¹

ΔCBSL_t	Total baseline carbon stock change at year t in the project area; tCO ₂ -e
EI	Ex ante estimated Effectiveness Index; %
t	1, 2, 3 ... T, a year of the proposed project crediting period; dimensionless

The following table show the results.

Table 42: Ex ante estimated net carbon stock change in the project area under the project scenario (Table 27 of VM0015 v1.1).

Year	Project year t	Total carbon stock decrease due to planned logging activities		Total carbon stock increase due to planned logging activities		Total carbon stock decrease due to unavoided unplanned deforestation		Total carbon stock change in the project case	
		annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative
		ΔCPLdPA_t	ΔCPLdPA	ΔCPLiPA_t	ΔCPLiPA	ΔCUDdPA_t	ΔCUDdPA	ΔCPSPA_t	ΔCPSPA
		tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
2021	0	0	0	0	0	0	0	0	0
2022	1	0	0	0	0	-5.048	-5.048	-5.048	-5.048
2023	2	14.841	14.841	1.265	1.265	-4.490	-9.537	-18.066	-23.114
2024	3	0	14.841	1.265	2.530	-3.809	-13.346	-2.544	-25.658
2025	4	0	14.841	1.265	3.795	-4.474	-17.820	-3.209	-28.867
2026	5	0	14.841	1.265	5.060	-4.007	-21.827	-2.742	-31.609
2027	6	0	14.841	1.265	6.325	-9.717	-31.544	-8.452	-40.061
2028	7	0	14.841	1.265	7.589	-4.661	-36.205	-3.396	-43.457
2029	8	0	14.841	1.265	8.854	-2.377	-38.582	-1.112	-44.569
2030	9	0	14.841	1.265	10.119	-1.687	-40.269	-422	-44.991
2031	10	0	14.841	1.265	11.384	-2.566	-42.835	-1.301	-46.292
2032	11	0	14.841	1.265	12.649	-3.976	-46.810	-2.711	-49.003
2033	12	0	14.841	1.265	13.914	-3.070	-49.880	-1.805	-50.808
2034	13	0	14.841	1.265	15.179	-2.198	-52.079	-934	-51.741
2035	14	0	14.841	1.265	16.444	-2.450	-54.529	-1.185	-52.927
2036	15	0	14.841	1.265	17.709	-2.323	-56.852	-1.058	-53.984
2037	16	0	14.841	1.265	18.974	-1.406	-58.258	-141	-54.126
2038	17	0	14.841	1.265	20.238	-3.024	-61.282	-1.759	-55.885
2039	18	0	14.841	1.265	21.503	-2.406	-63.688	-1.141	-57.026
2040	19	0	14.841	1.265	22.768	-2.120	-65.808	-855	-57.881
2041	20	0	14.841	1.265	24.033	-3.229	-69.036	-1.964	-59.845
2042	21	0	14.841	1.265	25.298	-3.116	-72.153	-1.851	-61.696
2043	22	23.835	38.677	4.052	29.350	-3.179	-75.331	-22.962	-84.658
2044	23	31.546	70.222	8.812	38.162	-2.467	-77.799	-25.201	-109.859
2045	24	0	70.222	8.812	46.975	-2.767	-80.565	6.046	-103.813
2046	25	29.251	99.473	12.376	59.350	-2.618	-83.183	-19.493	-123.306
2047	26	0	99.473	12.376	71.726	54	-83.129	12.430	-110.876

Year	Project year t	Total carbon stock decrease due to planned logging activities		Total carbon stock increase due to planned logging activities		Total carbon stock decrease due to unavoided unplanned deforestation		Total carbon stock change in the project case	
		annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative
		ΔCPLdPA_t	ΔCPLdPA	ΔCPLiPA_t	ΔCPLiPA	ΔCUDdPA_t	ΔCUDdPA	ΔCPSPA_t	ΔCPSPA
		tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
2048	27	28.336	127.809	16.652	88.378	46	-83.083	-11.638	-122.514
2049	28	16.266	144.075	19.107	107.484	41	-83.043	2.881	-119.633
2050	29	24.378	168.452	22.785	130.270	37	-83.006	-1.556	-121.189
2051	30	7.921	176.373	23.981	154.251	31	-82.975	16.091	-105.098

Although the efficiency index has been conservatively considered 90% for the ex-ante estimates, during this monitoring period the project activities were able to avoid all deforestation in the project area. As shown in Table 40, no forest logging occurred during this monitoring period. Thus, there were no project emissions related to planned logging activities and unavoided unplanned deforestation.

Table 43: Ex post estimated net carbon stock change in the project area under the project scenario (Table 27 of VM0015 v1.1)

Year	Project year t	Total carbon stock decrease due to planned logging activities		Total carbon stock increase due to planned logging activities		Total carbon stock decrease due to unavoided unplanned deforestation		Total carbon stock change in the project case	
		annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative
		ΔCPLdPA_t	ΔCPLdPA	ΔCPLiPA_t	ΔCPLiPA	ΔCUDdPA_t	ΔCUDdPA	ΔCPSPA_t	ΔCPSPA
		tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
2021	0	0	0	0	0	0	0	0	0
2022	1	0	0	0	0	0	0	0	0

Ex ante estimation of actual non-CO2 emissions from forest fires

Where forest fires have been included in the baseline scenario, non-CO2 emissions from biomass burning must be included in the project scenario. This was done by multiplying the baseline emissions by the factor $(1 - EI)$.

$$EBBPSPA_t = EBBBSPA_t * (1 - EI)$$

Where:

$EBBPSPA_t$ Total ex ante actual non-CO2 emissions from forest fires due to unavoided unplanned deforestation at year t in the project area; tCO2-e

$EBBBSPA_t$ Total non-CO2 emissions from forest fires at year t in the project area; tCO2-e

EI Ex ante estimated Effectiveness Index; %

t 1, 2, 3 ... T, a year of the proposed project crediting period; dimensionless

Table 44: Total ex ante estimated actual emissions of non-CO2 gases due to forest fires in the project area (Table 28 of VM0015 v1.1).

Year	Project year t	Total ex ante estimated actual non-CO2 emissions from forest fires in the Project area	
		annual	cumulative
		$EBBPSPA_t$	$EBBPSPA$
		tCO2-e	tCO2-e
2021	0	0	0
2022	1	344	344
2023	2	306	650
2024	3	260	909
2025	4	306	1,216
2026	5	275	1,491
2027	6	664	2,155
2028	7	320	2,475
2029	8	166	2,641
2030	9	119	2,760
2031	10	179	2,940
2032	11	276	3,215
2033	12	214	3,429
2034	13	154	3,583
2035	14	171	3,754
2036	15	161	3,915
2037	16	99	4,014
2038	17	208	4,222

Year	Project year t	Total ex ante estimated actual non-CO2 emissions from forest fires in the Project area	
		annual	cumulative
		EBBPSPA _t	EBBPSPA
		tCO ₂ -e	tCO ₂ -e
2039	18	167	4,389
2040	19	148	4,537
2041	20	223	4,760
2042	21	215	4,975
2043	22	220	5,195
2044	23	171	5,366
2045	24	191	5,557
2046	25	181	5,738
2047	26	0	5,738
2048	27	0	5,738
2049	28	0	5,738
2050	29	0	5,738
2051	30	0	5,738

The project activities were able to prevent any carbon decrease by forest fire inside the PA. Therefore, there was no non-CO2 emission in this monitoring period.

Table 45: Total ex post estimated actual emissions of non-CO2 gases due to forest fires in the project area (Table 28 of VM0015 v1.1).

Year	Project year t	Total ex ante estimated actual non-CO2 emissions from forest fires in the Project area	
		annual	cumulative
		EBBPSPA _t	EBBPSPA
		tCO ₂ -e	tCO ₂ -e
2021	0	0.0	0.0
2022	1	0.0	0.0

Table 46: Total ex ante estimated actual net carbon stock changes and emissions of non-CO2 gases in the project area (Table 29 of VM0015 v1.1).

Year	Project year t	Total ex ante carbon stock decrease due to planned activities		Total ex ante carbon stock increase due to planned activities		Total ex ante carbon stock decrease due to unavoided unplanned deforestation		Total ex ante net carbon stock change		Total ex ante estimated actual non-CO2 emissions from forest fires in the project area	
		annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative
		ΔCPA_{dPA_t} tCO ₂ -e	ΔCPA_{dPA} tCO ₂ -e	ΔCPA_{iPA_t} tCO ₂ -e	ΔCPA_{iPA} tCO ₂ -e	ΔCUD_{dPA_t} tCO ₂ -e	ΔCUD_{dPA} tCO ₂ -e	ΔCPS_{PA_t} tCO ₂ -e	ΔCPS_{PA} tCO ₂ -e	EBB_{PSPA_t} tCO ₂ -e	EBB_{PSPA} tCO ₂ -e
2021	0	0	0	0	0	0	0	0	0	0	0
2022	1	0	0	0	0	-5,048	-5,048	-5,048	-5,048	344	344
2023	2	14,841	14,841	1,265	1,265	-4,490	-9,537	-18,066	-23,114	306	650
2024	3	0	14,841	1,265	2,530	-3,809	-13,346	-2,544	-25,658	260	909
2025	4	0	14,841	1,265	3,795	-4,474	-17,820	-3,209	-28,867	306	1,216
2026	5	0	14,841	1,265	5,060	-4,007	-21,827	-2,742	-31,609	275	1,491
2027	6	0	14,841	1,265	6,325	-9,717	-31,544	-8,452	-40,061	664	2,155
2028	7	0	14,841	1,265	7,589	-4,661	-36,205	-3,396	-43,457	320	2,475
2029	8	0	14,841	1,265	8,854	-2,377	-38,582	-1,112	-44,569	166	2,641
2030	9	0	14,841	1,265	10,119	-1,687	-40,269	-422	-44,991	119	2,760
2031	10	0	14,841	1,265	11,384	-2,566	-42,835	-1,301	-46,292	179	2,940
2032	11	0	14,841	1,265	12,649	-3,976	-46,810	-2,711	-49,003	276	3,215
2033	12	0	14,841	1,265	13,914	-3,070	-49,880	-1,805	-50,808	214	3,429
2034	13	0	14,841	1,265	15,179	-2,198	-52,079	-934	-51,741	154	3,583
2035	14	0	14,841	1,265	16,444	-2,450	-54,529	-1,185	-52,927	171	3,754
2036	15	0	14,841	1,265	17,709	-2,323	-56,852	-1,058	-53,984	161	3,915
2037	16	0	14,841	1,265	18,974	-1,406	-58,258	-141	-54,126	99	4,014
2038	17	0	14,841	1,265	20,238	-3,024	-61,282	-1,759	-55,885	208	4,222
2039	18	0	14,841	1,265	21,503	-2,406	-63,688	-1,141	-57,026	167	4,389
2040	19	0	14,841	1,265	22,768	-2,120	-65,808	-855	-57,881	148	4,537
2041	20	0	14,841	1,265	24,033	-3,229	-69,036	-1,964	-59,845	223	4,760
2042	21	0	14,841	1,265	25,298	-3,116	-72,153	-1,851	-61,696	215	4,975
2043	22	23,835	38,677	4,052	29,350	-3,179	-75,331	-22,962	-84,658	220	5,195
2044	23	31,546	70,222	8,812	38,162	-2,467	-77,799	-25,201	-109,859	171	5,366
2045	24	0	70,222	8,812	46,975	-2,767	-80,565	6,046	-103,813	191	5,557
2046	25	29,251	99,473	12,376	59,350	-2,618	-83,183	-19,493	-123,306	181	5,738

Year	Project year t	Total ex ante carbon stock decrease due to planned activities		Total ex ante carbon stock increase due to planned activities		Total ex ante carbon stock decrease due to unavoided unplanned deforestation		Total ex ante net carbon stock change		Total ex ante estimated actual non-CO2 emissions from forest fires in the project area	
		annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative
		ΔCPAdPA_t	ΔCPAdPA	ΔCPAiPA_t	ΔCPAiPA	ΔCUDdPA_t	ΔCUDdPA	ΔCPSPA_t	ΔCPSPA	EBBPSPA_t	EBBPSPA
		tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
2047	26	0	99,473	12,376	71,726	54	-83,129	12,430	-110,876	0	5,738
2048	27	28,336	127,809	16,652	88,378	46	-83,083	-11,638	-122,514	0	5,738
2049	28	16,266	144,075	19,107	107,484	41	-83,043	2,881	-119,633	0	5,738
2050	29	24,378	168,452	22,785	130,270	37	-83,006	-1,556	-121,189	0	5,738
2051	30	7,921	176,373	23,981	154,251	31	-82,975	16,091	-105,098	0	5,738

Table 47: Total ex post estimated actual net changes in carbon stocks and emissions of GHG gases in the project area (Table 29 of VM0015 v1.1).

Year	Project year t	Total ex-post carbon stock decrease due to planned activities		Total ex-post carbon stock increase due to planned activities		Total ex-post carbon stock decrease due to unavoided unplanned deforestation		Total ex-post net carbon stock change		Total ex-post estimated actual non-CO2 emissions from forest fires in the project area	
		annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative
		ΔCPAdPA_t	ΔCPAdPA	ΔCPAiPA_t	ΔCPAiPA	ΔCUDdPA_t	ΔCUDdPA	ΔCPSPA_t	ΔCPSPA	EBBPSPA_t	EBBPSPA
		tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
2021	0	0	0	0	0	0	0	0	0	0	0
2022	1	0	0	0	0	0	0	0	0	0	0

3.2.3 Leakage

According to the methodology VM0015 v1.1, if leakage prevention measures include measures to enhance cropland and grazing land areas, a reduction in carbon stocks and/or an increase in GHG emissions may occur compared to the baseline case. If this decrease in carbon stocks or increase in GHG emissions is significant, it must be accounted for, and monitoring will be required, otherwise it must not be accounted for.

In the case of the Itaúba REDD+ Project, this component of the calculation is not applicable. The project activities do not involve a decrease in carbon stocks or increase in GHG emissions associated with leakage prevention activities. These activities do not involve any carbon stock reduction due to deforestation or additional emissions caused by increased grazing activities.

Still according to the methodology VM0015 v1.1, activities that will cause deforestation within the project area in the baseline case could be displaced outside the project boundary due to the implementation of the AUD project activity. If carbon stocks in the leakage belt area will decrease more during project implementation than projected in the baseline case, this will be an indication that leakage due to displacement of baseline activities has occurred. Leakage due to activity displacement can thus be estimated by ex post monitoring of deforestation in the leakage belt and comparing ex post observed deforestation with ex ante projected baseline deforestation.

Ex ante activity displacement leakage could only be guessed based on the anticipated combined effectiveness of the proposed leakage prevention measures and project activities. This was done by multiplying the estimated baseline carbon stock changes for the project area by a “Displacement Leakage Factor” (DLF) representing the percent of deforestation expected to be displaced outside the project boundary.

If deforestation agents do not participate in leakage prevention activities and project activities, the Displacement Factor shall be 100%. Where leakage prevention activities are implemented, the factor shall be equal to the proportion of the baseline agents estimated to be given the opportunity to participate in leakage prevention activities and project activities.

It is expected that 100% of potential deforestation agents in the Reference Region will be given the opportunity to participate in leakage prevention activities. Thus, the “Displacement Leakage Factor” (DLF) was conservatively considered as 0.05.

If emissions from forest fires have been included in the baseline, the ex-ante emissions from forest fires due to activity displacement leakage will be calculated by multiplying baseline forest fire emissions in the project area by the same DLF used to estimate the decrease in carbon stocks.

The equation for $\Delta CADLK$ (total ex ante estimated decrease in carbon stocks due to displaced deforestation) is set below:

$$\Delta CADLK_t = (CabBLSPA_t + CbbBSLPA_t - CwpBSLPA_t) \cdot (DLF)$$

Where:

$CabBLSPA_t$ tCO_2-e	Baseline carbon stock change in the above-ground biomass at year t in the project area;
$CbbBSLPA_t$ tCO_2-e	Baseline carbon stock change in the below-ground biomass at year t in the project area;
$CwpBSLP_t$	Baseline carbon stock change in the wood products at year t in the project area; tCO_2-e
DLF	Displacement Leakage Factor; %
t	1, 2, 3 ... T, a year of the proposed project crediting period; dimensionless

The equation for $\Delta EADLK_t$ (total ex ante estimated increase in GHG emissions due to displaced forest fires) is set below:

$$\Delta EADLK_t = (EBBBSLPA_t) * (DLF)$$

Where:

$EBBBSLPA_t$	Baseline non-CO2 emissions from forest fires at year t in the project area; tCO_2-e
DLF	Displacement Leakage Factor; %
t	1, 2, 3 ... T, a year of the proposed project crediting period; dimensionless

For this monitoring period, ex ante estimation of deforestation inside the leakage belt was 278 hectares, of which 224 ha in F1 and 54 ha in F2. Meanwhile, the total deforestation in this period was 172 ha, so there was no leakage due to activity displacement if we consider full period. From ex-post deforestation, 8 hectares of F2 was deforested in 2021 compared to the ex-ante projection of no deforestation. This deforestation that occurred in 2021 corresponds to 2% of total GHG emissions for the period, thus it is insignificant according to the "Tool for testing significance of GHG emissions in A/R CDM project activities" and has not been accounted for.

Table 48: Annual ex-ante estimated and ex post measured of deforestation in the Leakage Belt (Table 9.c of VM0015 v1.1)

Year	Total Ex-ante - LK					Total Ex-post - LK			
	Project year	F1	F2	annual ABSLLK _t	cummulative ABSLLK	F1	F2	annual ABSLLK _t	cummulative ABSLLK
		ha	ha	ha	ha	ha	ha	ha	ha
2021	0	0	0	0	0	0	8	8	8
2022	1	224	54	278	278	2	162	164	172
TOTAL				278				172	

Therefore, the leakage due to project activities was considered to be 0 for each year as shown below. This in turn resulted in 0 values for ex-post net carbon stock changes in the leakage belt, as shown in the following tables.

Table 49: Ex post estimated leakage due to activity displacement (Table 34 of VM0015 v1.1)

Year	Project year t	Total ex post measured decrease in carbon stocks due to displaced deforestation		Total ex post estimated increase in GHG emissions due to displaced forest fires	
		annual	cumulative	annual	cumulative
		ΔCADLK_t tCO ₂ -e	ΔCADLK tCO ₂ -e	EADLK_t tCO ₂ -e	EADLK tCO ₂ -e
2021	0	0	0	0	0
2022	1	0	0	0	0

Table 50: Ex post estimated total leakage (Table 35 of VM0015 v1.1)

Year	Project year t	Total ex post increase in GHG emissions due to displaced forest fires		Total ex post decrease in carbon stocks due to displaced deforestation		Total net carbon stock change due to leakage		Total net increase in emissions due to leakage	
		annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative
		EADLK_t tCO ₂ -e	EADLK tCO ₂ -e	ΔCADLK_t tCO ₂ -e	ΔCADLK tCO ₂ -e	ΔCLK_t tCO ₂ -e	ΔCLK tCO ₂ -e	ELK_t tCO ₂ -e	ELK tCO ₂ -e
2021	0	0	0	0	0	0	0	0	0
2022	1	0	0	0	0	0	0	0	0

3.2.4 Net GHG Emission Reductions and Removals

The net anthropogenic GHG emission reduction (STEP 9 of the VM0015 Methodology v1.1) of the proposed AUD project activity was calculated as follows:

$$\Delta REDD_t = (\Delta CBSLPA_t + EBBBSLPA_t) - (\Delta CPSPA_t + EBBPSPA_t) - (\Delta CLK + \Delta ELK_t)$$

Where:

$\Delta REDD_t$ Ex ante estimated net anthropogenic greenhouse gas emission reduction attributable to the AUD project activity at year t; tCO_{2e}

$\Delta CBSLPA_t$ Sum of baseline carbon stock changes in the project area at year t; tCO_{2e}

$EBBBSLPA_t$ Sum of baseline emissions from biomass burning in the project area at year t; tCO_{2e}

$\Delta CPSPA_t$ Sum of ex ante estimated actual carbon stock changes in the project area at year t; tCO_{2e}
of baseline emissions from biomass burning in the project area at year t; tCO_{2e}

$EBBPSPA_t$ Sum of (ex ante estimated) actual emissions from biomass burning in the project area at year t; tCO_{2e}

ΔCLK_t Sum of ex ante estimated leakage net carbon stock changes at year t; tCO_{2e}

t 1, 2, 3 ... T, a year of the proposed project crediting period; dimensionless

The number of Verified Carbon Units (VCUs) to be generated through the proposed AUD project activity at year t was calculated as follows:

$$VCU_t = REDD_t - VBC_t$$

$$VBC_t = (CBSLPA_t - CPSPA_t) * RF_t$$

Where:

VCU_t Number of Verified Carbon Units that can be traded at time t; tCO_{2-e}

$REDD_t$ Ex-ante estimated net anthropogenic greenhouse gas emission reductions attributable to the AUD project activity at year t; tCO_{2-e} ha⁻¹

VBC_t Number of Buffer Credits deposited in the VCS Buffer at time t; tCO_{2-e}

$CBSLPA_t$ Sum of baseline carbon stock changes in the project area at year t; tCO_{2e}

$CPSPA_t$ Sum of ex ante estimated actual carbon stock changes in the project area at year t; tCO_{2-e} ha⁻¹

RF_t Risk factor used to calculate VCS buffer credits; %

Note: RF_t is a risk factor determined using the latest version of the VCS-approved AFOLU Non-Permanence Risk Tool v4.0.

t 1, 2, 3 ... T, a year of the proposed project crediting period; dimensionless

In PD the risk rating according to the AFOLU Non-Permanence Risk Tool v4.0 was 20, being sub-divided into Internal Risk (17.0), External Risk (0) and Natural Risk (2.5), which is rounded up to 20 points. For this monitoring period, the risk decreased to 17 because the Natural Risk is now 2 and the Internal Risk is 15. The Natural Risk was considered lower because during the monitoring period there was no fire in the project area. The Internal Risk, on the other hand, was lower because of the decreased ratio of NPV_baseline_/NPV_project_with_VCUs, which decreases the opportunity cost.

Table 51: Ex ante estimated net anthropogenic GHG emission reductions ($\Delta REDD_t$).

Year	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions (tCO ₂ e)
2021	0	0	0	0
2022	53,913	5,391	2,696	45,826
2023	47,957	18,372	2,398	27,187
2024	40,690	2,804	2,034	35,851
2025	47,799	3,515	2,390	41,894
2026	42,820	3,017	2,141	37,662
2027	103,816	9,117	5,191	89,508
2028	49,809	3,716	2,490	43,602
2029	25,424	1,277	1,271	22,875
2030	18,065	542	903	16,620
2031	27,453	1,480	1,373	24,600
2032	42,511	2,986	2,126	37,399
2033	32,843	2,019	1,642	29,181
2034	23,522	1,087	1,176	21,259
2035	26,208	1,356	1,310	23,541
2036	24,838	1,219	1,242	22,377
2037	15,054	241	753	14,061
2038	32,314	1,966	1,616	28,731
2039	25,731	1,308	1,287	23,136
2040	22,676	1,003	1,134	20,539
2041	34,523	2,187	1,726	30,609
2042	33,313	2,066	1,666	29,581
2043	33,986	23,182	1,699	9,104
2044	26,383	25,371	1,319	-308
2045	29,579	-5,855	1,479	33,954

Year	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions (tCO ₂ e)
2046	27,992	19,674	1,400	6,918
2047	-543	-12,430	-27	11,914
2048	-457	11,638	-23	-12,073
2049	-408	-2,881	-20	2,494
2050	-365	1,556	-18	-1,903
2051	-306	-16,091	-15	15,800
Total	887,136	110,836	44,357	731,943

Table 52: Ex-ante Tradable VCUs and Buffer Credits.

Year	Project year t	Ex ante VCUs tradable		Ex ante buffer credits	
		annual	cumulative	annual	cumulative
		VCU _t tCO ₂ -e	VCU tCO ₂ -e	VBC _t tCO ₂ -e	VBC tCO ₂ -e
2021	0	0	0	0	0
2022	1	36,740	36,740	9,086	9,086
2023	2	21,820	58,561	5,366	14,452
2024	3	28,742	87,302	7,109	21,561
2025	4	33,588	120,891	8,306	29,867
2026	5	30,197	151,087	7,465	37,332
2027	6	71,764	222,851	17,744	55,077
2028	7	34,960	257,811	8,642	63,719
2029	8	18,344	276,156	4,531	68,249
2030	9	13,330	289,486	3,290	71,539
2031	10	19,729	309,214	4,872	76,411
2032	11	29,990	339,205	7,409	83,820
2033	12	23,402	362,607	5,779	89,599
2034	13	17,049	379,655	4,210	93,809
2035	14	18,878	398,534	4,663	98,473
2036	15	17,943	416,477	4,434	102,907
2037	16	11,277	427,754	2,784	105,691
2038	17	23,036	450,789	5,696	111,387
2039	18	18,553	469,342	4,584	115,970
2040	19	16,471	485,813	4,068	120,039
2041	20	24,545	510,357	6,065	126,104
2042	21	23,719	534,076	5,862	131,966
2043	22	7,339	541,415	1,765	133,731
2044	23	-308	541,107	0	133,731
2045	24	27,212	568,319	6,742	140,473

Year	Project year t	Ex ante VCUs tradable		Ex ante buffer credits	
		annual	cumulative	annual	cumulative
		VCU _t tCO ₂ -e	VCU tCO ₂ -e	VBC _t tCO ₂ -e	VBC tCO ₂ -e
2046	25	5,581	573,900	1,338	141,811
2047	26	9,537	583,437	2,377	144,188
2048	27	-12,073	571,364	0	144,188
2049	28	1,999	573,363	495	144,683
2050	29	-1,903	571,461	0	144,683
2051	30	12,643	584,103	3,157	147,840

Table 53: Ex-post estimated net anthropogenic GHG emission reductions.

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2021 (14 to 31 Dec.)	0	0	0	0
2022 (1 Jan. to 31 Dec.)	53,913	0	0	53,913
Total	53,913	0	0	53,913

Table 54: Ex-post estimated net anthropogenic GHG emission reductions and VCUs (Table 36 of VM0015).

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)	Buffer pool allocation	VCUs eligible for issuance
2021 (14 to 31 Dec.)	0	0	0	0	0	0
2022 (1 Jan. to 31 Dec.)	53,913	0	0	53,913	8,581	45,332
Total	53,913	0	0	53,913	8,581	45,332

The difference between ex-ante and ex-post emissions reduction is 17.65% as shown below. Since during this monitoring period the project there were no emissions from unavoided deforestation, forest fires and activity displacement leakage, there were no discounts in the ex-ante emissions reductions.

Table 55: Comparison between ex-ante emissions reductions and achieved emissions reductions.

Year	Ex-ante emissions reductions/ removals (tCO ₂ e)	Achieved emissions reductions/ removals (tCO ₂ e)	Percent difference	Justification for the difference
2021	0	0	-	-
2022	45,826	53,913	17.65%	No emissions due to unplanned deforestation, forest fires and activity displacement leakage.

3.3 Optional Criterion: Climate Change Adaptation Benefits

Does not apply. This project is not intended to be validated for the Gold Level of this section.

3.3.1 Activities and/or processes implemented for Adaptation (GL1.3)

Not applicable.

4 COMMUNITY

4.1 Net Positive Community Impacts

4.1.1 Community Impacts (CM2.1)

In the period 2021 to 2022, the internal project community received training and technical support in the implementation of activities related to monitoring the project area. During this period, the actions focused on strengthening the technical capabilities of handling the monitoring application by the farm worker and proponents. This system is the QuickCapture App, used by Carbonext to ensure the forest management of the areas. The training provided by the internal Geographical Information System department aimed to train them in the handling of the monitoring data collection application so that they become replicators of this knowledge. Instructing about the frequency and path of the monitoring according to the rainy season or summer, in addition to the accessibility and probability of occurrence of invasions.

With the additional resources to the training, a communication channel was created to facilitate the exchange of information among students, teachers and Carbonext's monitoring team, as a form of support in case there is any doubt or need for assistance in its handling. In addition, specific knowledge related to the basic concepts and benefits of the project was reformulated.

The external community, in turn, had actions implemented to engage in the project, where lectures and conversation rounds were held to clarify the carbon credit, forest preservation, and questions about the project, as well as visits to potential partners to develop activities. In order to gather more specific data, a questionnaire was implemented to find out information about the reality of the students in the schools, and a course to support teachers from the João Paulo II State School on socio-emotional issues.

Each of the expected impacts is presented below.

Community Group	Internal project community
Impact	Increased capacity for local territorial governance and natural resource management
Type of Benefit/Cost/Risk	Direct benefit projected on community groups that was obtained from the promotion of technical knowledge for forest governance and the development of a theoretical-practical workshop for the implementation of georeferenced monitoring
Change in Well-being	Training Practical and theoretical lesson on technical knowledge for maintenance and management of forest resources and land maintenance. Creation of a communication channel for questions and support. Informative Meeting Clarifications about basic concepts and benefits of the project

Community Group	External project community
Impact	Engagement and strengthening of institutional relationships
Type of Benefit/Cost/Risk	Direct benefit projected on school students that was obtained from the promotion of knowledge about carbon credit and preservation of forest resources. Besides the understanding of the importance of emotional balance for the development of activities and relationships in a professional environment for teachers.
Change in Well-being	Training A course to circumvent the post-pandemic circumstances of stress and overload reported by the teachers of the Papa João Paulo II State School, carrying out a program aimed at cultivating the causes for a more sustainable well-being in the work environment. Informative Meeting Understand students' perceptions of future perspectives, climate change and sustainability

4.1.2 Negative Community Impact Mitigation (CM2.2)

All project-related activities have only improved, developed, and protected the HCVs identified in the PD. For the monitoring period, no negative impact on the community was identified and therefore no measures or activities were implemented. However, considering the identified potential negative impacts projected by the community during the consultation meetings, mitigation strategies were established.

Table 56. Negative community impacts and mitigation strategies.

Potential Negative impacts	Mitigation strategy
Institutional problems and conflicts between communities	Implementation and dissemination of the channels for complaints, relationship, and technical questions
Inefficiency in the planning of project activities may negatively impact the community's calendar	Strategic plan and timeline for the implementation of the project activities respecting the activities developed on the farms and in the schools
Ineffective socialization processes and communication channels	Socialization meetings and workshops to ensure access to information for all community groups associated with the project.

Each activity is being carefully monitored by experts in each area, internal to Carbonext, and the precautionary principle will always be considered.

4.1.3 Net Positive Community Well-Being (CM2.3, GL1.4)

The net community impacts were essentially positive given the positive feedback of the implemented activities on community groups, in terms of engagement and territorial governance capacity, as well as increased perception of project benefits. In the previous section, it was evident that, for the monitoring period, no negative impacts on the community were identified and, therefore, no measures or activities were implemented. However, considering the potential negative impacts identified and projected by the community during the consultation meetings, strategies for their mitigation were established, as shown in the previous section.

In this sense, the impacts generated were positive for internal and external community, as well as other stakeholders involved in the project activities. As compared to the conditions foreseen in the scenario without the project, the implementation of the established activities would not be possible due to lack of resources to implement the environmental agenda in schools and local engagement for its promotion.

4.1.4 Protection of High Conservation Values (CM2.4)

No negative effects were evident due to the project activities on any of the high conservation values; on the contrary, the project activities are intended to increase local environmental awareness of the importance of the forest patches, and to encourage their legal guardians to effectively manage and care for their natural resources. The formulation of a mitigation project to reduce deforestation in the area is an instrument to promote protection of the proposed HCVs.

Below are the main results of the actions implemented during the monitoring period to protect the HCVs.

HCV 2 – Landscape-level Ecosystems, Ecosystem Mosaics

High Conservation Values	Municipal Natural Park of Itaúba
Monitoring period results	Articulation with local environmental secretary and environmental awareness activities in schools
Area	Out of Project Area

HCV 4 – Ecosystem Services

High Conservation Values	Forest Fragments
Monitoring period results	Development of activities on forest monitoring and environmental awareness activities for farm workers
Area	Project Area

4.2 Other Stakeholder Impacts

4.2.1 Mitigation of Negative Impacts on Other Stakeholders (CM3.2)

For the monitoring period, no negative impacts on other stakeholders were identified and therefore no measures or activities were implemented. However, considering the identified potential negative impacts projected by the community, local authorities and institutions in the influence area, active communication strategies and socialization visits were established.

In the table below are the main stakeholders visited in the actions during the monitoring period.

Table 57. List of other stakeholders visited during the monitoring period.

Stakeholder	Sector	Departments	People consulted
Municipal Government of Itaúba	Government	7	7
Municipal Government of Marcelândia	Government	8	8
EMBRAPA - Brazilian Public Agricultural Research Company	Government	1	1

EMPAER - Mato-Grossense Research, Assistance and Rural Extension Company	Government	1	4
UFMT - Federal University of Mato Grosso	Government	2	2
Association of Brazil nut Collectors from Itaúba	Private	1	2
Chestnut sellers from Itaúba	Private	1	3
Total	-	21	27

4.2.2 Net Impacts on Other Stakeholders (CM3.3)

The project activities have not resulted in net negative impacts on the well-being of other stakeholders. Each of the net impacts are presented below.

Community Group	Government stakeholders
Impact	Increased capacity for local territorial governance and involvement in the environmental agenda
Type of Benefit/Cost/Risk	Projected direct benefit for government stakeholders through information dissemination among the project executing entities about carbon market and REDD+ project development in the region
Change in Well-being	Information and consultation meetings Four meetings in the framework of the stakeholder consultation and explanation of project steps and timeline

Community Group	Private stakeholders
Impact	Mobilization for the possibility of partnerships to develop project activities
Type of Benefit/Cost/Risk	Direct implementation designed for conservation and sustainable management activities associated with the project.
Change in Well-being	<u>Hiring for the development of activities</u> Cost benefit analysis of activities and verification of possible activities to be implemented

4.3 Community Impact Monitoring

4.3.1 Community Monitoring Plan (CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

The community impact monitoring plan was designed within a time frame of six years, with the possibility of modifying or including new indicators according to the dynamics of the project and the achievement of the objectives.

For the monitoring period 2021-2022, the achievements and benefits of the outcome indicators have been presented; however, it's important to mention that part of outcomes will only be achieved in the next verification period. During this monitoring period there have been changes based on experience and lessons learned, thanks to active communication work with farm workers, schools, and other stakeholders.

As the first monitoring output for the 2021-2022 period, this plan identified the communities, community groups and other stakeholders to be monitored, the variables, types of measurements and sampling methods, as well as determined the frequency of monitoring and reporting for each type and method.

Considering the guidelines of the projected Monitoring Plan, the output and outcome indicators for which supporting information was available for the period 2021-2022 were followed. These outputs were based primarily on minutes and attendance lists available from each of the entities consulted by the project and the measured reports.

Below are the main output and outcome indicators per strategic line with the progress in the monitoring period. The impact indicators and output indicators that require evaluation by the survey method will be evaluated upon validation of the project, according to the guidelines of the monitoring plan. The results presented here contribute to the achievement of the short-term objectives set by the project, which were contemplated in the analysis of the theory of change for each of its activities.

Below are the main results of the actions implemented during the monitoring period.

Table 58. The main results of the actions implemented during the monitoring period.

Indicator/ Source of verification	Result	Gender	Key action of project	Effect	Impacted community group
Number of channels enabled to share information about the project	Three main channels (Complaint Channel, Relation Channel and Technical Questions Channel)	Does not apply	Design and implementation of information dissemination strategies Definition of first mechanisms for dealing with complaints and grievances (attention in the field and telephone contact).	Positive	Representatives of environmental institutions, local and regional public institutions, and other strategic actors in the area
Number of meetings for dissemination of the project and clarification of doubts	Two activities that included a community socialization and diagnostic component	258 females and 236 males	Develop the practice of socialization as a strategic way to partner in activities. Understanding the local scenario with its opportunities and challenges	Positive	Representatives from environmental institutions, local and regional public institutions, internal and external community, and other strategic players in the area
Number of training courses held for capacity building in forest monitoring	One activity that included the use of a cell phone application for georeferenced monitoring of forest fragments	3 males (1 from internal community and 2 landowners)	Develop and improve territorial management capabilities. Developing technical knowledge replicators	Positive	Internal community (Farmworkers)
Number of training sessions held to strengthen working, studying and socializing skills	6 modules on the importance of emotional well-being for personal and work social interactions	14 females	Strengthen interaction and good relationship practices in schools. Improve the working environment for teachers and students	Positive	External community (Schools)

- Internal diagnosis of schools

In order to identify opportunities and challenges, a questionnaire on the socio-educational issue was applied to the teachers and students of the schools. Respecting their will to participate in the survey, the application occurred only with those who felt comfortable to answer, not discriminating the authors of the answers.

STUDENTS

For the student's questionnaire, information was collected on school infrastructure, prospects, approach to environmental education, encouragement of cultural practices (reading and arts), and school dropout problems.

- *Pedro Bianchini State School*

In the elementary school, 40 students from the 7th and 8th grades answered the questionnaire. It was reported that the school has internet access, as well as computers for use, in addition to sports activities and recreational and competitive activities. Regarding the environment, the school structure is good, but the sports court is not. However, the teachers encourage the students with competitions and awards. In cases of school dropout, the school searches actively to find out the reason, most often work-related, and this follow-up is monitored by the guardianship council in serious cases.

When asked about environmental education, the students affirmed that the subject is taught in an interdisciplinary way, especially in geography. Before the presence of the project, the students had no knowledge of what REDD+ projects were, nor of carbon credits. In the students' collective opinion, in order to improve their performance, it would be interesting to have more practical classes in the subjects, as well as a study about emotional education. Their biggest collective dreams are to finish school without failing, to improve their lives, and to go to college.

In the high school students, 40 students belonging to different school years, answered the questionnaire. It was reported that the school has internet access, as well as computers for use, and offers cultural practices such as theater, dance, and handicrafts. Regarding the environment, the school's structure is good, and there is an incentive to prepare the students to enter university. All students affirm that they think it is important to take a higher education course. However, it is reported that the main reason for dropping out of school is the need to work. Another point reported by the students is the presence of early pregnancy, which is a topic addressed within the school through sex education.

When asked about environmental education, the students said that the subject is taught in an interdisciplinary way, where there is no school project that is focused only on the theme. However, topics such as environmental conservation and climate change are raised in the classroom. Before the presence of the project, the students did not know what REDD+ projects were, nor carbon credits. In the students' collective opinion, the biggest collective dream is the need for a psychologist for students and teachers to improve their educational performance, as well as the need to implement selective collection.

- *Papa João Paulo II State School*

In the elementary school, 61 students from the 7th and 8th grades answered the questionnaire. The school does not have internet access, but the library has two computers available for use. As for the cultural practices, students were informed that theater, dance and judo are offered to them. In relation to the

environment, some said they found a good structure, others not so much. It was mentioned that the teachers encourage the students, but there is a low engagement. The main cause of dropping out of school reported is due to work or lack of interest. There are cases of early pregnancy, and when this happens studies are interrupted.

When asked about environmental education, students said that the topic is addressed at school, mainly in the subjects of science and geography, talking about environmental conservation and climate change. Before the presence of the project, the students did not know what REDD+ projects were, nor carbon credits. In the students' collective opinion, to improve their educational performance they would need more field classes, as well as a school with better infrastructure. Their greatest collective dreams are to play interactive games with other schools, change the school menu, or even move from their current school.

In the high school students, 80 high school students participated in the questionnaire, in the morning period, belonging to 5 classes, with different years of schooling. It was reported that the school has no internet, and there are no computers for use, besides the cultural practices being scarce. Regarding the environment, it was reported that the majority does not think the school structure is good, and only a minority prepares for university entrance. However, most of the students say they think it is important to take a higher education course. The main reason for dropping out of school is the need to work, besides early pregnancy, which in most cases, the students do not return to school.

When questioned about environmental education, the students affirmed that the subject is not addressed in the classroom and that before the presence of the project, the students had no knowledge of what REDD+ projects would be, nor of carbon credits. In the students' collective opinion, the biggest collective dream is the need for a psychologist for students and teachers as a way to improve their educational performance.

In the afternoon, the questionnaire was applied to 43 high school students from different school years. Like the morning students, the afternoon students reported that the school does not have Internet access, but they think it is important to create a computer lab. As for the environment, most of them don't think the school's structure is good, but they are preparing for university entrance. The students from the afternoon period consider it important to enroll in higher education but encounter difficulties because the chosen course does not exist in public universities or because the course will take place in another city. The main reason for dropping out of school reported is the need to work, but they pointed out that there is a lack of motivation and interest, because of the poor job prospects. It was mentioned that cases of early pregnancy rarely occur in their class, however there is sexual education through lectures offered by the municipality.

When asked about environmental education, the students said that the topic is rarely addressed, being usually introduced in geography, where it involves the topics of environmental conservation and climate change. Before the presence of the project, the students did not know what REDD+ projects were, nor carbon credits. In the students' collective opinion, the biggest collective dream is to have practical classes to help their school performance, as well as reinforcement in the subject of mathematics.

- *Paulo Freire State School*

In the elementary school, 88 students from the 7th and 8th grades answered the questionnaire. It was reported that the school has internet as well as computers for use through a government project, however, there is little available for cultural activities. Regarding the environment, some believe that the school has a good infrastructure, while others do not. The school is going through a period of renovation, which has

not yet been completed. It was mentioned that the teachers encourage the students, but there is a low engagement. The main reason for dropping out of school is due to work or pregnancy. Where these students stop studying to work in the commerce or on the farms. Another point mentioned, is that due to the COVID-19 pandemic many students lost their bond with the school.

When asked about environmental education, the students affirmed that the subject is taught in an interdisciplinary way, but nothing very specific. Before the presence of the project, the students did not know what REDD+ projects were, nor carbon credits. In the students' collective opinion, to improve their performance, it would be interesting to have projects that encourage students to study, because currently there is a lack of interest. Their greatest collective dreams are to go on a school trip, go on a camping trip, or do activities that awaken their interest.

In the high school students, 45 students from different school years answered the questionnaire. It was reported that the school has internet, but of poor quality, and that cultural practices are not common. Regarding the environment, most of the students think the school structure is good, and there is an incentive to prepare the students to enter university. All students affirm that they think it is important to take higher education courses.

The students did not point out the main reason for dropping out of school, however, they claim to have dropouts. It was also mentioned that there are cases of early pregnancy and that they believe that they should have sex education instructions at school. A sensitive point raised by the students was the issue of school lunch, which is often missing and that the resource is very little. About R\$0.45 cents/per student is invested in meals for students.

When asked about environmental education, students said that the subject is taught in geography, but only for climate change issues. Before the presence of the project, the students had no knowledge of what REDD+ projects would be, nor carbon credits. In the collective opinion of the students, the greatest collective dream is that there are improvements in the teaching method, in the distribution of snacks, besides the restructuring of the school and motivation of students and teachers.

TEACHERS

For the teachers' questionnaire, information was collected on age, gender, ethnic group, education level, as well as information on the perception of the work environment.

A total of 15 questionnaires were applied in which the age group with the highest number of representatives was between 40 and 49 years old, with 7 individuals. This was followed by the 30 to 39 age brackets, with 6 individuals, and, to a lesser extent, the 25 to 29 age brackets, with 2 individuals. 60% of the group identified themselves as female, in which the most expressive ethnic group was white, followed by black and brown with the same value. 93% of the teachers affirmed having some kind of graduate degree, and 100% said they had already participated in some continuing education activity, whether it was an update or training. 47% of the teachers say that they do not feel their profession is undervalued, and 80% claim to have a heavy workload.

Regarding the work environment, 100% say they like the environment, where relationships are described as friendly to very friendly. However, 30% say they do not feel safe in the work environment, and that in general the schools have a good infrastructure, but there is room for improvement. 100% affirm that there was a lag of students because of the COVID-19 pandemic, where 40% of the professionals believe that

there should be an accelerated recovery process for students. Most teachers report that the textbooks used in their subjects are of regular to good quality and that all students have access to the textbooks.

- *Cultivating Emotional Balance Course*

The Cultivating Emotional Balance course was offered to Papa João Paulo II State School, where given the post pandemic circumstances of stress and overload reported by teachers at school, bringing a program focused on emotional wellbeing was highly important. This was an abbreviated course, with a total duration of 15h, in which the first class was in face-to-face format and the others was via the Zoom platform.

Some teachers preferred to take the classes at school, using the computer in the secretary's office, so the image resource was impaired in some situations. However, it was the only way they could participate, which was quite valid given the technological limitations. Some people watched the classes from their own homes on their computers or cell phones. It was analyzed that part of class dynamics would work better if each participant had their own screen and in their own space, but there were also surprises that came up from new the possibilities created.

The course had an initial interest of 24 people, yet only 18 people ended up enrolling. Within this course period, the average engagement was over 50% (Figure 15). The main reasons for dropping out of the course were reported to be overload of school activities and fatigue. A total of 14 people completed the course, all of whom were women.

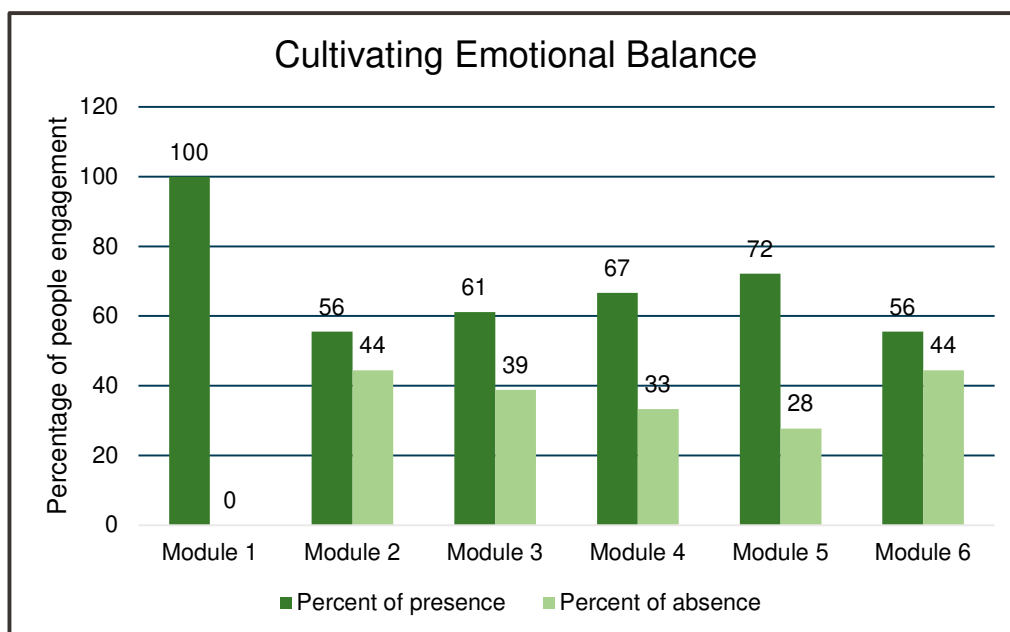


Figure 15. Graphic of percentage of participation

The lecturer's report informs that initially everyone expects magic formulas, which are mainly quick and non-existent, but eventually change their minds and allow a natural interest to emerge. And that's how eventually the group of students got organized. Some participants didn't understand that there would be 6 modules, and so they gave up halfway through. Two teachers reported that they could initially attend on the class day and time, but in the end then didn't show up.

Nevertheless, the group that remained was open and interested in the proposed themes, some more than others, but on average they were involved with the course. The main impact and difficulties reported by each module are reported in the table below.

	Module 1	Module 2	Module 3	Module 4	Module 5	Module 6
Main Impact	Active listening	Self-Analysis	Self-reflection	Self-knowledge	Self-compassion	Orientation
Difficulties	Unreported	Unreported	Unreported	Fatigue	Unreported	Unreported

Table 59. Main course impacts and difficulties.

When asked if they had ever attended a course on emotional wellbeing, most participants reported that they had not (Figure 16). After taking the course there were several positive reports such as: "The course was very fruitful, because in a very broad way we learned how to manage our feelings/emotions, and, when it comes to our profession it is more than necessary, it is indispensable!", as well as "I want to leave here my gratitude for the great course, for the teachings transmitted, for your charisma and wisdom, for sure they are learnings that we will take for our whole life, thank you very much". From the testimonials collected it is possible to affirm that in general the course had a good reception and good development.

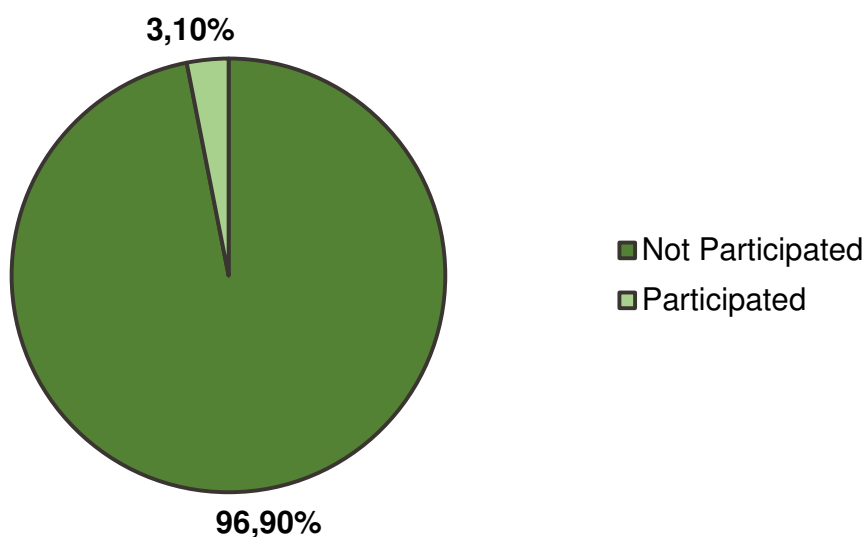


Figure 16. Percentage of people that had previously attended a course on emotional well-being.

- *Capacity building in forest monitoring*

The training for the use of the ArcGIS QuickCapture app for monitoring was given by Carbonext's internal georeferencing department with the people in charge of the farms. The objective of the training was to train them in the handling of the application to collect data for monitoring the areas, facilitating the territorial management of the project.

The training lasted about 2 to 3 hours (Figure 17) and the choice of the people who received this training was based on the function performed in the farm and its rotation within the perimeter. The idea is that the people trained will serve as replicators of this information for new people being hired in the farm and by demand. From the app installation until the moment of this report, more than 18 rounds have been conducted using the app.

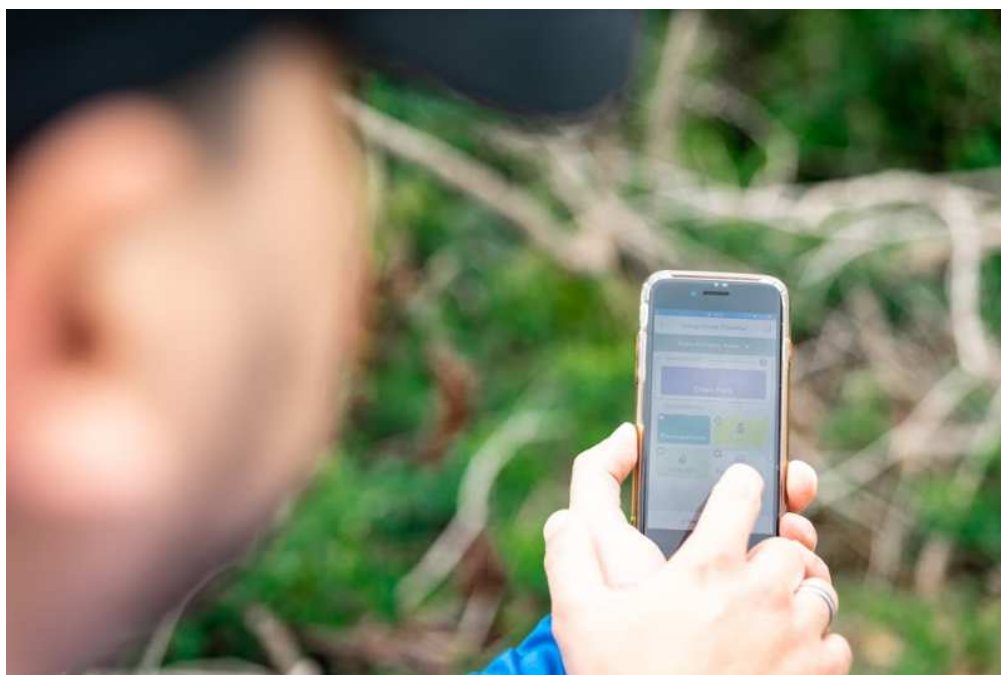


Figure 17. Training instruction for internal foremen. Place: Santa Maria Farm, Marcelândia – MT.

4.3.2 Monitoring Plan Dissemination (CM4.3)

The initial identification and consultation processes with local communities and other parties interested by the project's actions refer to the initial validation period; however, socialization meetings and progress of project activities were maintained throughout the whole verification period.

During the first monitoring period, continuous communication was maintained with the project's internal and external communities, as well as with other stakeholders. In addition, at each visit it was clarified and

informed about the steps related to the project activities, including the delivery of project brochures, as well as development stages and communication channels.

All results will be publicly available on the Internet and will be intended to develop summaries that will be communicated to the communities and other stakeholders through the appropriate means defined in Section 2.3.2 of the PD. In addition, all documents and information on monitoring and verification results will be published on the VCS and CCB standards platforms following validation and verification of the project.

4.4 Optional Criterion: Exceptional Community Benefits

Does not apply. This project is not intended to be validated for the Gold Level of this section.

4.4.1 Short-term and Long-term Community Benefits (GL2.2)

Does not apply.

4.4.2 Marginalized and/or Vulnerable Community Groups (GL2.4)

Does not apply.

4.4.3 Net Impacts on Women (GL2.5)

Does not apply.

4.4.4 Benefit Sharing Mechanisms (GL2.6)

Does not apply.

4.4.5 Governance and Implementation Structures (GL2.8)

Does not apply.

4.4.6 Smallholders/Community Members Capacity Development (GL2.9)

Does not apply.

5 BIODIVERSITY

5.1 Net Positive Biodiversity Impacts

5.1.1 Biodiversity Changes (B2.1)

In the project area, the following changes in biodiversity occurred in this first monitoring period:

Table 60. Changes in Biodiversity in the first monitoring period.

Change in Biodiversity	Habitat / Biodiversity Conservation (Fauna and Flora)
Monitored Change	Changes: actual, positive, and direct
Justification of Change	The ITAÚBA REDD+ Project avoided the deforestation of 152 hectares of the project area during this first monitoring period. The prevention was based on the practices of deforestation and degradation monitoring with MonitoraCarbon and land patrol, resulting in a positive impact on habitat and preserved the fauna and flora of the project area, preventing habitat loss, fragmentation, invasions, and other disturbances in the forest and to the biodiversity.

Change in Biodiversity	Environmental awareness
Monitored Change	Changes: actual, positive, and direct
Justification of Change	The project promoted environmental awareness with students from the Itaúba and Marcelândia schools and with workers from the properties that are within the project area. This were an effective introduction to one of the goals of the project, which aims to build an environmental awareness about conservation of biodiversity and forests.

5.1.2 Mitigation Actions (B2.3)

The activities most likely to negatively impact the biodiversity in the project area would be deforestation for cattle, soy farming or logging. To avoid these activities and mitigate the impacts, the ITAÚBA REDD+ Project worked, in this monitoring period, mainly on surveillance and monitoring of the area, and on environmental awareness for the community.

To protect and monitor the Project Area, the Deforestation Alert System (Carbonext System Deforestation Monitoring-MonitoraCarbon) was implemented. In addition, surveillance was reinforced with land patrols. This land patrol was automated through the QuickCapture app, which allows the employee in the field to mark georeferenced points of interest (fire outbreaks, deforestation, natural events, and invasion) and take pictures with his cell phone. Carbonext receives field data immediately, and in case of occurrences it can act more quickly and efficiently.

Lectures were given in schools and on farms to present the Itaúba REDD+ Project and its impacts, clarify doubts about the project and other related issues, such as how the carbon credit market works and the benefits of forest preservation. These activities aimed to involve the whole community in the project's objectives and introduce them to environmental issues, as well as help raise awareness about the environment, nature and species conservation, and the importance of forests.

Each activity is being carefully monitored by experts in each area, internal to Carbonext, and the precautionary principle will always be considered.

In this first monitoring period, negative impacts on biodiversity due to the project were not identified. The measures taken for maintenance or enhancement of the HCV attributes are associated to the reduction of deforestation, leading to no habitat loss and no biodiversity loss in the project area.

5.1.3 Net Positive Biodiversity Impacts (B2.2, GL1.4)

The impacts on biodiversity have been positive compared to the scenario without the project once the entire project area was protected with surveillance and monitoring actions during this monitoring period. This has prevented the conversion and degradation of land use for increased cattle ranching, soy farming, and logging, which are constant deforestation drivers in the reference region due to their economic attractiveness. Consequently, habitat loss, fragmentation, and invasions in the project area were avoided. Therefore, it can be said that the net impact on biodiversity is positive in this first monitoring report.

5.1.4 High Conservation Values Protected (B2.4)

The project's activities aim to avoid deforestation and maintain forest cover, creating environmental awareness and preparing local communities and landowners to deal with eventual degradation episodes.

The table below shows the HCV's identified in the project area, the actions taken to avoid negative impact and the actual results. Considering that the project contributed to conservation by avoiding deforestation and degradation of the area, the HCV's related to biodiversity are not negatively affected by the project activities.

Table 61: Biodiversity HCV identified in the ITAÚBA REDD+ Project, activity to avoid negative impact and the actual results.

HCV	Activity to avoid Negative Impact	Actual Results
HCV 1 - Diversity of Species HCV 3 - Ecosystem & Habitats	Increased environmental awareness and monitoring/surveillance of the project area, avoiding the deforestation of 152 ha of forest, habitat and species loss in this monitoring period.	Deforestation within the project area was less than projected in the scenario without the project, and consequently the negative impact on biodiversity also decreases.

5.1.5 Invasive Species (B2.5)

This item is not applicable to this project, since no species were introduced in the project area.

5.1.6 Impacts of Non-native Species (B2.6)

This item is not applicable to this project, since no species were introduced in the project area.

5.1.7 GMO Exclusion (B2.7)

This item is not applicable to this project, since GMO's were not used in the project.

5.1.8 Inputs Justification (B2.8)

This item is not applicable to this project, since no chemical inputs were used in the project.

5.2 Offsite Biodiversity Impacts**5.2.1 Negative Offsite Biodiversity Impacts (B3.1) and Mitigation Actions (B3.2)**

In this monitoring period, the Itaúba REDD+ Project did not result in negative impacts on biodiversity outside the project area, as the project activities were directed to protect the forest area within the project area and starting to introduce about environmental issues in the community.

5.2.2 Net Offsite Biodiversity Benefits (B3.3)

As mentioned in item 5.2.1, there were no net negative impacts outside the project area during this monitoring period. In case of any detection of negative impact, the project is prepared to mitigate and resolve any unexpected, after technical analysis made by Carbonext.

The project activities provide positive impacts for biodiversity due to the decrease of deforestation, helping in the conservation of fauna and flora. The environmental education activities will also help to spread environmental awareness about conservation of the forest and its living organisms outside the project area and a positive impact is expected in the long term.

5.3 Biodiversity Impact Monitoring**5.3.1 Biodiversity Monitoring Plan (B4.1, B4.2, GL1.4, GL3.4)**

In this first monitoring period, the focus of the project was to implement and improve monitoring. The project areas have been monitored remotely by the MonitoraCarbon system since June 2022. This system emits fire and deforestation alerts quickly, allowing early detection in case of occurrence.

The land patrols have been made by a local employee (forest monitor) using a motorcycle, and this patrolling has been done since the beginning of the Project. In November 2022, it was automated using the QuickCapture application, which allows the monitor to patrol and mark points of deforestation, illegal invasions, fire, and natural events. In addition to marking georeferenced points, the application also allows the recording of photos. This data is collected offline and at the first internet connection the application sends the data to Carbonext, which receives data from the field quickly and, in case of occurrences, can act immediately and effectively. The Operational Procedure "Using QuickCapture for Monitoring" has been developed to implement the use of this tool. The frequency and route of the land patrol was determined after investigating in the field the conditions of the roads, accesses, season (rainy/dry) and the most pressured borders. According to these characteristics, weekly, bi-weekly and monthly routes and frequencies are described in the operational procedure "Itauba Forestry Integrity Plan".



Figure 18. Training carried out in November 2022 to implement the application QuickCapture.

In addition, a mapping of border neighbors' contacts with the project farms was carried out, in order to create a community network to be activated in case of any occurrence. This way, action in the field can be faster, and neighbors can share resources such as water, trucks, and employees to act in case of forest fires, for example.

These actions to intensify and automate monitoring already allow the monitoring of biodiversity indicators B.9, B.10, B.12 and B.13 (following table). These actions are also part of the operational procedure of the project's "Forest Integrity Plan".

Table 62. Biodiversity monitoring plan.

Indicator ID	Variables to be monitored	Type	Area to be monitored	Monitoring frequency	Type of Measurement	Sampling Method	Objective	Result MR1
B.1	Total number of detected fauna species	State	PA	Every 5 years	Systematic campaigns during wet/dry season, using binoculars, playbacks of vocalizations, photographs and/or interviews with local residents.	Identification by: visualization, traces and/or vocalization,	Maintenance (or increase) in number of total fauna species	Not applicable to MR1
B.2	Number of detected endangered fauna species	State	PA	Every 5 years	Systematic campaigns during wet/dry season, using binoculars, playbacks of vocalizations, photographs and/or interviews with local residents.	Identification by: visualization, traces and/or vocalization,	Maintenance (or decrease) of number of endangered species or it decrease	Not applicable to MR1
B.3	Number of detected invasive fauna species	State	PA	Every 5 years	Systematic campaigns during wet/dry season, using binoculars, playbacks of vocalizations, photographs and/or interviews with local residents.	Identification by: visualization, traces and/or vocalization,	Maintenance (or decrease) of number of invasive species	Not applicable to MR1
B.4	Number of new fauna species detected	State	PA	Every 5 years	Systematic campaigns during wet/dry season, using binoculars, playbacks of vocalizations, photographs and/or interviews with local residents.	Identification by: visualization, traces and/or vocalization,	N/A	Not applicable to MR1

Indicator ID	Variables to be monitored	Type	Area to be monitored	Monitoring frequency	Type of Measurement	Sampling Method	Objective	Result MR1
B.5	Total number of detected flora species	State	PA	Every 6 years	Fauna inventory carried out by a third party	Through plot allocations	Maintenance (or increase) in number of total fauna species	Not applicable to MR1
B.6	Number of detected endangered flora species	State	PA	Every 6 years	Fauna inventory carried out by a third party	Through plot allocations	Maintenance (or decrease) of number of endangered species or it decrease	Not applicable to MR1
B.7	Number of detected invasive flora species	State	PA	Every 6 years	Fauna inventory carried out by a third party	Through plot allocations	Maintenance (or decrease) of number of invasive species	Not applicable to MR1
B.8	Number of new flora species detected	State	PA	Every 6 years	Fauna inventory carried out by a third party	Through plot allocations	N/A	Not applicable to MR1
B.9	Extent of natural vegetation as measured in satellite imagery (ha)	State	PA and LB	Every year	Measured through GIS software	N/A	Maintenance of the Project Area Forest integrity	Avoided deforestation in 152 ha
B.10	Extent of forest area impacted by forest fires (ha)	Pressure	PA and LB	Every year	Satellite sensors constantly monitor the incidence of fire focal points. Based on the data, field expeditions are held to measure the extension of the impact when significative. The sum	N/A	Avoid significant forest fire impacts inside the Project Area	0 ha

Indicator ID	Variables to be monitored	Type	Area to be monitored	Monitoring frequency	Type of Measurement	Sampling Method	Objective	Result MR1
					of the significant impacted area is the total area considered to be impacted by forest fires.			
B.11	Loss of vegetation cover due to sustainable forest management (ha)	Pressure	Annual Production Units (areas planned for forest management activities)	Every year	Measurement of forest area opened for road, log decks and skid trails construction	Through plot allocations	Keep the opening of areas for infrastructure within the limit established by law ⁸²⁵¹	0 ha
B.12	Frequency of guard patrols (Patrols per month)	Response	PA	Every year	Collection of evidence for every patrol using the QuickCapture Application. Measurement of frequency utilizing only valid patrols according to respective operating procedure.	N/A	Execution of guard patrols 4 times per month	The frequency of land patrols has occurred as planned (Table 4).
B.13	Constancy of MonitoraCarbon deforestation alert system (Days functioning/days not functioning in a given monitoring period)	Response	N/A	Every year	Collection of evidence of the systems functionality through the period	N/A	Constant functionality of the deforestation alert system (Monitora Carbon)	The MonitoraCarbon system has worked constantly since its implementation.

Due to actions regarding indicators B.9, B.10, B.12 and B.13, it was possible to avoid 152 hectares of forest from deforestation, fires, and invasions in this monitoring period. The MonitoraCarbon system has worked constantly since its implementation. The frequency and dates of land patrols has occurred as planned, as shown in section 2.2.7. In addition, there was no sustainable forest management during this monitoring period. All this has contributed to the maintenance of the forests and consequently with the biodiversity of fauna and flora.

For the Itaúba REDD+ Project, the inventories of fauna and flora were not planned for this monitoring period and will occur in 2025 (fauna) and 2026 (flora).

5.3.2 Biodiversity Monitoring Plan Dissemination (B4.3)

The biodiversity monitoring plan will be made public on Carbonext' social medias and Verra's website. Summaries of the monitoring results will be presented for the communities during the next onsite visit. For other stakeholders the results will be sent by email.

5.4 Optional Criterion: Exceptional Biodiversity Benefits

Does not apply. This project is not intended to be validated for the Gold Level of this section.

5.4.1 Trigger Species Population Trends (GL3.3)

Not applicable.

6 ADDITIONAL PROJECT IMPLEMENTATION INFORMATION

Not additional information.

7 ADDITIONAL PROJECT IMPACT INFORMATION

Not additional information.

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